Contents

Exact Generating-Function Analysis of a Two-Class Non-Preemptive Priority $M/M/1$ Queue with Jockeying and Abandonment

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Abstract

We study the stationary queue-length distribution of a two-class, non-preemptive priority $M/M/1$ queue augmented with two waiting-room mechanisms: *jockeying*, in which a waiting customer switches to the other queue, and *abandonment*, in which a waiting customer leaves the system. Jockeying conserves the total number of customers, whereas abandonment does not, and combining both yields a model (Model-X) that is analytically intractable. We therefore present a collection of simpler models: Model-A (baseline), Model-B (bidirectional jockeying), Model-B_2 (jockeying from the priority class to the non-priority class) and Model-C_2 (abandonments in the priority class). We also study two head-of-line models: Model-B_2^{H} (only the first customer in the priority queue can jockey) and Model-C_2^{H} (only the first customer in the priority queue can abandon).

The central object is the bivariate Probability Generating Function (PGF) $P(x, y)$ of the two queue lengths, obtained by analytical and, where the structure allows, probabilistic arguments. The purpose of this work is to present a unified framework for deriving the functional equation of each model and to expose the mathematical challenges intrinsic to each. Model-A is classical and collapses to an algebraic expression by finding a kernel root inside the unit disc; Model-B_2 is solved by imposing analyticity at an interior point and Model-C_2 by imposing boundedness at the unit-disc boundary $x = 1$. The head-of-line models, like the baseline Model-A, also yield an algebraic expression for the PGF. Two models are left open: Model-B reduces to an integral expression for the PGF involving a Volterra-type integral equation, and Model-X compounds this with nonlinear characteristic curves.

Every closed form is validated against a truncated continuous-time Markov Chain (CTMC) that recovers the baseline model as the mechanism rates vanish. The comparison reveals a key difference between jockeying and abandonment: the former merely redistributes delay between classes, whereas the latter substantially shortens the queue lengths.

1 Introduction

Motivation

Priority queues arise naturally wherever a common server must satisfy heterogeneous demand. The primary motivation for this thesis is healthcare operations, specifically the management of waiting lists for elderly-care facilities, where medical severity dictates a user's —hereafter *customer*— priority class. Here prolonged waiting may itself alter the system's dynamics: a lack of appropriate care can deteriorate a waiting customer's health, leading them to change priority status (jockeying) or leave the system (abandonment), whether to seek alternative care or through death. Despite this motivation, the work is fundamentally modelling-driven: we characterise the long-run behaviour of queueing systems in which jockeying and/or abandonment play a role.

This thesis analyses a 2-class $M/M/1$ queue with non-preemptive priority: each class arrives according to a Poisson process (rates λ_1 and λ_2), service times are exponential at a common rate μ, and the single server completes the current service even when a higher-priority customer arrives. To this we add the mechanisms of jockeying and abandonment. The model and its mechanisms are described in more detail in §4.

How these two mechanisms interact within a priority discipline, and how that interaction affects the difficulty of obtaining a closed-form Probability Generating Function (PGF), is the main focus of this thesis.

The analytical challenge

In full generality, the bivariate PGF for Model-X satisfies a functional equation with terms of three kinds: algebraic coefficients, partial derivatives in x —from jockeying and abandonment in class-1— and partial derivatives in y —from the same mechanisms in class-2. When both kinds of derivative term are present, the equation belongs to a family of integro-differential equations whose solution would require more advanced tools. The problem is moreover asymmetric: derivative terms from the non-priority class complicate it more than those from the priority class alone.

We therefore solve less complex models obtained by setting some of the jockeying and abandonment parameters to zero. Table 1 lists the models studied here and indicates whether a PGF expression is obtained for each:

Model	**Active mechanism**	**Equation type**	**Status**
X	jockeying and abandonment	PDE, transcendental characteristics	open
A	none	algebraic PGF equation	solved
B	bidirectional jockeying	PDE, Volterra family	open
B_2	one-way jockeying ($1 \to 2$)	ODE in x, Kummer	solved
C_2	class-1 abandonment	ODE in x, Kummer	solved
B_2^{H}	one-way jockeying, head-of-line	algebraic PGF equation	solved
C_2^{H}	class-1 abandonment, head-of-line	algebraic PGF equation	solved

Table 1: Models studied in this thesis, with the active mechanism, the resulting functional-equation type, and the solution status. The head-of-line variants B_2^{H} and C_2^{H} restrict the mechanism to the customer at the head of Queue 1, which collapses the derivative term to an algebraic one and renders the model solvable in closed form.

Some models —notably Model-X and Model-B— are left incomplete because they involve derivative terms in y; others, such as a potential Model-B_1, Model-C or Model-C_1, are not considered for the same reason, as they all reduce to a Volterra integral equation lying beyond the scope of this thesis. Model-B serves as our representative example of how such problems reduce to that integral equation; the remaining models are solvable. The last two entries, Models B_2^{H} and C_2^{H} (§§9 and 10), are *head-of-line* simplifications in which only the customer at the head of Queue 1 may jockey or abandon. The mechanism then acts at the constant rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than the length-proportional $\gamma_1 n_1$ or $\theta_1 n_1$, so the fundamental equation stays algebraic and is solved by the Kernel Method exactly as in Model-A.

Methods

Three complementary proof strategies are used throughout.

Kernel method / ODE with integrating factor. For Model-A, which has no derivative terms, the algebraic coefficient of $P(x, y)$ vanishes on a curve (the *kernel*); evaluating the equation there determines the remaining unknowns. Model-B_2 and Model-C_2, which do carry derivative terms in x, reduce to a first-order linear ODE in x with an explicit integrating factor, and their unknowns follow from a regularity argument at a singularity of that factor. We present both strategies together, as they share the same underlying idea: the latter extends the former to the case where the unknowns are independent of the derivative terms and can be taken outside the integral.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, class-2 arrivals are Poisson, so the PASTA property holds and we can apply the Pollaczek–Khinchine formula together with the *effective service time* seen by a class-2 customer. This is an $M/G/1$ queue whose service times equal the busy period of the class-1 subsystem —an $M/M/1$ queue for Model-A and an $M/M/1+M$ system for Model-C_2. In both cases the argument independently confirms the analytic results. Since jockeying destroys the Poisson structure, it does not apply to the other models.

Partial PGF (PPGF) in the alternative state space $\widetilde{S}$. For the baseline Model-A we introduce an alternative state space indexed by the number of class-2 customers —whose distribution is hard— and the total number of customers —which is easy, since the total is conserved and behaves like a regular $M/M/1$ queue. This yields a PPGF with a non-homogeneous term. Cohen's recursion argument [Coh56] was developed for the regular state space; we explore, with limited success, the challenges of carrying it over to this alternative space and assess whether Cohen's trick can deliver a feasible rough approximation.

Structure of the thesis

Section 2 reviews the relevant literature and Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines the full model, Model-X, and derives the general balance equations and the fundamental PGF equations, both for the regular and the alternative state space. The analysis then proceeds from baseline to synthesis: Section 5 solves the baseline Model-A in closed form; Section 6 shows that bidirectional jockeying (Model-B) reduces the problem to a Volterra-type integral equation and is left open. Sections 7 and 8 then recover tractability in the one-way-jockeying (Model-B_2) and class-1-abandonment (Model-C_2) specialisations, while Sections 9 and 10 show how the head-of-line variants, Models B_2^{H} and C_2^{H}, collapse the functional equation to an algebraic one solved in closed form. Finally, Section 11 validates every closed form against an exact CTMC and compares the solved models (§11.2), and Section 12 concludes by abstracting the obstruction taxonomy and setting out the future work.

2 Literature

The study of two-class priority queues with departure mechanisms sits at the intersection of three well-developed streams: the exact generating-function analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying and dynamic priority changes. The applied motivation throughout is the management of access to care — elderly-care waiting lists, emergency departments, and transplant registries — in which the two mechanisms studied here carry direct clinical readings: a waiting patient whose condition deteriorates changes priority class (jockeying), while one who leaves the list by death or transfer reneges (abandonment). This thesis contributes an exact, single-server treatment that admits jockeying and abandonment within a unified bivariate-PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

The direct methodological ancestor of this work is Cohen [Coh56], who studied a single server loaded by two independent Poisson streams and introduced the partial-generating-function recursion — the "Cohen trick" — that resolves the lower-priority queue length by transforming only one coordinate of the joint distribution while keeping the other discrete. That device is exactly the one reused, in modern notation, in the partial-PGF analysis of Model-A on the total-count state space (§5.2). The stationary queue-length law of the two-class non-preemptive $M/M/1$ priority queue — the content of Theorem 1 (Model-A) — is in this sense classical, recoverable by several routes; the contribution of the present work is not that result in isolation but the *unified framework* built around it, into which jockeying and abandonment are introduced one mechanism at a time and solved by a common functional-equation method.

The most recent exact closed-form treatment in this line is that of Zuk & Kirszenblat [ZK23], who obtain the joint and marginal queue-length distributions of a non-preemptive $M/M/c$ queue with two priority classes and a common service rate by generating-function methods and a quadratic recurrence; their single-server specialisation ($c = 1$) coincides with Model-A. A complementary strand studies dynamic rather than static priority: Stanford et al. [STZ14] analyse the *accumulating-priority* $M/G/1$ queue, in which each class earns priority linearly in its waiting time so that older customers eventually overtake newer ones, and derive exact waiting-time distributions through a maximum-priority process and Laplace–Stieltjes transforms. Accumulating priority is a continuous-time analogue of the discrete, rate-γ_i jockeying studied here; like the present work it is exact and single-server, but its object is the sojourn time rather than the bivariate queue-length PGF.

Priority queues with abandonment

Abandonment enters queueing theory through the Erlang-A ($M/M/c + M$) model, in which each waiting customer holds an independent exponential patience clock; the modern treatment and its heavy-traffic scaling are due to Garnett et al. [GMR02], and the single-server case $M/M/1 + M$ supplies the busy-period structure on which Model-C_2 rests (§3.3). The analytic signature of impatience is well documented: per-customer patience clocks make the transition rates state-inhomogeneous, so the generating-function solutions, where they close at all, close in terms of hypergeometric series.

Kapodistria [Kap11] exhibits this pattern at its sharpest in a single-server queue with *synchronised* abandonments — all waiting customers abandon simultaneously, each with probability p, at Poisson opportunity epochs — solving the stationary law exactly through basic (q-)hypergeometric series and recovering, in the no-synchronisation limit $p \to 0^+$, the $M/M/1+M$ queue-length PGF as a ratio of Kummer functions: precisely the special-function structure that carries the boundary function of Model-C_2.

Exact stationary results for priority queues *with* reneging are scarce and almost always multi-server. Jouini & Roubos [JR14] treat an $M/M/s+M$ queue with two non-preemptive classes and per-class exponential patience, deriving Laplace–Stieltjes transforms for conditional and unconditional waiting times together with the probabilities of service and of abandonment — the multi-server analogue of Model-C_2, but with the waiting time, not the queue-length PGF, as its object. Where exact transforms are unavailable the literature turns to fluid and diffusion approximations or to many-server asymptotics, in which the Erlang-A stationary law and its descendants are expressed through confluent hypergeometric and incomplete-gamma functions [ZM05]; the same special-function structure appears exactly, and at finite scale, in the closed form for Model-C_2.

The exact multi-server priority results that do exist carry no class-asymmetric reneging — those of Wang et al. [WBSW15] and Zuk & Kirszenblat [ZK23] are abandonment-free — so Model-C_2 appears to be the first exact, single-server, queue-length treatment of *class-asymmetric* abandonment, with impatience acting on the priority class alone. The clinical reading that licenses this asymmetry is long-standing: Zenios [Zen99] models the organ-transplant waiting list as a queue whose reneging is patient mortality, precisely the regime in which losing high-priority demand is the outcome of interest.

Jockeying and priority changes

The closest predecessor on the jockeying side is de Waal [dW92], who studies an $M/G/1$ priority queue in which impatient customers *change priority class* while waiting. Its entry in Table 2 is marked approximate not by choice of method but by necessity: with general service times the governing equations do not close, and the analysis delivers Laplace–Stieltjes-based *approximations* of the waiting-time distributions rather than an exact stationary law. The healthcare incarnation of the same idea is the model of Hu, Chan & Dong [HCD22], who let customers transition between a moderate and an urgent class as their condition deteriorates or improves and characterise a near-optimal $c\mu/\theta$-type scheduling index in heavy traffic; the system is multi-server and its results are asymptotic, so it too is approximate in the sense of Table 2. Against this backdrop Model-B_2 is the exact, single-server counterpart of the approximate multi-server "priority-change" literature: it retains the deterioration mechanism — a waiting class-1 customer relocating to the class-2 queue at rate γ_1 — but pays for exactness with the single-server, Markovian, one-directional restriction, returning a closed-form boundary function in place of an approximation.

Multi-server extensions and performance design

The last reference bounds the present work from the side of scale. Wang et al. [WBSW15] give the exact queue-length PGF of an $M/M/c$ queue with two *preemptive* priority classes and class-dependent service rates, a useful contrast to the non-preemptive, common-rate discipline used throughout this thesis. That the result carries no departure mechanism is symptomatic of the multi-server literature as a whole: once $c > 1$ is combined with jockeying or abandonment, the analysis moves to approximations and asymptotics, confirming that exact, mechanism-rich analysis is essentially a single-server phenomenon.

Methodological lineage

The functional-equation technique used in Sections 7 and 8 belongs to the kernel-method and boundary-value tradition for two-dimensional Markovian queues. The systematic theory — reducing a bivariate balance equation to the curve on which its kernel vanishes, and from there to a boundary-value problem for the unknown boundary functions — is developed by Fayolle, Iasnogorodski & Malyshev [FIM99] for random walks in the quarter-plane. That programme still delivers explicit two-class results: Devos, Walraevens, Fiems & Bruneel [DWFB20] obtain the closed-form joint queue-length distribution of a discrete-time two-class queue with randomly alternating service by analysing the kernel of just such a functional equation and pinning its two unknown boundary functions through analytic continuation.

The pinning arguments of this thesis are the elementary, one-dimensional residue of that programme. In Model-B_2 the boundary function is fixed by demanding *analyticity* of the PGF at the interior point where the kernel root meets the diagonal (§7), and in Model-C_2 by demanding mere *boundedness* at the unit-disc boundary (§8); both select the analytic solution of the reduced equation, exactly the role the boundary conditions play in the general theory.

Two narrower tools recur. The confluent hypergeometric (Kummer) function that carries the Model-C_2 boundary function is the same special function through which the Erlang-A and reneging-queue stationary laws are expressed [ZM05, Kap11, DLM]; and the selection of the decaying solution of the three-term recurrence behind the partial-PGF analysis is governed by Pincherle's theorem, for which we cite the recurrence-relation treatment of Gautschi [Gau67] alongside the DLMF [DLM].

Position of this thesis

Table 2: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
This thesis (Model-A)	**1**	✓	–	–	✓	✓
This thesis (Model-B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model-C_2)	**1**	✓	–	✓	✓	✓

Table 2 places this work against its closest neighbours, and reading it by row makes the gap precise. The Cohen (1956) row is the single-server, exact, non-preemptive queue-length baseline with neither extra mechanism — the ancestor of Model-A. de Waal (1992) adds jockeying (priority change) at a single server, but only approximately and for waiting times rather than queue lengths. Jouini & Roubos (2014) add abandonment exactly, but at c servers and again for waiting times. Wang et al. (2015) is exact and queue-length-focused, yet multi-server, preemptive, and mechanism-free. Stanford et al. (2014) is single-server and exact with a dynamic-priority (jockeying-like) mechanism, but its object is the sojourn time. Zuk & Kirszenblat (2023) is the exact multi-server queue-length result whose $c = 1$ case is Model-A, with neither jockeying nor abandonment. Hu, Chan & Dong (2022) carries both mechanisms — jockeying-as-deterioration and abandonment — but multi-server and only approximately.

No row above combines, at a single server, the exact bivariate queue-length PGF with either jockeying or abandonment — which is precisely the column pattern of the three "This thesis" rows. The gap can therefore be stated in one sentence: prior to this work there is no exact, single-server, queue-length analysis that places non-preemptive priority together with jockeying or abandonment inside a single PGF framework.

The non-claims must be stated with equal precision. The framework is unified but not complete: the full model (Model-X, Remark 3), in which jockeying and abandonment act together, and the bidirectional-jockeying model (Model-B, §6) are left open, each reducing to a Volterra-type integral equation that this thesis characterises but does not solve in closed form. The exact contributions are confined to the three "This thesis" rows of Table 2 — the baseline Model-A, the one-way jockeying Model-B_2, and the class-1 abandonment Model-C_2, together with their head-of-line simplifications. The claim is exactness and unification at a single server with one mechanism active at a time; it is explicitly not a solution of the combined or bidirectional problems, which remain, as in the multi-server literature, beyond exact reach.

3 Preliminaries

This section collects the background material used throughout the derivations, so that the later arguments can stay focused on their primary objectives.

The central object throughout is the (P)PGF of the model in question, together with the system's idle probability for some corollaries.

3.1 $M/M/1$ queue

The canonical $M/M/1$ queue is the foundation of queueing theory and is treated in many textbooks; we follow [AR02]. Customers arrive according to a Poisson process —i.e. the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-serve (FCFS) —also known as first-in-first-out (FIFO)— order. The system is stable if and only if the load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ $(n \geq 0)$ together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0.$$

By the PASTA property (§3.2), this also corresponds to the distribution seen by arriving customers. Let L be the number of customers in the system, including any in service. Its expectation is

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time between a customer's arrival and their departure after service—follows by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

A less common but highly relevant concept is the Busy Period (BP): the time between a customer arriving to an idle system and the departure that next leaves it empty. Its Laplace–Stieltjes Transform (LST) and expectation are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{3.1}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3.2}$$

3.2 Non-preemptive priority and the PASTA property

Throughout this thesis the two customer classes are served under a *non-preemptive* priority discipline: whenever the server becomes free it admits a waiting class-1 customer if any are present, and a class-2 customer only otherwise, but a customer already in service is *never* interrupted—a class-1 arrival that finds a class-2 customer in service waits until that service completes. Priority therefore governs only the order in which the queues are drained, not the service in progress. Because the service rate μ is common to both classes, the *memorylessness* of the exponential service time makes the residual service of the customer in progress $\mathrm{Exp}(\mu)$ regardless of its class; hence the class of the in-service customer does not influence the future evolution of the queue lengths, which is why a single in-service customer of either class suffices to mark the server busy.

We also invoke repeatedly the *PASTA* property (*Poisson Arrivals See Time Averages*) [AR02]: if a system is fed by a Poisson arrival process that satisfies the *lack-of-anticipation assumption* (at each instant the future increments of the arrival process are independent of the current system state), then the long-run fraction of arrivals that find the system in a given state equals the stationary probability of that state, so the distribution seen by an arriving customer coincides with the time-stationary one. Both hypotheses hold here: the exogenous class-i arrivals are independent Poisson processes at rates λ_i, and they are independent of the system's internal exponential service, jockeying and abandonment clocks, so they cannot anticipate the future. By Poisson *superposition* the merged arrival stream is itself Poisson at rate $\lambda = \lambda_1 + \lambda_2$, so PASTA applies to the aggregate state; and since each class stream is an independent Poisson process—equivalently, an i.i.d. λ_i/λ *thinning* of the merged stream—PASTA applies to each class separately.

3.3 $M/M/1{+}M$ queue (Erlang-A)

Another model relevant to this thesis is the $M/M/1{+}M$ queue, also known as the *Erlang-A* model. It extends the $M/M/1$ queue —Poisson arrivals at rate λ, exponential service at rate μ— with customer impatience: each *waiting* customer (the one in service is not impatient) independently abandons after an exponentially distributed patience time with rate θ. Crucially, unlike the $M/M/1$ it does not require $\rho = \lambda/\mu < 1$ for stability: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady-state distribution follows from its balance equations, where n denotes the number of customers in the *system*:

$$\lambda\,\pi_n = (\mu + n\theta)\,\pi_{n+1}, \qquad n \geq 0.$$

Iterating yields

$$\pi_n = \pi_0\,\frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1)$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)]^{-1}$ (see §3.5 for the Kummer function ${}_1F_1$), and summing the geometric-like terms shows that the queue-length PGF is correspondingly a *ratio* of Kummer functions, $\Pi(z) = {}_1F_1(1;\, \mu/\theta;\, (\lambda/\theta)z)\,/\,{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)$. The same closed form is derived by Kapodistria [Kap11, Thm. 3] as the no-synchronisation limit of a queue with synchronised abandonments, there in the sojourn-time-impatience convention (the customer in service also abandons), which amounts to replacing μ by $\mu + \theta$.

Unlike the textbook facts collected above, the busy-period identity that follows is the one preliminary we derive from scratch, as Model-C_2 rests on it directly. The *Busy Period* of this $M/M/1 + M$ is far more complex than its $M/M/1$ counterpart. We record it for the class-1 subsystem used in Model-C_2, with arrival rate λ_1, service rate μ and abandonment rate θ_1, so that the relevant idle probability is $\pi_0 = [{}_1F_1(1;\, \mu/\theta_1;\, \lambda_1/\theta_1)]^{-1}$. Writing B_C for this busy period, its LST and expectation are

$$\widetilde{B}_C(s) = \cfrac{\mu}{\lambda_1 + \mu + s - \cfrac{\lambda_1(\mu + \theta_1)}{\lambda_1 + \mu + \theta_1 + s - \cfrac{\lambda_1(\mu + 2\theta_1)}{\lambda_1 + \mu + 2\theta_1 + s - \cdots}}}, \tag{3.3}$$

$$\mathbb{E}[B_C] = \frac{1}{\lambda_1}\Big({}_1F_1\big(1;\, \mu/\theta_1;\, \lambda_1/\theta_1\big) - 1\Big) \tag{3.4}$$

The continued fraction (3.3) is generated by a first-passage recursion. Let $\phi_n(s)$ denote the LST of the first-passage (sub-busy-period) time from n customers in the system down to $n-1$. Conditioning on the first transition out of state n—an arrival (rate λ_1) to state $n+1$, or a departure by service completion or abandonment (combined rate $\mu + (n-1)\theta_1$: one customer in service plus $n-1$ waiting, each impatient at rate θ_1) to state $n-1$—the strong Markov property gives

$$\phi_n(s) \;=\; \frac{\mu + (n-1)\theta_1}{\lambda_1 + \mu + (n-1)\theta_1 + s - \lambda_1\,\phi_{n+1}(s)}, \qquad n \geq 1, \qquad \widetilde{B}_C(s) = \phi_1(s). \tag{3.5}$$

Unrolling (3.5) from $n = 1$ reproduces the continued fraction (3.3) term by term. The continued fraction also sums in closed form, and the argument is short enough to give in full. Fix $s > 0$ and scale the rates by θ_1, setting

$$a = \frac{\mu}{\theta_1}, \qquad b = \frac{s}{\theta_1}, \qquad c = \frac{\lambda_1}{\theta_1},$$

so that (3.5) reads $\phi_n(s) = (a+n-1)/\big((a+n-1) + b + c - c\,\phi_{n+1}(s)\big)$. Consider the integrals

$$I(p) \;=\; \int_0^1 t^{\,p}\,(1-t)^{\,b-1}\,e^{-ct}\,dt, \qquad p > -1,$$

which by the Euler representation (3.10) are Beta-function multiples of Kummer values, $I(p) = B(p+1,\, b)\,{}_1F_1(p+1;\, p+1+b;\, -c)$. Integrating the exact derivative $\frac{d}{dt}\big[t^{\,p}(1-t)^{\,b}e^{-ct}\big]$ over $[0,1]$ — the boundary terms vanish for $p, b > 0$ — and splitting $(1-t)^b = (1-t)^{b-1} - t\,(1-t)^{b-1}$ in the two terms where it appears yields the three-term recurrence

$$(p+b+c)\,I(p) \;=\; p\,I(p-1) \;+\; c\,I(p+1), \tag{3.6}$$

a relation of contiguous type for ${}_1F_1$ (obtainable equivalently by combining the contiguous relations of [DLM, §13.3]). Dividing (3.6) at $p = a+n-1$ by $I(a+n-2)$ shows that the ratios $R_n := I(a+n-1)/I(a+n-2)$ satisfy exactly the recursion of ϕ_n. Two further facts pin the identification. First, by Laplace's method at the endpoint $t = 1$, $I(p) \sim e^{-c}\,\Gamma(b)\,p^{-b}$ as $p \to \infty$: the solution $\{I(p)\}$ decays only polynomially, whereas the dominant solution of (3.6) grows superexponentially (its consecutive ratio grows like p/c), so $\{I(p)\}$ is the *minimal (recessive)* solution. Second, by *Pincherle's theorem* the convergent continued fraction equals precisely the ratio built from the minimal solution. Hence

$$\widetilde{B}_C(s) \;=\; \phi_1(s) \;=\; \frac{I(a)}{I(a-1)} \;=\; \frac{B(a+1,\,b)\;{}_1F_1\big(a+1;\,a+1+b;\,-c\big)}{B(a,\,b)\;{}_1F_1\big(a;\,a+b;\,-c\big)}, \tag{3.7}$$

a contiguous ratio of the form (3.11). This identity is used, with $s = \lambda_2(1-y)$, in the boundedness route for Model-C_2 (§8, Equation (8.11)).

3.4 Pollaczek–Khinchine formula

The $M/G/1$ queue [AR02] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho_B = \lambda\,\mathbb{E}[B] < 1$ is required (written ρ_B to keep the symbol ρ for the two-class load of this thesis). Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\,\widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z)) - z}. \tag{3.8}$$

3.5 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* appears in several derivations in this thesis; it reads as follows:

$${}_1F_1(a;\,b;\,z) \;=\; \sum_{n=0}^{\infty} \frac{a^{(n)}}{b^{(n)}\,n!}\,z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1}(a+k), \quad a^{(0)} = 1, \tag{3.9}$$

where b is not 0 or a negative integer —i.e. $b \notin \{0,-1,-2,\dots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\,b;\,z)$ is the unique solution of *Kummer's equation* [DLM]:

$$z\,u'' + (b-z)\,u' - a\,u = 0,$$

normalised by ${}_1F_1(a;\,b;\,0) = 1$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$${}_1F_1(a;\,a+b;\,z) \;=\; \frac{1}{B(a,b)}\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt, \tag{3.10}$$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, *contiguous ratios*: ratios of two ${}_1F_1$ values whose parameters differ by integer shifts. Two patterns arise in this thesis. The *simultaneous shift*

$$\frac{{}_1F_1(a+1;\,a+1+b;\,z)}{{}_1F_1(a;\,a+b;\,z)}, \tag{3.11}$$

in which both parameters move by one so that their difference b is preserved, is by the Euler representation (3.10) a ratio of two weighted Beta integrals differing only in the exponent of t; it appears as the closed-form value of the Erlang-A busy-period continued fraction (§3.3, Equation (3.7)) and hence in the boundary function of Model-C_2. The *first-parameter shift* ${}_1F_1(a+1;\,b;\,z)\big/{}_1F_1(a;\,b;\,z)$, with the second parameter fixed, appears instead in the boundary function of Model-B_2 (§7).

3.6 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms appear in the derivations, requiring us to solve first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.6.1 Integrating factor

This method is standard to solve —for each fixed y— a *first-order linear ODE in x*. Such an equation has standard form

$$\frac{dP}{dx} + p(x;\, y)\, P \;=\; q(x;\, y). \tag{3.12}$$

The *integrating factor* $\mu_I(x;y) = \exp\bigl(\int_0^x p(t;y)\,dt\bigr)$ converts (3.12) to $\frac{d}{dx}[P\cdot\mu_I] = q\cdot\mu_I$. Integrating from 0 to x gives the general solution

$$P(x,y) \;=\; \frac{1}{\mu_I(x;\, y)}\left[C_0(y) + \int_0^x q(t;\, y)\,\mu_I(t;\, y)\,dt\right], \tag{3.13}$$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.6.2 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x, y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$a\frac{\partial P}{\partial x} + b\frac{\partial P}{\partial y} = cP + d \tag{3.14}$$

is solved by tracing the characteristic curve $(x(t), y(t))$ satisfying

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b. \tag{3.15}$$

By the chain rule, $\frac{d}{dt}P(x(t), y(t)) = a\,P_x + b\,P_y$, so (3.14) reduces to

$$\frac{dP}{dt} = cP + d, \tag{3.16}$$

a first-order linear ODE in t. Its homogeneous part $\frac{dP}{dt} = cP$ is solved at once by separation of variables; the inhomogeneous term d is then absorbed either by the integrating factor (§3.6.1) or, once that homogeneous solution is in hand, by variation of parameters (§3.6.3). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \;:=\; b\,x - a\,y$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of (3.16) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x, y) via (3.15) gives the general solution of (3.14). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x, y)$ for $|x|, |y| \le 1$ uniquely determines it.

3.6.3 Variation of parameters

The integrating factor of §3.6.1 builds a solution from scratch by constructing μ_I. Frequently, however, we *already* hold a nonzero solution of the homogeneous equation —most importantly, the one the Method of Characteristics (§3.6.2) produces along each characteristic curve. *Variation of parameters* is the technique that turns any such known homogeneous solution directly into the general solution of the inhomogeneous equation, with no further integrating factor required. We describe it in the form in which it is used later.

Consider a first-order linear ODE written as

$$\frac{dP}{dx} = f(x)\,P + h(x), \tag{3.17}$$

and suppose a nonzero homogeneous solution P_h is known, that is, $\frac{dP_h}{dx} = f(x)\,P_h$. The homogeneous equation then has general solution $C\,P_h(x)$ for an arbitrary constant C. To solve the full equation (3.17), we *vary the constant*: we promote C to a function $C(x)$ and look for a solution of the form

$$P(x) = C(x)\,P_h(x).$$

Differentiating by the product rule and using $P_h' = f\,P_h$,

$$\begin{aligned}\frac{dP}{dx} &= C'(x)\,P_h(x) + C(x)\,P_h'(x)\\ &= C'(x)\,P_h(x) + f(x)\,\big[C(x)\,P_h(x)\big] = C'(x)\,P_h(x) + f(x)\,P(x).\end{aligned}$$

Substituting this back into (3.17), the common term $f(x)\,P(x)$ cancels from both sides and only the derivative of the varied constant survives:

$$C'(x)\,P_h(x) = h(x) \qquad \Longrightarrow \qquad C'(x) = \frac{h(x)}{P_h(x)}.$$

Integrating from a base point x_0 gives $C(x) = F + \int_{x_0}^{x} h(t)/P_h(t)\,dt$ with F an arbitrary constant, and hence the general solution

$$P(x) = P_h(x)\left[F + \int_{x_0}^{x} \frac{h(t)}{P_h(t)}\,dt\right]. \tag{3.18}$$

The known homogeneous solution P_h thus does double duty: it is the overall prefactor and, as $1/P_h$, the weight inside the integral. The constant F is fixed by a boundary condition. Taking in particular $P_h = 1/\mu_I$ reproduces the integrating-factor formula (3.13) of §3.6.1.

This is exactly the step that completes the Method of Characteristics for an *inhomogeneous* PDE. Once §3.6.2 has reduced the PDE to an ODE of the form (3.17) along a characteristic and supplied its homogeneous solution P_h, formula (3.18) delivers the full solution along that characteristic, the constant F becoming an arbitrary function $F(\xi)$ of the characteristic label. Model-B (§6) is solved by precisely this pairing.

4 The general model (Model-X) and its fundamental equation

This section defines the general two-class system with both jockeying (γ_i) and abandonment (θ_i); we call this full model *Model-X*. The simpler variants are recovered as special cases by zeroing the appropriate parameters, as summarised in Table 3:

Model	Jockeying	Abandonment	Parameter restriction
X	both $(1\leftrightarrow 2)$	both $(1,2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1\leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1\to 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$
B_2^{H}	$1\to 2$, head-of-line	none	$\gamma_1\,\mathbf{1}_{\{n_1\geq 1\}} > 0,\ \gamma_2 = \theta_1 = \theta_2 = 0$
C_2^{H}	none	class 1, head-of-line	$\theta_1\,\mathbf{1}_{\{n_1\geq 1\}} > 0,\ \gamma_1 = \gamma_2 = \theta_2 = 0$

Table 3: Nested family of models. Model-X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes. The two head-of-line variants B_2^{H} and C_2^{H} are listed separately, as their zeroing mechanism involves $\mathbf{1}_{\{n_1\geq 1\}}$ rather than being proportional to the queue length.†

Figure 1 gives a visual overview of Model-X and each variant we consider. It omits the two head-of-line variants, which are less conventional and for which such a diagram would add little:

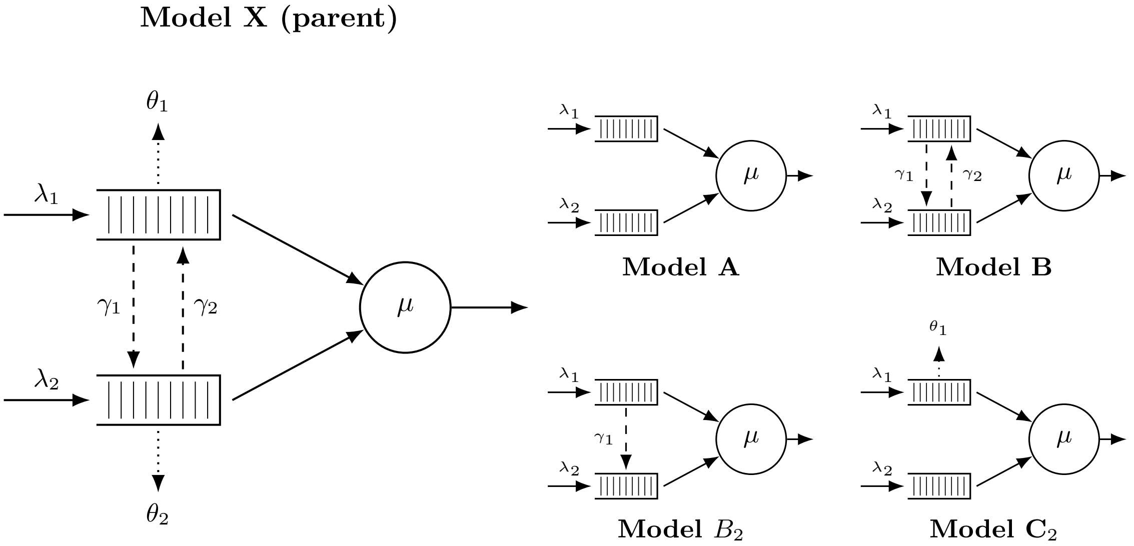

Figure 1: Physical queue layout for Model-X (left) and its four specialisations (right, 2×2 grid). Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows are for Poisson arrivals (λ_i) and exponential service (μ); dashed arrows are for jockeying(γ_i); and dotted arrows are for abandonments (θ_i)

The model in this section is an $M/M/1$ queue with two priority classes. Customers are served First-Come, First-Served (FCFS) within their own class, and class-1 (the priority class) has non-preemptive priority over class-2: a service in progress is never interrupted, so a class-2 customer in service is not displaced by an arriving class-1 customer. Beyond these basic dynamics, Model-X adds two queue-leaving mechanisms. First, *jockeying*: a waiting customer leaves its own queue to *join the other queue*, at per-customer rate γ_1 for direction $1 \to 2$ and γ_2 for $2 \to 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer leaves the *system* at per-customer rate θ_i, decreasing the total by one. Concretely, each waiting class-i customer (not the one in service) holds an independent $\text{Exp}(\theta_i)$ patience clock and abandons when it expires, so the aggregate class-i abandonment rate in a state with n_i waiting customers is the linear term $\theta_i n_i$ in the balance equations. This is precisely the linear-reneging mechanism of the $M/M/1 + M$ queue from §3.3.

In Kendall's notation this is an $M/M/1$-type model: Poisson arrivals at rate λ_i for class-i, exponentially distributed service times at rate μ, and a single server. To define the stochastic processes that drive this system, we introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that N_1, N_2 and N count customers in the *queues*, not the *system*. Since the service rate is common to both classes, we write the idle state as (0) and all other states as (n_1, n_2), so $(0, 0)$ means someone is in service and the queues are empty. Alternatively, an auxiliary state space $\widetilde{S}$ tracks only n_2 and the total $n = n_1 + n_2$, subject to $n \geq n_2 \geq 0$. Formally:

$$\begin{aligned} S &:= \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}, \\ \widetilde{S} &:= \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \end{aligned} \tag{4.1}$$

We define the long-run probabilities of finding the system in each state:

- π_0 for the idle state (0), it belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

Arrivals are Poisson for each class and service times exponential (common across classes), with the following notation:

- λ_1, λ_2: arrival rates of classes 1 and 2, respectively;
- μ: service rate of a customer, regardless of class;
- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class $i \in \{1, 2\}$;
- γ_1, γ_2: per-customer jockeying rates from Class $1 \to 2$ and $2 \to 1$, respectively; jockeying transfers a customer between queues, preserving the total number of customers;
- θ_1, θ_2: per-customer abandonment (reneging) rates from classes 1 and 2, respectively; abandonment removes a customer from the system, reducing the total n by one.

For the aggregate system parameters, we define:

- $\lambda = \lambda_1 + \lambda_2$: total arrival rate;
- $\rho = \frac{\lambda}{\mu}$: total system load.

Stability. Jockeying does not affect stability (customers are merely relocated). Model-X has abandonments from both classes, so the departure rate $\mu + \theta_1 n_1 + \theta_2 n_2$ grows unbounded, ensuring positive recurrence for all loads. In contrast, models without abandonments require $\rho < 1$ (they behave like M/M/1). Mixed cases, such as Model-C_2, are discussed separately (§8).

Computation of π_0 and $\pi(0,0)$. Balance between state (0) and $(0,0)$ gives $\lambda \pi_0 = \mu\, \pi(0,0)$. Hence,

$$\pi(0,0) \;=\; \rho\, \pi_0. \tag{4.2}$$

This holds regardless of jockeying and abandonments (neither mechanism acts when there are no waiting customers). However, π_0 itself depends on the abandonment model. For models *without* abandonment, $\pi_0 = 1 - \rho$ (M/M/1 reduction). For models *with* abandonment, the departure rate depends on the split (n_1, n_2), so π_0 must be determined from the bivariate PGF (§4.2).

4.1 Balance equations

This subsection presents the state-transition diagrams of Model-X on both state spaces S and $\widetilde{S}$, and derives the corresponding sets of balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model-X on S is given in Figure 2.

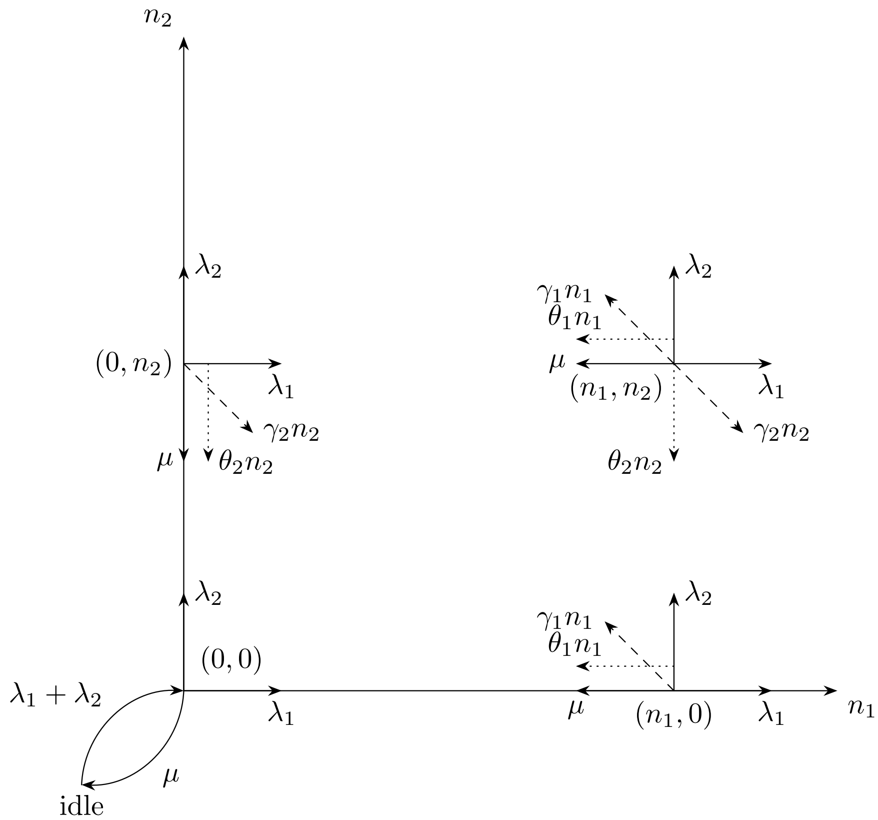

Figure 2: State transition diagram on state space S for the model with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1 + (\gamma_2 + \theta_2)n_2]\,\pi(n_1, n_2) \\
&\quad = \quad \mu\pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1\pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2\pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1(n_1 + 1)\pi(n_1 + 1, n_2 - 1) \\
&\qquad + \gamma_2(n_2 + 1)\pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1(n_1 + 1)\pi(n_1 + 1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(n_1, n_2 + 1) \qquad\qquad \text{for} \quad n_1 \geq 1, n_2 \geq 1
\end{aligned}
\tag{4.3}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n_2]\,\pi(0, n_2) \\
&\quad = \quad \lambda_2\pi(0, n_2 - 1) \\
&\qquad + \mu\pi(1, n_2) \\
&\qquad + \mu\pi(0, n_2 + 1) \\
&\qquad + \gamma_1\pi(1, n_2 - 1) \\
&\qquad + \theta_1\pi(1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(0, n_2 + 1) \qquad\qquad \text{for} \quad n_1 = 0, n_2 \geq 1
\end{aligned}
\tag{4.4}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1]\,\pi(n_1, 0) \\
&\quad = \quad \lambda_1\pi(n_1 - 1, 0) \\
&\qquad + \mu\pi(n_1 + 1, 0) \\
&\qquad + \gamma_2\pi(n_1 - 1, 1) \\
&\qquad + \theta_1(n_1 + 1)\pi(n_1 + 1, 0) \\
&\qquad + \theta_2\pi(n_1, 1) \qquad\qquad \text{for} \quad n_1 \geq 1, n_2 = 0
\end{aligned}
\tag{4.5}
$$

$$
[\lambda_1 + \lambda_2]\,\pi(0, 0) = (\mu + \theta_1)\pi(1, 0) + (\mu + \theta_2)\pi(0, 1) \tag{4.6}
$$

$$
(\lambda_1 + \lambda_2)\pi_0 = \mu\pi(0, 0) \tag{4.7}
$$

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in (4.1).

For readability we denote its limiting probabilities by $\widetilde{\pi}(n_2, n)$, for $(n_2, n) \in \widetilde{S}$. Figure 3 shows the corresponding state transition diagram.

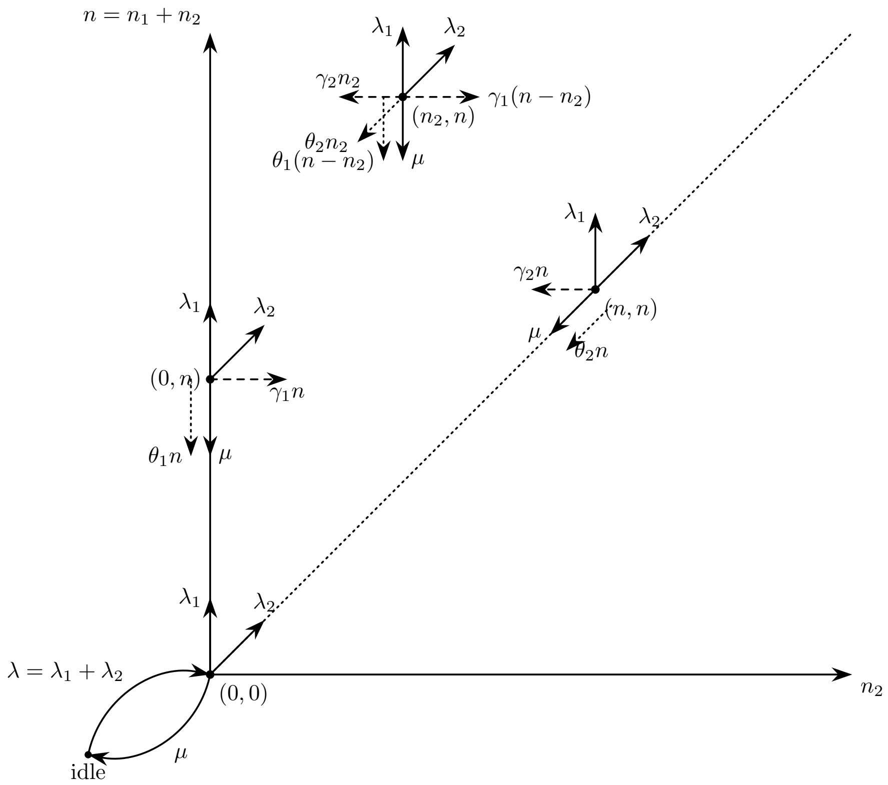

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)(n - n_2) + (\gamma_2 + \theta_2)n_2]\, \widetilde{\pi}(n_2, n) \\
&\quad = \lambda_1 \widetilde{\pi}(n_2, n-1) \\
&\qquad + \lambda_2 \widetilde{\pi}(n_2 - 1, n-1) \\
&\qquad + \mu \widetilde{\pi}(n_2, n+1) \\
&\qquad + \gamma_1(n - n_2 + 1)\widetilde{\pi}(n_2 - 1, n) \\
&\qquad + \gamma_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n) \\
&\qquad + \theta_1(n + 1 - n_2)\widetilde{\pi}(n_2, n+1) \\
&\qquad + \theta_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n+1)
\end{aligned} \qquad \text{for} \quad 1 \le n_2 \le n-1,\ n \ge 2 \qquad (4.8)$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n]\, \widetilde{\pi}(n, n) \\
&\quad = \lambda_2 \widetilde{\pi}(n-1, n-1) \\
&\qquad + \mu \widetilde{\pi}(n+1, n+1) \\
&\qquad + \mu \widetilde{\pi}(n, n+1) \\
&\qquad + \gamma_1 \widetilde{\pi}(n-1, n) \\
&\qquad + \theta_1 \widetilde{\pi}(n, n+1) \\
&\qquad + \theta_2(n+1)\widetilde{\pi}(n+1, n+1)
\end{aligned} \qquad \text{for} \quad n_2 = n,\ n \ge 1 \qquad (4.9)$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n]\, \widetilde{\pi}(0, n) \\
&\quad = \lambda_1 \widetilde{\pi}(0, n-1) \\
&\qquad + \mu \widetilde{\pi}(0, n+1) \\
&\qquad + \gamma_2 \widetilde{\pi}(1, n) \\
&\qquad + \theta_1(n+1)\widetilde{\pi}(0, n+1) \\
&\qquad + \theta_2 \widetilde{\pi}(1, n+1)
\end{aligned} \qquad \text{for} \quad n_2 = 0,\ n \ge 1 \qquad (4.10)$$

$$\begin{aligned}&[\lambda_1 + \lambda_2]\,\widetilde{\pi}(0,0) \\ &\quad = \quad (\mu+\theta_1)\widetilde{\pi}(0,1) + (\mu+\theta_2)\widetilde{\pi}(1,1)\end{aligned} \tag{4.11}$$

$$(\lambda_1 + \lambda_2)\widetilde{\pi}_0 = \mu\widetilde{\pi}(0,0) \tag{4.12}$$

4.2 Derivation of equations for (Partial) Probability Generating functions

To study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint queue lengths N_1 and N_2:

$$P(x,y) = \mathbb{E}[x^{N_1} y^{N_2}] = \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(n_1,n_2) x^{n_1} y^{n_2}.$$

The derivation also uses the boundary terms, where at least one queue is empty:

$$P_x(x) = P(x,0) = \sum_{n_1=0}^{\infty} \pi(n_1,0) x^{n_1}$$

$$P_y(y) = P(0,y) = \sum_{n_2=0}^{\infty} \pi(0,n_2) y^{n_2}.$$

Note that both terms include the busy-but-empty state $\pi(0,0)$, whereas the fully idle state is treated separately.

A further term arises in the balance equations, describing the boundary dynamics when the class-1 queue holds exactly one waiting customer:

$$P_1(y) = \sum_{n_2=1}^{\infty} \pi(1,n_2) y^{n_2}.$$

This is an intermediate term; as the derivation proceeds, it will be expressed in terms of the primary boundary functions $P_x(x)$ and $P_y(y)$.

In the same fashion, we introduce the **Partial Probability Generating Function (PPGF)**. This technique, notably used by Cohen [Coh56], transforms only a subset of the random variables while keeping the discrete indices of the rest. For a priority queue it is particularly advantageous: it lets us exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and concentrate the analytical effort on the non-trivial PGF of the non-priority class.

Cohen developed this framework for multi-server systems; we apply it here to the single-server case. To avoid confusion from mid-20th-century notation, we give a modern formulation consistent with the variables defined above.

We define the PPGF with respect to N_2, conditioned on the first queue being in state n_1:

$$\widetilde{P}(n_1,y) = \mathbb{E}\left[y^{N_2} \nVdash_{\{N_1=n_1\}}\right] = \sum_{n_2=0}^{\infty} \pi(n_1,n_2) y^{n_2}. \tag{4.13}$$

Similarly, the transform with respect to N_1, keeping the second queue in discrete state n_2, is:

$$\widetilde{P}(x,n_2) = \mathbb{E}\left[x^{N_1} \nVdash_{\{N_2=n_2\}}\right] = \sum_{n_1=0}^{\infty} \pi(n_1,n_2) x^{n_1}.$$

These partial transforms relate to the joint PGF by:

$$P(x,y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1,y) x^{n_1} = \sum_{n_2=0}^{\infty} \widetilde{P}(x,n_2) y^{n_2}.$$

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S = \{(n_1,n_2) : n_1, n_2 \geq 0\}$, with Poisson arrival rates λ_1, λ_2, exponential service rate μ, jockeying rates γ_1, γ_2 and abandonment rates θ_1, θ_2, where n_1 and n_2*

*count the class-*1 *and class-*2 *customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution* $\{\pi(n_1, n_2)\}$ *exists and the joint PGF*

$$P(x,y) = \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(n_1,n_2)\, x^{n_1} y^{n_2}$$

is analytic on the closed unit polydisc $|x| \le 1,\ |y| \le 1$. *Then* $P(x,y)$, *together with the boundary function* $P_y(y) = P(0,y) = \sum_{n_2=0}^{\infty} \pi(0,n_2) y^{n_2}$, *satisfies the first-order linear partial differential equation*

$$\begin{aligned} &\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) \\ &+ xy\left[\gamma_1(x-y) + \theta_1(x-1)\right] \frac{\partial}{\partial x} P(x,y) \\ &+ xy\left[\gamma_2(y-x) + \theta_2(y-1)\right] \frac{\partial}{\partial y} P(x,y) \\ &\quad = \quad \mu(x-y)P_y(y) - \mu x(1-y)\,\pi(0,0). \end{aligned} \tag{4.14}$$

Remark 1 (Reading the fundamental equation). Equation (4.14) repays a moment's mechanical reading, since its structure dictates the entire analysis that follows. The bracketed algebraic coefficient multiplying $P(x,y)$ is exactly the kernel of the baseline Model-A: the flux balance between the arrival contributions $\lambda_1 x^2 y$ and $\lambda_2 x y^2$, the service contribution μy, and the total outflow $(\lambda_1+\lambda_2+\mu)xy$.

The two convection coefficients carry the added mechanisms. In the x-convection coefficient $xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]$, the jockeying factor $(x-y)$ vanishes on the diagonal $x = y$ — the analytic signature that jockeying merely relocates a customer and so conserves the total count — while the abandonment factor $(x-1)$ vanishes at $x = 1$, the signature that abandonment respects normalisation ($P(1,1) = 1$) yet removes a customer and thereby destroys conservation. The y-convection coefficient $xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]$ states the symmetric facts for the class-2 mechanisms, with $(y-x)$ vanishing on the diagonal and $(y-1)$ at $y = 1$.

The right-hand side carries the two boundary unknowns that any solution must pin down: the boundary function $P_y(y) = P(0,y)$, the generating function of the states with an empty priority queue, and the scalar $\pi(0,0)$, the probability that the server is busy with both queues empty.

The position of these vanishing factors fixes, model by model, where the operative singularity sits and hence which technique applies. With both convection terms absent (Model-A, §5) the equation is algebraic and $P_y(y)$ is recovered from the kernel root $x^*(y)$ inside the unit disc. With one-way jockeying alone (Model-B_2, §7) the factor $(x-y)$ places the singularity at the interior point $x = y$ with a negative exponent, and $P_y(y)$ is pinned by analyticity there. With class-1 abandonment alone (Model-C_2, §8) the factor $(x-1)$ places it on the boundary $x = 1$ with a positive exponent, and $P_y(y)$ is pinned by mere boundedness. Restricting a mechanism to the head of the queue removes the derivative outright, collapsing the equation to the algebraic Model-A form (Models B_2^{H} and C_2^{H}, §9). These four regimes are gathered side by side in Table 5 of §11.2, and are synthesised, together with the two irreducible obstructions, as the obstruction taxonomy of §12.

Proof. We multiply each balance equation by $x^{n_1} y^{n_2}$ and sum over its domain.

From the interior equation (4.3):

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&\quad=\quad \mu\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\lambda_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\lambda_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)x^{n_1}y^{n_2}\\
&\qquad+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2}\\
&\qquad+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\qquad+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2}\\
&LHS_1+LHS_2+LHS_3+LHS_4+LHS_5\\
&\quad=\quad RHS_1+RHS_2+RHS_3+RHS_4+RHS_5+RHS_6+RHS_7
\end{aligned}
$$

Evaluating each summand:

$$
\begin{aligned}
&LHS_1\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^{0}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right]\\
&\quad-\ (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned}
$$

$$
\begin{aligned}
&LHS_2\\
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&= \gamma_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}\\
&= \gamma_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right]\\
&= \gamma_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right]\\
&= \gamma_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]\\
&LHS_3\\
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&= \gamma_2 \sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} - 0\cdot\pi(n_1,0)x^{n_1}\right]\\
&= \gamma_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right]\\
&= \gamma_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]\\
&LHS_4\\
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&= \theta_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}\\
&= \theta_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right]\\
&= \theta_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right]\\
&= \theta_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]\\
&LHS_5\\
&= \theta_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&= \theta_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right]\\
&= \theta_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]
\end{aligned}
$$

RHS_1

$$
\begin{aligned}
&= \mu \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
&= \mu x^{-1} \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1+1} y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
&= \mu x^{-1} \left[\sum_{m=2}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=2}^{\infty} \pi(m,0) x^m \right] \\
&= \mu x^{-1} \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - x \sum_{n_2=0}^{\infty} \pi(1, n_2) y^{n_2} - \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2} \right] \\
&- \mu x^{-1} \left[\sum_{m=0}^{\infty} \pi(m,0) x^m - x\pi(1,0) - \pi(0,0) \right] \\
&= \mu x^{-1} \left[P(x,y) - xP_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right]
\end{aligned}
$$

RHS_2

$$
\begin{aligned}
&= \lambda_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1} y^{n_2} \\
&= \lambda_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1-1} y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
&= \lambda_1 x \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=0}^{\infty} \pi(m,0) x^m \right] \\
&= \lambda_1 x \left[P(x,y) - P_x(x) \right]
\end{aligned}
$$

RHS_3

$$
\begin{aligned}
&= \lambda_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2-1} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} \\
&= \lambda_2 y \left[\sum_{n_1=0}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} - \sum_{m=0}^{\infty} \pi(0,m) y^m \right] \\
&= \lambda_2 y \left[P(x,y) - P_y(y) \right]
\end{aligned}
$$

RHS_4

$$\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2}\\
&= \gamma_1 y \sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty}(n_1+1)\pi(n_1+1,m_2)x^{n_1}y^{m_2}\\
&= \gamma_1 y \sum_{m_1=2}^{\infty}\sum_{m_2=0}^{\infty}m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2}\\
&= \gamma_1 y \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty}m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2} - \sum_{m_2=0}^{\infty}\pi(1,m_2)y^{m_2}\right]\\
&= \gamma_1 y \left[\frac{\partial}{\partial x}P(x,y) - P_y(y|n_1=1)\right]
\end{aligned}$$

RHS_5

$$\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2}\\
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{m_2=2}^{\infty}m_2\pi(n_1-1,m_2)x^{n_1}y^{m_2-1}\\
&= \gamma_2 x \left[\sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty}m_2\pi(n_1-1,m_2)x^{n_1-1}y^{m_2-1} - \sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1-1}\right]\\
&= \gamma_2 x \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty}m_2\pi(m_1,m_2)x^{m_1}y^{m_2-1} - \sum_{m_1=0}^{\infty}\pi(m_1,1)x^{m_1}\right]\\
&= \gamma_2 x \left[\frac{\partial}{\partial y}P(x,y) - P_x(x|n_2=1)\right]
\end{aligned}$$

RHS_6

$$\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&= \theta_1 \sum_{m_1=2}^{\infty}\sum_{n_2=1}^{\infty}m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2}\\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=1}^{\infty}m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\right]\\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=0}^{\infty}m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{m_1=0}^{\infty}m_1\pi(m_1,0)x^{m_1-1}\right]\\
&\quad - \theta_1 \left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2} - \pi(1,0)\right]\\
&= \theta_1 \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) - P_y(y|n_1=1) + \pi(1,0)\right]
\end{aligned}$$

$$
\begin{aligned}
& RHS_7 \\
= \quad & \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1, n_2+1) x^{n_1} y^{n_2} \\
= \quad & \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} \\
= \quad & \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1, 1) x^{n_1} \right] \\
= \quad & \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2 \pi(0, m_2) y^{m_2-1} \right] \\
& - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1, 1) x^{n_1} - \pi(0,1) \right] \\
= \quad & \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

Collecting and multiplying through by xy to clear the x^{-1} factor:

$$
\begin{aligned}
& (\lambda_1 + \lambda_2 + \mu) \left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0) \right] \\
& + \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
& + \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
& + \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
& + \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
& \quad = \quad \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right] \\
& \qquad + \lambda_1 x \left[P(x,y) - P_x(x) \right] \\
& \qquad + \lambda_2 y \left[P(x,y) - P_y(y) \right] \\
& \qquad + \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right] \\
& \qquad + \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right] \\
& \qquad + \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right] \\
& \qquad + \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right]P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right]P_y(y)\\
&\qquad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\qquad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+\mu\pi(1,0)\\
&\qquad+\theta_1\pi(1,0)\\
&\qquad+\theta_2\pi(0,1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 x y^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 x y^2\right]P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{4.15}
$$

The x-boundary equation (4.5), treated analogously, gives:

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad=\quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}\\
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^{m}+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^{m}\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^{m}+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^{m}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad+\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad+\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad=\quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad+\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad-(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad-\theta_2 xy\,\pi(0,1)\\
&\qquad+xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{4.16}
$$

The y-boundary equation (4.4) gives:

$$
\begin{aligned}
&\sum_{n_2=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n_2\right]\pi(0,n_2)y^{n_2}\\
&\quad=\quad \lambda_2\sum_{n_2=1}^{\infty}\pi(0,n_2-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(0,n_2+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{\infty}\pi(1,n_2-1)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{\infty}(n_2+1)\pi(0,n_2+1)y^{n_2}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(0,n_2)y^{n_2}\\
&+(\gamma_2+\theta_2)y\sum_{n_2=1}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu y^{-1}\sum_{m=2}^{\infty}\pi(0,m)y^m\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{m=2}^{\infty}m\pi(0,m)y^{m-1}\\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)
\end{aligned}
$$

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad [\mu+\gamma_1 y+\theta_1]\,P_y(y|n_1=1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad [\mu+\gamma_1 y+\theta_1]\,P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad [\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}\tag{4.17}$$

Substituting (4.16) into (4.15) eliminates all P_x terms:

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)
\end{aligned}\tag{4.18}$$

Isolating $[\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)$ from (4.17) and substituting into (4.18):

$$\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\Big\{\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)\\
&\qquad\quad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)+\mu x(1-y)\,\pi(0,0)\Big\}\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}$$

□

4.2.2 Equation for PPGF in state space $\widetilde{S}$

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space $\widetilde{S}=\{(n_2,n):0\le n_2\le n\}$, where n is the total number of customers* waiting *in the queues and n_2 the number of waiting class-2 customers. Assume positive recurrence, so that the stationary distribution $\{\widetilde{\pi}(n_2,n)\}$ exists, and define the partial PGF (transforming only n_2, keeping n discrete)*

$$\widetilde{P}(y,n)=\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)\,y^{n_2},$$

a polynomial of degree n in y. Then, for every $n \geq 1$, the family $\{\widetilde{P}(y,n)\}_{n\geq 0}$ satisfies the differential–difference equation

$$\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n)\\
&= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)\\
&\quad + (\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)+\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1).
\end{aligned}$$

Proof. Multiply the interior equation (4.8) by y^{n_2} and sum over $1 \leq n_2 \leq n-1$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n-1}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}
\end{aligned}$$

Adding the diagonal equation (4.9) multiplied by y^n extends most summation limits to n; the λ_1- and γ_2-terms on the RHS remain restricted to $n_2 \leq n-1$, and a spare $\mu\widetilde{\pi}(n+1,n+1)y^n$

survives:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\quad +\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\quad +\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\quad +\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\quad +\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\quad +\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}$$

Evaluating each summand:

LHS_1

$$\begin{aligned}
&= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right]\\
&= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]
\end{aligned}$$

LHS_2

$$\begin{aligned}
&= \gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \gamma_1\left[n\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\right]\\
&= \gamma_1\left[n\left(\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right)-y\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2-1}\right]\\
&= \gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned}$$

$$
\begin{aligned}
&LHS_3 \\
&= \gamma_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \gamma_2 \left[y \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2-1} \right] \\
&= \gamma_2 y \frac{d}{dy} \widetilde{P}(y,n) \\
&LHS_4 \\
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2) \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y \frac{d}{dy} \widetilde{P}(y,n) \right] \\
&LHS_5 \\
&= \theta_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2} \\
&= \theta_2 y \frac{d}{dy} \widetilde{P}(y,n) \\
&RHS_1 \\
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2} \\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right] \\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) y^{0} \right] \\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) \right] \\
&RHS_2 \\
&= \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2} \\
&= \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2-1} \right] \\
&= \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2} \right] \\
&= \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right) \right] \\
&= \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} \right] \\
&RHS_3 \\
&= \mu \widetilde{\pi}(n+1,n+1) y^{n} + \mu \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2} \\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2} \right] \\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1) y^{n_2} - \widetilde{\pi}(0,n+1) y^{0} - \widetilde{\pi}(n+1,n+1) y^{n+1} \right] \\
&= \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1) y^{n} (1-y) \right]
\end{aligned}
$$

RHS_4

$$\begin{aligned}
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
&= \gamma_1 \left[y \sum_{n_2=1}^{n} (n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1} \right] \\
&= \gamma_1 \left[y \sum_{k=0}^{n-1} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \\
&= \gamma_1 \left[y \sum_{k=0}^{n} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \quad \text{(since the } k=n \text{ term is 0)} \\
&= \gamma_1 y \left[n \sum_{k=0}^{n} \widetilde{\pi}(k,n)y^{k} - y \sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} \right] \\
&= \gamma_1 y \left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned}$$

RHS_5

$$\begin{aligned}
&= \gamma_2 \sum_{n_2=1}^{n-1} (n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
&= \gamma_2 \sum_{k=2}^{n} k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \\
&= \gamma_2 \left[\sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^{0} \right] \\
&= \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n) \right]
\end{aligned}$$

RHS_6

$$\begin{aligned}
&= \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&= \theta_1 \left[(n+1) \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n+1)y^{n_2} \right] \\
&= \theta_1 \left[(n+1) \left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1) - \widetilde{\pi}(n+1,n+1)y^{n+1} \right) \right] \\
&\quad - \theta_1 \left[y \left(\sum_{n_2=0}^{n+1} n_2\widetilde{\pi}(n_2,n+1)y^{n_2-1} - (n+1)\widetilde{\pi}(n+1,n+1)y^{n} \right) \right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
&\quad - \theta_1 \left[y\frac{d}{dy}\widetilde{P}(y,n+1) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1) \right]
\end{aligned}$$

RHS_7

$$
\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2} \\
&= \theta_2 \sum_{k=2}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} \quad (\text{letting } k = n_2+1) \\
&= \theta_2 \left[\sum_{k=1}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} - 1\cdot\widetilde{\pi}(1,n+1)y^0\right] \\
&= \theta_2 \left[\frac{d}{dy}\widetilde{P}(y,n+1) - \widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Collecting:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right] \\
&+\gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\gamma_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&+\theta_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\theta_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&\qquad = \lambda_1\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n-\widetilde{\pi}(0,n-1)\right] \\
&\qquad + \lambda_2 y\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n\right] \\
&\qquad + \mu\left[\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)+\widetilde{\pi}(n+1,n+1)y^n(1-y)\right] \\
&\qquad + \gamma_1 y\left[n\widetilde{P}(y,n)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\qquad + \gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n)-\widetilde{\pi}(1,n)\right] \\
&\qquad + \theta_1\left[(n+1)\big(\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)\big)-y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\qquad + \theta_2\left[\frac{d}{dy}\widetilde{P}(y,n+1)-\widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Regrouping:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\qquad + \{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)-\lambda_1\widetilde{\pi}(0,n-1)-[\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1) \\
&\qquad\quad -\gamma_2\widetilde{\pi}(1,n)-\theta_2\widetilde{\pi}(1,n+1)\} \\
&\qquad - y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

The y-boundary equation (4.10) makes the bracketed terms vanish; since $\widetilde{\pi}(n,n-1)=0$ (outside $\widetilde{S}$):

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\qquad + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

$\square$

Remark 2 (The $\widetilde{S}$ recursion is a failed simplification). The partial-PGF recurrence of Lemma 2 was introduced in the expectation that generating only over the class-2 index, while carrying the total count n as a discrete parameter, would *simplify* the analysis — collapsing the two-dimensional balance on S to a one-dimensional recurrence in n. It does the opposite. The forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ ties each level n to the unknown diagonal sequence $\{\pi(0,n)\}$, so the recurrence does not close on itself: solving it would require precisely the boundary probabilities it was meant to deliver. Only for the baseline Model-A does that forcing decay fast enough to be dropped, and even there the result is an *approximation*, accurate only in a restricted parameter region (Approximation 1). For every model with an active mechanism the $\widetilde{S}$ route is strictly harder than the S analysis it was meant to replace. We therefore record it as a deliberate negative result: the $\widetilde{S}$ coordinatisation is retained only for Model-A, and the S formulation is used throughout the remainder of this thesis.

5 Analysis of Model-A ($\gamma_1 = \gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model-A is the baseline of this collection: all jockeying and abandonment parameters are zero, leaving an ordinary two-class $M/M/1$ queue with non-preemptive priority. We analyse it in both state spaces S and $\widetilde{S}$: on the former we give two complete proofs, on the latter a conditionally accurate approximation, evaluated in Approximation 1.

The balance equations are obtained by plugging $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ into Equations (4.3), (4.5), (4.4), (4.6), and (4.7) for state space S, and Equations (4.8), (4.9), (4.10), (4.11), and (4.12) for state space $\widetilde{S}$. The corresponding starting points are Lemma 1 for S and Lemma 2 for $\widetilde{S}$.

Stability. Since there is no abandonment, the system is positive recurrent if and only if the total load satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1. \tag{5.1}$$

Figure 4 shows the queueing diagram of Model-A.

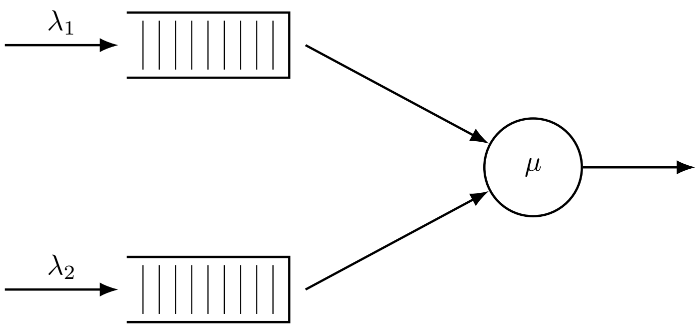

Figure 4: Queue layout for Model-A. Only Poisson arrivals and exponential service are present; all active mechanisms rates are zero.

5.1 Analysis of Model-A: state space S

We begin from Equation (4.14) with $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, in which the boundary function $P_y(y) = P(0,y)$ is still to be determined:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0). \tag{5.2}$$

The following theorem summarises the main result of this subsection; two standalone proofs are given, one analytical and one probabilistic.

Theorem 1. *Consider a queueing system with two priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class 2, and exponentially distributed service times with rate μ independently of the class. Under the stability condition $\rho < 1$ (5.1), the joint PGF $P(x,y) = \mathbb{E}[x^{N_1}y^{N_2}]$ is*

$$P(x,y) = \left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} \cdot y(1-y)\left[\frac{x - x^*(y)}{x^*(y) - y}\right] \cdot \rho(1-\rho), \tag{5.3}$$

where

$$x^*(y) = \frac{[\mu + \lambda_1 + \lambda_2(1-y)] - \sqrt{[\mu + \lambda_1 + \lambda_2(1-y)]^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{5.4}$$

Corollary 1. *Under the stability condition $\rho < 1$ (5.1), the empty-state probabilities for Model-A are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho). \tag{5.5}$$

In particular, $P(1,1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof of Corollary 1. Setting $x = y = z$ in the fundamental equation (5.2), the boundary term $\mu(x-y)P_y(y)$ vanishes:

$$\left[(\lambda_1+\lambda_2+\mu)z^2 - \mu z - \lambda_1 z^3 - \lambda_2 z^3\right] P(z,z) = -\mu z(1-z)\pi(0,0).$$

Dividing both sides by z and factoring the bracket:

$$-(1-z)\left[\mu - (\lambda_1+\lambda_2)z\right] P(z,z) = -\mu(1-z)\pi(0,0).$$

Cancelling $(1-z) \neq 0$ for $z \in [0,1)$ and rearranging:

$$P(z,z) = \frac{\mu\,\pi(0,0)}{\mu - \lambda z} = \frac{\pi(0,0)}{1-\rho z},$$

where $\lambda = \lambda_1 + \lambda_2$ and the second equality follows by dividing numerator and denominator by μ.

Setting $z = 1$ and using the normalisation $P(1,1) = 1 - \pi_0$:

$$1 - \pi_0 = \frac{\pi(0,0)}{1-\rho}. \tag{5.6}$$

The global balance equation for the idle state gives $\lambda\pi_0 = \mu\pi(0,0)$, i.e. $\pi(0,0) = \rho\pi_0$. Substituting into (5.6):

$$1 - \pi_0 = \frac{\rho\pi_0}{1-\rho} \implies (1-\pi_0)(1-\rho) = \rho\pi_0 \implies \pi_0 = 1-\rho,$$

and therefore $\pi(0,0) = \rho(1-\rho)$. □

5.1.1 Analytical approach for Model-A in state space S

Proof of Theorem 1, analytical approach. We begin from the fundamental equation for Model-A (5.2):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0).$$

We apply the *Kernel Method*, used in [Coh56]: fix $y \in (0,1]$ and find $x^*(y)$ such that the LHS of the expression above is zero. Since $P(x,y)$ is a PGF, it is bounded. At any such root the LHS vanishes, so the RHS must also vanish:

$$\mu\left[(x^*(y) - y)\,P_y(y) - x^*(y)(1-y)\,\pi(0,0)\right] = 0.$$

To find the roots of the coefficient of $P(x,y)$, divide by $y \neq 0$ and set

$$\begin{aligned} 0 &= (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 x y \\ 0 &= \lambda_1 x^2 - \left[\mu + \lambda_1 + \lambda_2(1-y)\right] x + \mu, \end{aligned}$$

whose roots are

$$x^{\pm}(y) = \frac{\left[\mu+\lambda_1+\lambda_2(1-y)\right] \pm \sqrt{\left[\mu+\lambda_1+\lambda_2(1-y)\right]^2 - 4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y = 1$:

$$x^*(1) = 1 \quad \text{and} \quad x^+(1) = \mu/\lambda_1 > 1.$$

We require our root to lie inside the closed unit disc for $y \in [0,1)$. Computing the derivative of $x^*(y)$:

$$\frac{d}{dy}x^*(y) = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1+\lambda_2(1-y)}{\sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}\right] > 0, \tag{5.7}$$

since $(\mu+\lambda_1+\lambda_2(1-y))^2 > (\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu$ whenever $\lambda_1\mu > 0$, by squaring numerator and denominator in the inner fraction. So $x^*(y)$ is strictly increasing in $[0,1]$, and since $x^*(1) = 1$ we have $x^*(y) < 1$ for $y \in [0,1)$. We make use of Vieta's formulas to discard the other root for the kernel polynomial $\lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]x + \mu = 0$: the product of the two roots satisfies $x^*(y)\, x^+(y) = \mu/\lambda_1 > 1$, so $x^+(y) = \mu/(\lambda_1 x^*(y)) > 1$ throughout $[0,1)$. So the only root strictly inside the unit disc for $y \in [0,1)$ is $x^*(y)$:

$$x^*(y) = \frac{(\mu + \lambda_1 + \lambda_2(1-y)) - \sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{5.8}$$

Returning to the fundamental equation (5.2) at $x = x^*(y)$, the RHS also equals zero, giving

$$\mu \cdot [(x^*(y) - y)P_y(y) - x^*(y)(1-y)\pi(0,0)] = 0 \quad \Longrightarrow \quad P_y(y) = \frac{x^*(y)(1-y)}{x^*(y) - y}\pi(0,0). \tag{5.9}$$

By Corollary 1, $\pi(0,0) = \rho(1-\rho)$. Substituting this and Equation (5.9) into the fundamental equation (5.2):

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho)$$

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho)$$

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu(1-y)\left[\frac{y(x - x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)$$

$$\left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) = y(1-y)\left[\frac{x - x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ recovers $P(x,y)$ in Theorem 1 and concludes the proof. □

5.1.2 Probabilistic approach for Model-A in state space S

Proof of Theorem 1, probabilistic approach. This proof mirrors the analytical one, but derives the boundary function $P_y(y)$ by means of probabilistic arguments. We again begin from the fundamental equation for Model-A (Equation (5.2)):

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu\left[(x-y)P_y(y) - x(1-y)\pi(0,0)\right]. \tag{5.10}$$

Class-2 conditional distribution via $M/G/1$. Since class 1 has non-preemptive priority, class-2 customers are not taken into service whenever the priority queue is not empty ($N_1 > 0$); their *effective service time* B is therefore the duration of a complete busy period of the single-class ordinary $M/M/1$ sub-queue with arrival rate λ_1 and service rate μ. From class-2's viewpoint the system is thus an $M/G/1$ queue with arrival rate λ_2 and generally distributed service times B and mean $\mathbb{E}[B]$. The LST and mean of B are presented in §3.1 (Equations (3.1)–(3.2)), now with λ_1 instead of the generic λ:

$$\widetilde{B}(s) = \frac{(\mu + \lambda_1 + s) - \sqrt{(\mu + \lambda_1 + s)^2 - 4\mu\lambda_1}}{2\lambda_1},$$
$$\mathbb{E}[B] = \frac{1}{\mu(1-\rho_1)}.$$

Applying the Pollaczek–Khinchine (PK) formula (§3.4, Equation (3.8)) to this $M/G/1$ system, with $\widetilde{B}(\lambda_2(1-y)) = x^*(y)$ (cf. (5.8)) and $\lambda_2\mathbb{E}[B] = \rho_2/(1-\rho_1)$:

$$L_2(y) = \frac{(1 - \lambda_2\mathbb{E}[B])(1-y)\widetilde{B}(\lambda_2(1-y))}{\widetilde{B}(\lambda_2(1-y)) - y} = \frac{1-\rho}{1-\rho_1} \cdot \frac{x^*(y)(1-y)}{x^*(y)-y}.$$

By the PASTA property, the time-average distribution of N_2 conditional on (busy, $N_1 = 0$) coincides with the $M/G/1$ steady-state distribution, so $\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = L_2(y)$.

By the law of total expectation, $P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0]$. So

$$P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = \mathbb{P}(\text{busy}, N_1 = 0) \cdot L_2(y).$$

Under non-preemptive priority, the priority queue operates autonomously (memorylessness of residual service time), so $\mathbb{P}(N_1 = 0 \mid \text{busy}) = 1-\rho_1$ by the geometric law of the ordinary $M/M/1(\lambda_1, \mu)$. Combined with $\mathbb{P}(\text{busy}) = \rho$, we have

$$\mathbb{P}(\text{busy}, N_1 = 0) = \rho(1-\rho_1);$$

and therefore

$$P_y(y) = \rho(1-\rho_1)\cdot L_2(y) = \rho(1-\rho_1)\cdot \frac{1-\rho}{1-\rho_1}\cdot \frac{x^*(y)(1-y)}{x^*(y)-y} = \frac{x^*(y)(1-y)}{x^*(y)-y}\cdot(1-\rho)\rho \tag{5.11}$$

Substituting (5.11) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 1) into (5.10):

$$\begin{aligned}
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho)\\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho)\\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)\\
\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) &= y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ to isolate $P(x,y)$ recovers Theorem 1 and concludes the proof. $\square$

5.1.3 Partial-PGF recurrence approach for Model-A in state space S

This proof follows Cohen [Coh56]'s method. We keep the class-1 index n_1 as a discrete parameter and generate only with respect to N_2, via the PPGF $\widetilde{P}(n_1, y) = \sum_{n_2\geq 0}\pi(n_1,n_2)\,y^{n_2}$ from (4.13). Multiplying the balance equations by y^{n_2} and summing over n_2 turns them into a second-order linear recurrence in n_1, whose characteristic roots fix the geometric structure of the solution in the priority-class index.

Proof. We start from the Model-A balance equations, obtained by setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the interior equation (4.3) and the x-boundary equation (4.5):

$$\begin{aligned}
(\lambda_1+\lambda_2+\mu)\,\pi(n_1,n_2) &= \mu\,\pi(n_1+1,n_2) + \lambda_1\,\pi(n_1-1,n_2) + \lambda_2\,\pi(n_1,n_2-1), && n_1\geq 1,\ n_2\geq 1,\\
(\lambda_1+\lambda_2+\mu)\,\pi(n_1,0) &= \mu\,\pi(n_1+1,0) + \lambda_1\,\pi(n_1-1,0), && n_1\geq 1,\ n_2 = 0.
\end{aligned}\tag{5.12}$$

Now we take transforms by summing all over their states, but only with respect to n_2, and we substitute our PPGF $\widetilde{P}(n_1,y) = \sum_{n_2=0}^{\infty}\pi(n_1,y)y^{n_2}$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\
&\quad= \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2} + \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2} + \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2}\\
&LHS_1 \quad = \quad RHS_1 + RHS_2 + RHS_3\\
&LHS_1\\
&\quad= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2}\\
&\quad= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2} - \pi(n_1,0)y^0\right]\\
&\quad= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(n_1,y) - \pi(n_1,0)\right]
\end{aligned}$$

$$
\begin{aligned}
&RHS_1\\
&\quad= \mu \sum_{n_2=1}^{\infty} \pi(n_1+1,n_2) y^{n_2}\\
&\quad= \mu \left[\sum_{n_2=0}^{\infty} \pi(n_1+1,n_2) y^{n_2} - \pi(n_1+1,0) y^0 \right]\\
&\quad= \mu \left[\widetilde{P}(n_1+1,y) - \pi(n_1+1,0) \right]\\
&RHS_2\\
&\quad= \lambda_1 \sum_{n_2=1}^{\infty} \pi(n_1-1,n_2) y^{n_2}\\
&\quad= \lambda_1 \left[\sum_{n_2=0}^{\infty} \pi(n_1-1,n_2) y^{n_2} - \pi(n_1-1,0) y^0 \right]\\
&\quad= \lambda_1 \left[\widetilde{P}(n_1-1,y) - \pi(n_1-1,0) \right]\\
&RHS_3\\
&\quad= \lambda_2 \sum_{n_2=1}^{\infty} \pi(n_1,n_2-1) y^{n_2}\\
&\quad= \lambda_2 y \sum_{n_2=1}^{\infty} \pi(n_1,n_2-1) y^{n_2-1}\\
&\quad= \lambda_2 y \sum_{m=0}^{\infty} \pi(n_1,m) y^{m}\\
&\quad= \lambda_2 y\, \widetilde{P}(n_1,y)\\
&(\lambda_1+\lambda_2+\mu)\big[\widetilde{P}(n_1,y) - \pi(n_1,0)\big]\\
&\quad= \mu\big[\widetilde{P}(n_1+1,y) - \pi(n_1+1,0)\big]\\
&\qquad+ \lambda_1\big[\widetilde{P}(n_1-1,y) - \pi(n_1-1,0)\big]\\
&\qquad+ \lambda_2 y\, \widetilde{P}(n_1,y)
\end{aligned}
$$

$$
\begin{aligned}
&\big[\mu+\lambda_1+\lambda_2(1-y)\big]\widetilde{P}(n_1,y) - \mu\widetilde{P}(n_1+1,y) - \lambda_1\widetilde{P}(n_1-1,y)\\
&\quad= (\lambda_1+\lambda_2+\mu)\pi(n_1,0) - \mu\pi(n_1+1,0) - \lambda_1\pi(n_1-1,0)
\end{aligned}
\tag{5.13}
$$

By the x-boundary balance equation (5.12), the right-hand side of (5.13) vanishes completely, giving the homogeneous recurrence

$$
[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{P}(n_1,y) = \mu\widetilde{P}(n_1+1,y) + \lambda_1\widetilde{P}(n_1-1,y), \qquad n_1 \ge 1. \tag{5.14}
$$

Characteristic roots and mode selection. Equation (5.14) is a *second-order linear recurrence* in n_1 with constant coefficients, so its general solution is a linear combination of two geometric sequences $[\widetilde{x}^{\pm}(y)]^{n_1}$, where $\widetilde{x}^{\pm}(y)$ solve the characteristic equation

$$
\mu\,\widetilde{x}^2 - [\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{x} + \lambda_1 = 0, \tag{5.15}
$$

that is,

$$
\widetilde{x}^{\pm}(y) = \frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big) \pm \sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2 - 4\lambda_1\mu}}{2\mu}.
$$

The *characteristic polynomial* (Equation (5.15)) is the reciprocal to that of the bivariate PGF approaches (i.e. the reciprocal of $\lambda_1 x^2 - [\mu+\lambda_1+\lambda_2(1-y)]x+\mu$), thus its roots are the reciprocals of that kernel's roots: the positive root is the reciprocal of the negative one ($\widetilde{x}^{+}(y) = 1/x^{*}(y)$), and conversely $\widetilde{x}^{-}(y) = 1/x^{+}(y)$. By Vieta's formulas, $\widetilde{x}^{-}(y)\,\widetilde{x}^{+}(y) = \lambda_1/\mu = \rho_1 < 1$. Recall from §5.1.1 that $x^{*}(y) < 1 < x^{+}(y)$ for $y \in [0,1)$. We select the negative-sign root because the sequence $\{\widetilde{P}(n_1,y)\}_{n_1\ge 0}$ must be summable:

$$
\sum_{n_1=0}^{\infty} \widetilde{P}(n_1,y) = P(1,y) \le P(1,1) = 1-\pi_0 = \rho < \infty.
$$

At $y = 1$, we have

$$\widetilde{x}^-(1) = \rho_1 \quad \text{and} \quad \widetilde{x}^+(1) = 1,$$

the latter making the geometric sequence $[\widetilde{x}^+(y)]^{n_1}$ non-summable. We therefore define $\widetilde{x}^*(y) := \widetilde{x}^-(y)$,

$$\widetilde{x}^*(y) := \widetilde{x}^-(y) = \frac{\big(\mu + \lambda_1 + \lambda_2(1-y)\big) - \sqrt{\big(\mu + \lambda_1 + \lambda_2(1-y)\big)^2 - 4\lambda_1\mu}}{2\mu} = \rho_1\, x^*(y). \tag{5.16}$$

The last identity holds because $\widetilde{x}^*$ and x^* share the same numerator and differ only in the denominator (2μ versus $2\lambda_1$). Next,

$$\widetilde{P}(n_1, y) = \widetilde{P}(0, y)\, [\widetilde{x}^*(y)]^{n_1}, \qquad n_1 \geq 0, \tag{5.17}$$

where the relation also holds at $n_1 = 0$ trivially. Recall that at $y = 1$, we have $\widetilde{x}^*(1) = \lambda_1/\mu = \rho_1$. Thus, $\widetilde{P}(n_1, 1) = \widetilde{P}(0, 1)\, \rho_1^{n_1}$ and $\widetilde{P}(0, 1) = P_y(1) = \rho(1-\rho_1)$. The constant $\widetilde{P}(0, y)$ is resolved by its definition

$$\widetilde{P}(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2)\, y^{n_2} = P(0, y) = P_y(y),$$

as determined in the bivariate PGF approaches. Summing the geometric law (Equation (5.17)) over $n_1 \geq 0$ for $x \in (0, 1]$ recovers

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y)\, x^{n_1} = P_y(y) \sum_{n_1=0}^{\infty} \big[\widetilde{x}^*(y)\, x\big]^{n_1} = \frac{P_y(y)}{1 - \widetilde{x}^*(y)\, x}. \tag{5.18}$$

To bring this to the form of Theorem 1, we define the kernel of the fundamental equation, divided by μ:

$$K(x, y) := (1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2 = -y\rho_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{5.19}$$

where $x^*(y)$ and $x^+(y)$ are the roots found in §5.1.1. By Vieta's product formula, $x^*(y)\, x^+(y) = \mu/\lambda_1 = 1/\rho_1$. This, together with $\widetilde{x}^*(y)$ (Equation (5.16)), gives $\widetilde{x}^*(y) = \rho_1 x^*(y) = 1/x^+(y)$. Using the factorisation of $K(x, y)$ (5.19), the denominator of the reconstructed PGF (5.18) becomes

$$1 - \widetilde{x}^*(y)\, x = 1 - \frac{x}{x^+(y)} = \frac{x^+(y) - x}{x^+(y)} = \frac{K(x, y)}{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}.$$

Substituting this into our reconstructed $P(x, y)$ (Equation (5.18)), using $P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)$ and also noting that $\rho_1\, x^*(y)\, x^+(y) = 1$, we obtain

$$\begin{aligned}
P(x, y) &= \frac{x^*(y)(1-y)}{x^*(y) - y}\, \rho(1-\rho) \cdot \frac{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}{K(x, y)} \\
&= \underbrace{\rho_1\, x^*(y)\, x^+(y)}_{=1}\, \frac{1}{K(x, y)}\, y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\, \rho(1-\rho) \\
&= \big[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\big]^{-1} \cdot y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\, \rho(1-\rho),
\end{aligned}$$

which recovers Equation (5.3) from Theorem 1 and concludes the proof. □

5.2 Analysis of Model-A on $\widetilde{S}$: scope and limitations of the partial-PGF recursion

Setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the $\widetilde{S}$ balance equations (4.8)–(4.12) and forming the PPGF $\widetilde{P}(y, n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2, n)\, y^{n_2}$ yields the inhomogeneous recurrence:

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu]\, \widetilde{P}(y, n) \\
&= \quad (\lambda_1 + \lambda_2 y)\widetilde{P}(y, n-1) + \mu\widetilde{P}(y, n+1) + \mu y^n (1-y)\widetilde{\pi}(n+1, n+1)
\end{aligned} \tag{5.20}$$

The forcing term in (5.20) is controlled by the diagonal probabilities $\widetilde{\pi}(n+1, n+1) = \pi(0, n+1)$, whose decay we first establish from the exact S-solution.

Lemma 3 (Geometric decay of the boundary probabilities)**.** *For Model-A the boundary probabilities $\pi(0,n)$ satisfy $\pi(0,n) = O(r^{-n})$ for every $r < 1/\rho$; equivalently they decay geometrically with rate ρ.*

Proof. By definition $P_y(y) = P(0,y) = \sum_{n\geq 0} \pi(0,n)\, y^n$ is the generating function of $\{\pi(0,n)\}$, and from (5.9) with $\pi(0,0) = \rho(1-\rho)$,

$$P_y(y) = \rho(1-\rho)\,\frac{x^*(y)(1-y)}{x^*(y)-y}.$$

Evaluating the kernel quadratic (5.4) at $x = y$ gives $\lambda_1 y^2 - [\mu+\lambda_1+\lambda_2(1-y)]y+\mu = (\lambda y-\mu)(y-1)$ with $\lambda = \lambda_1+\lambda_2$, so $x^*(y) = y$ holds only at $y = 1$ and $y = \mu/\lambda = 1/\rho$. At $y = 1$ the simple zeros of $(1-y)$ and $(x^*(y)-y)$ cancel and P_y is analytic with $P_y(1) = \rho$; the branch point of x^* lies beyond $1/\rho$. Hence the singularity of P_y nearest the origin is the simple pole at $y = 1/\rho > 1$, so P_y is analytic on the disc $|y| < 1/\rho$. By the Cauchy coefficient estimates, for every $r < 1/\rho$ there is $C_r < \infty$ with $\pi(0,n) \leq C_r\, r^{-n}$, i.e. geometric decay at rate ρ. □

Approximation 1 (Heuristic PPGF on $\widetilde{S}$)**.** *Neglecting the boundary-diagonal forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ in (5.20) — which is exponentially small by Lemma 3 — the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y,n) \;\approx\; \rho(1-\rho)\,[\widetilde{y}^*(y)]^n,$$

where

$$\widetilde{y}^*(y) = \frac{(\lambda+\mu) - \sqrt{(\lambda+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}, \qquad \lambda = \lambda_1+\lambda_2.$$

At $y = 1$ this recovers $\widetilde{P}(1,n) = (1-\rho)\rho^{n+1}$, consistent with the $M/M/1(\lambda,\mu)$ steady-state marginal for n customers waiting.

Justification of Approximation 1. The inhomogeneous term $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ prevents the direct application of a geometric-sequence ansatz. The diagonal probabilities $\widetilde{\pi}(n,n) = \pi(0,n)$ correspond to states where all n waiting customers are class 2, and by Lemma 3 they decay geometrically in n at rate ρ; the forcing term is therefore exponentially small, and neglecting it yields an approximation whose accuracy is quantified below. The resulting homogeneous recurrence is

$$(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n) = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1).$$

The characteristic polynomial of this recurrence is

$$\mu[\widetilde{y}(y)]^2 - (\lambda_1+\lambda_2+\mu)\widetilde{y}(y) + (\lambda_1+\lambda_2 y) = 0,$$

with roots

$$\widetilde{y}^{\pm}(y) = \frac{(\lambda_1+\lambda_2+\mu) \pm \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}.$$

At $y = 1$ the roots are $\widetilde{y}_+(1) = 1$ and $\widetilde{y}_-(1) = \rho$. The root $\widetilde{y}_+ = 1$ would produce a non-summable geometric series as $n \to \infty$, so only the negative-sign root is admissible:

$$\widetilde{y}^*(y) = \frac{(\lambda_1+\lambda_2+\mu) - \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}.$$

The geometric ansatz gives $\widetilde{P}(y,n) = \widetilde{P}(y,0)\cdot[\widetilde{y}^*(y)]^n$. Using $\widetilde{P}(y,0) = \widetilde{\pi}(0,0) = \pi(0,0) = \rho(1-\rho)$ (Corollary 1):

$$\widetilde{P}(y,n) = \rho(1-\rho)\cdot[\widetilde{y}^*(y)]^n.$$

Since n counts customers *waiting* (the customer in service is not counted), $n+1$ is the total number in the system, so evaluating at $y = 1$:

$$\widetilde{P}(1,n) = \rho(1-\rho)\cdot\rho^n = (1-\rho)\rho^{n+1}, \qquad \square$$

which matches the $M/M/1(\lambda,\mu)$ stationary distribution for $n+1$ customers in the system [AR02].

Failure mechanism. The neglected forcing $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ feeds in the diagonal states, in which all waiting customers are class 2. Using $\widetilde{\pi}(n,n)=\pi(0,n)$ and the rate proved in Lemma 3, this term decays in n like ρ^n (modulated by y^n), whereas the homogeneous solution decays like $[\widetilde{y}^*(y)]^n$. The approximation is therefore accurate exactly when the forcing decays faster, i.e. $\rho<\widetilde{y}^*(y)$ on the relevant range of y; this holds when the priority class carries most of the load (ρ_1/ρ large, ρ moderate), precisely the empirical validity region of the approximation.

The obstruction is structural and recurs throughout this thesis: an unknown boundary/diagonal trace — here the sequence $\pi(0,n)$ — couples into an otherwise solvable recursion, exactly as the unknown diagonal $P(z,z)$ enters the Volterra forcing of Models B and X (§6 and Remark 3). A rigorous closure would retain the forcing and solve (5.20) by variation of constants in n, which reintroduces the very diagonal sequence $\{\pi(0,n)\}$ as an unknown to be determined self-consistently; see §12.5.

6 Analysis of Model-B ($\gamma_1,\gamma_2>0$, $\theta_1=\theta_2=0$)

In this section, we derive a general solution to the fundamental equation of Model-B. However, this is provided in terms of the unknown boundary function $P_y(y)$, which requires more advanced mathematical tools and is thus left open, but included for illustrative purposes to conclude that having derivative terms in y in our fundamental equation is a non-trivial obstacle. So, this section not only shows the difficulty of Model-B but also justifies not carrying out an analogous derivation for Model-X. Moreover, since jockeying breaks our Poisson arrivals and renewal structure, the probabilistic argument explained in § 5.1.2 is no longer available.

Model-B introduces bidirectional jockeying ($\gamma_1,\gamma_2>0$) and no abandonment ($\theta_1=\theta_2=0$). Figure 5 shows the queue layout.

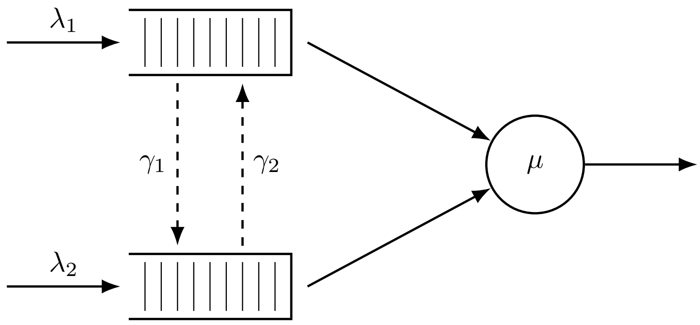

Figure 5: Queue layout for Model-B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a class-1 customer down to Queue 2 and γ_2 moves a class-2 customer up to Queue 1. Both transfers conserve the total n.

Stability. $\rho=(\lambda_1+\lambda_2)/\mu<1$. Jockeying preserves the total customer count and does not affect this condition, as the arriving customers can also be re-allocated.

Setting $\theta_1=\theta_2=0$ in the general fundamental equation (4.14) (Lemma 1) removes both abandonment factors $(x-1)$ and $(y-1)$ and leaves the *Model-B fundamental equation*

$$\begin{aligned}\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\big]\,P(x,y)&\\+\gamma_1\,xy(x-y)\,\frac{\partial P}{\partial x}(x,y)+\gamma_2\,xy(y-x)\,\frac{\partial P}{\partial y}(x,y)&\\=\mu(x-y)\,P_y(y)-\mu\,x(1-y)\,\pi(0,0).&\end{aligned}\tag{6.1}$$

This equation governs the entire analysis of the section, and we refer to it throughout in place of the general equation (4.14). Its only derivative terms are the two jockeying convection terms, both carrying the diagonal factor $(x-y)$ that makes the Method of Characteristics applicable.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models. Since Model-B is not solved, however, no theorem is attached to it.

Corollary 2. *The empty-system probability π_0 and the empty-queues probability $\pi(0,0)$ of Model-B coincide with those of the Model-A baseline,*

$$\pi_0=1-\rho,\qquad \pi(0,0)=\rho(1-\rho),\tag{6.2}$$

and the diagonal of the PGF is $P(z,z) = \pi(0,0)/(1-\rho z)$.

The proof follows immediately by setting $x = y = z$ in the Model-B fundamental equation (6.1): both jockeying terms, which carry the factor $(x-y)$, vanish, dragging the term $\mu(x-y)P_y(y)$ with them. The surviving diagonal equation is therefore *identical* to that of Model-A with $x = y = z$, so the proof of Corollary 2 reduces to that of Corollary 1.

For Model-B we do not provide a solution; however, working out the problem exposes the precise analytical obstacle, which we record as a claim. For the sake of readability, we will *not* define here the auxiliary functions; they are introduced throughout the proof. The following claim is made:

Claim 1. *Assume stability* $\rho < 1$*. Along the characteristic lines* $\gamma_2 x + \gamma_1 y = C_1$ *of the Model-B fundamental equation* (6.1)*, parametrised by* $y(x) = (C_1 - \gamma_2 x)/\gamma_1$*, the bivariate PGF of Model-B admits the following integral representation:*

$$P\big(x, y(x)\big) \;=\; P_h\big(x, y(x)\big) \int_0^x \frac{h\big(t; C_1\big)}{P_h\big(t, y(t)\big)}\, dt, \tag{6.3}$$

where P_h *is the homogeneous solution and the forcing function* h *depends linearly on the unknown boundary trace* $P_y(y) = P(0,y)$*. Evaluating* (6.3) *along the diagonal* $x = y = z$*, where* $P(z,z) = \pi(0,0)/(1-\rho z)$ *is known from Corollary 2, reduces the determination of* P_y *to a* Volterra integral equation of the first kind

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt. \tag{6.4}$$

This model is therefore left open.

Proof of Claim 1. The proof consists of five steps: (i) homogeneous solution via the Method of Characteristics, (ii) sign analysis of the exponent function E, (iii) particular solution via the Method of Variation of Parameters, (iv) determining the arbitrary function from the boundary condition, and (v) evaluation at $x = y$ yielding a Volterra-type equation.

Step (i): Homogeneous solution via the Method of Characteristics.

The Model-B fundamental equation (6.1) is a first-order linear PDE in $P(x,y)$. Setting the right-hand side to zero and dividing by y yields the homogeneous PDE

$$\gamma_1 x(x-y)\,\frac{\partial P}{\partial x} - \gamma_2 x(x-y)\,\frac{\partial P}{\partial y} = -\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P.$$

Following the Method of Characteristics (§3.6.2), we associate to this PDE the characteristic system

$$\frac{dx}{\gamma_1 x(x-y)} = \frac{dy}{-\gamma_2 x(x-y)} = \frac{dP}{-\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P}.$$

Taking the ratio of the first two differentials, the $x(x-y)$ factors cancel, giving $dx/\gamma_1 = dy/(-\gamma_2)$. Integrating yields our characteristic constant:

$$C_1 = \gamma_2 x + \gamma_1 y.$$

The characteristic curves are therefore the straight lines of slope $-\gamma_2/\gamma_1$ in the (x,y)-plane. We parametrise y by setting $y(x) = (C_1 - \gamma_2 x)/\gamma_1$. To extract the equation for P along this curve, we need to define our coefficient $A(x,y)$:

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy. \tag{6.5}$$

Substituting $y = (C_1 - \gamma_2 x)/\gamma_1$ so that $\lambda_2 xy = (\lambda_2 C_1/\gamma_1)x - (\lambda_2\gamma_2/\gamma_1)x^2$, we obtain, collecting by powers of x,

$$\begin{aligned} A\big(x, y(x)\big) &= (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \frac{\lambda_2 C_1}{\gamma_1}\,x + \frac{\lambda_2\gamma_2}{\gamma_1}\,x^2 \\ &= -\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1}\,x^2 + \frac{\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1}{\gamma_1}\,x - \mu. \end{aligned}$$

Similarly, $x - y(x) = [(\gamma_1+\gamma_2)x - C_1]/\gamma_1$, so $\gamma_1 x(x - y(x)) = x[(\gamma_1+\gamma_2)x - C_1]$. The dP/P equation from the characteristic system therefore now reads as follows:

$$\frac{dP}{P} = \frac{-A(x,y(x))}{\gamma_1 x(x-y(x))}\,dx = \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu}{\gamma_1 x\big[(\gamma_1+\gamma_2)x - C_1\big]}\,dx.$$

To integrate this, we decompose the integrand via partial fractions. Writing the integrand as $\frac{1}{\gamma_1} I(x)\,dx$,

$$I(x) = K + \frac{M}{x} + \frac{N(C_1)}{(\gamma_1+\gamma_2)x - C_1};$$

satisfying

$$\begin{aligned} & K(\gamma_1+\gamma_2)x^2 + \big[M(\gamma_1+\gamma_2) + N - KC_1\big]x - MC_1 \\ = \quad & (\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu. \end{aligned}$$

The x^2 and x^0 coefficient matches give K and M immediately; the x^1 match determines N after simplification:

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1+\lambda_2}{\gamma_1+\gamma_2}C_1 - (\lambda_1+\lambda_2+\mu) + \frac{\mu(\gamma_1+\gamma_2)}{C_1}\right]. \tag{6.6}$$

For N, the x^1 coefficient match gives

$$\begin{aligned} N &= -\gamma_1(\lambda_1+\lambda_2+\mu) + \lambda_2 C_1 + KC_1 - M(\gamma_1+\gamma_2) \\ &= -\gamma_1(\lambda_1+\lambda_2+\mu) + \lambda_2 C_1 + \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)\,C_1}{\gamma_1+\gamma_2} + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1} \\ &= -\gamma_1(\lambda+\mu) + C_1\,\frac{\lambda_2(\gamma_1+\gamma_2) + \gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2} + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1} \\ &= -\gamma_1(\lambda+\mu) + \frac{\gamma_1\lambda}{\gamma_1+\gamma_2}\,C_1 + \frac{\gamma_1\mu(\gamma_1+\gamma_2)}{C_1}; \end{aligned}$$

factoring γ_1 out recovers $N(C_1)$ from (6.6).

Integrating $dP/P = (1/\gamma_1)\,I(x)\,dx$ term by term gives

$$\ln P = \frac{K}{\gamma_1}x - \frac{\mu}{C_1}\ln x + \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\ln\big[(\gamma_1+\gamma_2)x - C_1\big] + \ln f(C_1), \tag{6.7}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now note that

$$(\gamma_1+\gamma_2)x - C_1 = (\gamma_1+\gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x-y),$$

so $\ln[(\gamma_1+\gamma_2)x - C_1] = \ln\gamma_1 + \ln(x-y)$.

Exponentiating (6.7) term by term,

$$P \;=\; f(C_1)\,\exp\!\Big(\tfrac{K}{\gamma_1}x\Big)\cdot x^{-\mu/C_1}\cdot \gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}\,(x-y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))};$$

the factor $\gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}$ depends on x and y only through C_1 and can be absorbed, together with f, into a redefined arbitrary function $F(C_1)$:

$$P \;=\; F(C_1)\cdot\exp\!\Big(\tfrac{K}{\gamma_1}x\Big)\cdot x^{-\mu/C_1}\cdot(x-y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}.$$

Reading off the exponent of $(x-y)$, with $C_1 = \gamma_2 x + \gamma_1 y$, defines

$$E(x,y) := \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)} = \frac{\lambda_1+\lambda_2}{(\gamma_1+\gamma_2)^2}(\gamma_2 x + \gamma_1 y) - \frac{\lambda_1+\lambda_2+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{\gamma_2 x + \gamma_1 y}. \tag{6.8}$$

The general homogeneous solution is therefore $F(C_1)\,P_h(x,y)$, with F an arbitrary, differentiable function of the characteristic constant C_1 and P_h the particular homogeneous solution

$$P_h(x,y) = \exp\!\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x + \gamma_1 y)}\cdot(x-y)^{E(x,y)} \tag{6.9}$$

that will serve as the integrating factor for variation of parameters in Step (iii). We will see in Step (iv) that F must vanish.

Step (ii): The exponent $E(x,y)$ is strictly positive in $(0,1)^2$.

Recall $\lambda = \lambda_1 + \lambda_2$. Along any fixed characteristic, $C_1 = \gamma_2 x + \gamma_1 y$ is constant, so the exponent function (6.8) depends only on C_1, which ranges over $(0, \gamma_1+\gamma_2)$ as $(x,y) \in (0,1)^2$:

$$E(C_1) = \frac{\lambda}{(\gamma_1+\gamma_2)^2}\,C_1 - \frac{\lambda+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{C_1}.$$

Showing that $E(C_1) > 0$ on $C_1 \in (0, \gamma_1 + \gamma_2)$, follows from two facts. First, E vanishes at the right endpoint,

$$E(\gamma_1 + \gamma_2) = \frac{\lambda - (\lambda + \mu) + \mu}{\gamma_1 + \gamma_2} = 0.$$

Second, E is strictly decreasing in $(0, \gamma_1 + \gamma_2]$, using $u \leq \gamma_1 + \gamma_2$ and $\mu > \lambda$ (i.e. $\rho < 1$):

$$\begin{aligned} E'(u) &= \frac{\lambda}{(\gamma_1 + \gamma_2)^2} - \frac{\mu}{u^2} \\ &\leq \frac{\lambda}{(\gamma_1 + \gamma_2)^2} - \frac{\mu}{(\gamma_1 + \gamma_2)^2} = \frac{\lambda - \mu}{(\gamma_1 + \gamma_2)^2} < 0, \end{aligned}$$

Since E decreases strictly to 0 at its right endpoint, it is strictly positive at every point to its left, so:

$$E(C_1) > E(\gamma_1 + \gamma_2) = 0 \qquad \text{for all } C_1 \in (0, \gamma_1 + \gamma_2),$$

that is, $E(x, y) > 0$ for all $(x, y) \in (0, 1)^2$. For this proof, the crucial boundary behaviour is that $E \to 0$ as $(x, y) \to (1, 1)$; the blow-up caused by μ/C_1 as $C_1 \to 0^+$ is irrelevant.

Step (iii): Particular solution via variation of parameters.

This step consists in upgrading the homogeneous solution P_h (Equation (6.9)) with its inhomogeneous part, along the characteristic. We start by rewriting our fundamental equation for Model-B (Equation (6.1)). The coefficient of P on the LHS can be written as $yA(x, y)$ —where $A(x, y)$ is defined as in Step (i) (Equation (6.5))—; and we define the RHS of the fundamental equation as follows:

$$G(x, y) := \mu(x - y)\, P_y(y) - \mu\, x(1 - y)\, \pi(0, 0); \tag{6.10}$$

this forcing involves the unknown boundary trace $P_y(y) = P(0, y)$ together with the known constant $\pi(0, 0) = \rho(1 - \rho)$ of Corollary 2. Separating the terms that involve derivatives and the ones that do not, we have

$$\gamma_1 xy(x - y)\, \partial_x P + \gamma_2 xy(y - x)\, \partial_y P = -\, y\, A(x, y)\, P + G(x, y),$$

which is the convection form (3.14) of the Method of Characteristics. So, along the characteristic $C_1 = \gamma_2 x + \gamma_1 y$ found in Step (i) (parametrised by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$), §3.6.2 reduces it to a first-order linear ODE of the form (3.17),

$$\frac{dP}{dx} = f(x; C_1)\, P + h(x; C_1). \tag{6.11}$$

The homogeneous part $\frac{dP}{dx} = f(x; C_1)\, P$ is P_h (Equation (6.9)), so there is no need to write f out; and its forcing function is the right-hand side divided by the coefficient of $\partial_x P$,

$$h(x; C_1) = \frac{G\big(x, y(x)\big)}{\gamma_1\, x\, y(x)\, \big(x - y(x)\big)}. \tag{6.12}$$

Equation (6.11) is an instance of (3.17) with the known homogeneous solution P_h. So the variation-of-parameters formula (3.18) of §3.6.3 delivers its general solution, with the arbitrary constant becoming a function $F(C_1)$ of the characteristic label:

$$P(x, y(x)) = P_h(x, y(x)) \left[F(C_1) + \int_{x_0}^{x} \frac{h(t; C_1)}{P_h(t, y(t))}\, dt \right]. \tag{6.13}$$

Step (iv): Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic with constant $C_1 > 0$, the endpoint at $x = 0$ is $y(0) = C_1/\gamma_1$, giving

$$P\left(0, \frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which is finite, being a value of the boundary generating function P_y. However, P_h (Equation (6.9)) contains the factor $x^{-\mu/C_1} \to +\infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$ for $C_1 > 0$), while the integral in (6.13) vanishes as $x_0 \to 0$, since the integration interval shrinks. Therefore

$$\lim_{x \to 0^+} P(x, y(x)) = \lim_{x \to 0^+} P_h(x, y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can only be finite if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (6.13) yields the integral representation of Claim 1:

$$P(x, y(x)) = P_h(x, y(x)) \int_0^x \frac{h(t; C_1)}{P_h(t, y(t))}\, dt. \tag{6.14}$$

Step (v): Reduction to the Volterra equation.

By Corollary 2 the empty-state probabilities are $\pi_0 = 1 - \rho$ and $\pi(0,0) = \rho(1-\rho)$, with diagonal value $P(z,z) = \pi(0,0)/(1-\rho z)$. We plug those, along with our inhomogeneous part G (Equation (6.10)) into our forcing function (Equation (6.12))

$$\begin{aligned} h(t; C_1) &= \frac{\mu(t - y(t))\, P_y(y(t)) - \mu t(1-y(t))\, \pi(0,0)}{\gamma_1\, t\, y(t)\, (t - y(t))} \\ &= \frac{\mu\, P_y(y(t))}{\gamma_1\, t\, y(t)} - \frac{\mu(1-y(t))\, \pi(0,0)}{\gamma_1\, y(t)\, (t-y(t))}. \end{aligned} \tag{6.15}$$

We will evaluate Equation (6.14) along the diagonal —i.e. $x = y = z$, with points of the form (z,z)—, which has $C_1 = (\gamma_1 + \gamma_2)z$ and thus the following parametrisation, for fixed z:

$$y(t; z) = \frac{(\gamma_1 + \gamma_2)z - \gamma_2 t}{\gamma_1}, \tag{6.16}$$

so $y(0; z) = (\gamma_1 + \gamma_2)z/\gamma_1 > z$ and $y(z; z) = z$. Thus $y(t; z) > t$ for all $t \in [0, z)$, which implies $t - y(t; z) < 0$ inside the integration interval. Note the change of sign of the second summand when writing $t - y(t; z) = -(y(t; z) - t)$ in the denominator of (6.15):

$$h\big(t;\, (\gamma_1+\gamma_2)z\big) = \frac{\mu}{\gamma_1\, t\, y(t;z)}\, P_y\big(y(t;z)\big) + \frac{\mu\,(1 - y(t;z))}{\gamma_1\, y(t;z)\, (y(t;z) - t)}\, \pi(0,0). \tag{6.17}$$

From Step (ii), $E > 0$, so the factor $(x - y(x;z))^E$ in P_h vanishes as $x \to z^-$, giving $P_h\big(x, y(x;z)\big) \to 0$. At the same time, the integrand in (6.14) has a singularity near $t = z$: the forcing function (6.17) contributes with a $(y(t;z) - t)^{-1}$ factor; and the denominator $P_h\big(t, y(t;z)\big)$, with a $(y(t;z)-t)^{-E}$. So together $h/P_h \sim (y(t;z) - t)^{-1-E}$; since $-1 - E < -1$, the integral diverges as $x \to z^-$. Nevertheless, the product of the two factors is finite: by Step (iv), $P_h\big(x, y(x;z)\big) \int_0^x \frac{h}{P_h}\, dt$ equals the PGF value $P\big(x, y(x;z)\big)$, which converges to $P(z,z)$ as $x \to z^-$. This behaviour aligns with the fact that we are working in the context of PGFs, which are always bounded. By Corollary 2 that value is $\pi(0,0)/(1-\rho z)$, so the $0 \cdot \infty$ limit resolves to

$$P(z,z) = \frac{\pi(0,0)}{1-\rho z} = \lim_{x \to z^-} P_h\big(x, y(x;z)\big) \int_0^x \frac{h\big(t; (\gamma_1+\gamma_2)z\big)}{P_h\big(t, y(t;z)\big)}\, dt.$$

Substituting (6.17) and separating the term containing the unknown boundary function P_y from the known remainder, this limit splits into

$$\begin{aligned} \frac{\pi(0,0)}{1-\rho z} &= \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x;z)\big)}{\gamma_1} \int_0^x \frac{P_y\big(y(t;z)\big)}{t \cdot y(t;z) \cdot P_h\big(t, y(t;z)\big)}\, dt \\ &\quad + \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x;z)\big)\, \pi(0,0)}{\gamma_1} \int_0^x \frac{1 - y(t;z)}{y(t;z) \cdot (y(t;z) - t) \cdot P_h\big(t, y(t;z)\big)}\, dt. \end{aligned} \tag{6.18}$$

We evaluate the two limits in (6.18) separately. Three facts make the diagonal characteristic exactly solvable.

First, its geometry is linear:

$$y(t;z) - t = \frac{(\gamma_1+\gamma_2)z - \gamma_2 t}{\gamma_1} - t = \frac{\gamma_1 + \gamma_2}{\gamma_1}\,(z - t). \tag{6.19}$$

Second, the exponent of Step (ii) is constant along the characteristic ($C_1 = (\gamma_1+\gamma_2)z$) and can be factored as follows:

$$E_z := E\big((\gamma_1+\gamma_2)z\big) = \frac{\lambda z^2 - (\lambda+\mu)z + \mu}{(\gamma_1+\gamma_2)z} = \frac{(\mu - \lambda z)(1-z)}{(\gamma_1+\gamma_2)z} > 0, \qquad z \in (0,1).$$

Third, every factor of P_h in (6.9) except the exponential, the power of x, and $(x-y)^E$ depends on (x,y) only through C_1, which is fixed on the characteristic. The ratio therefore splits nicely into a smooth part and a single power:

$$\frac{P_h\big(x,y(x;z)\big)}{P_h\big(t,y(t;z)\big)} = R(x,t)\Big(\frac{z-x}{z-t}\Big)^{E_z}, \qquad R(x,t) := e^{\frac{K}{\gamma_1}(x-t)}\Big(\frac{t}{x}\Big)^{\mu/((\gamma_1+\gamma_2)z)}, \tag{6.20}$$

with R smooth and $R(z,z)=1$.

We apply that to the *second* limit of Equation (6.18). Substituting (6.20) and (6.19), its integrand near $t=z$ is $w(z)(z-t)^{-1-E_z}$ with $w(z)=\mu\pi(0,0)(1-z)/((\gamma_1+\gamma_2)z)$. The two singular factors then balance exactly: the prefactor vanishes like $(z-x)^{E_z}$ while the truncated integral diverges like $(z-x)^{-E_z}/E_z$. Hence their product converges to $w(z)/E_z$:

$$\begin{aligned}&\lim_{x\to z^-}(z-x)^{E_z}\int^x w(z)(z-t)^{-1-E_z}dt\\ &= \quad \frac{w(z)}{E_z} = \frac{\mu\pi(0,0)(1-z)}{(\gamma_1+\gamma_2)z}\cdot\frac{(\gamma_1+\gamma_2)z}{(\mu-\lambda z)(1-z)} = \frac{\mu\pi(0,0)}{\mu-\lambda z} = \frac{\pi(0,0)}{1-\rho z}.\end{aligned}$$

The *first* limit vanishes: its integrand behaves as $(z-t)^{-E_z}$, one power less singular than the second limit's $(z-t)^{-1-E_z}$, and is killed by the vanishing prefactor $(z-x)^{E_z}$. The *second* limit reproduces the full diagonal value from $\pi(0,0)$. Thus the diagonal fixes no information about P_y at leading order; determining P_y requires the next order in $(z-x)$, which is the open problem.

Collecting the P_y-term of (6.18) into a kernel and the remaining $\pi(0,0)$-term into a known datum gives

$$\mathcal{K}(z,t) := \frac{\mu}{\gamma_1}\,\frac{R(z,t)}{t\,y(t;z)}\,(z-t)^{-E_z}, \tag{6.21}$$

$$\mathcal{F}(z) := \frac{\pi(0,0)}{1-\rho z} - \frac{\mu\,\pi(0,0)}{\gamma_1}\int_0^z \frac{R(z,t)\,\big(1-y(t;z)\big)}{y(t;z)\,\big(y(t;z)-t\big)}\,(z-t)^{-E_z}\,dt, \tag{6.22}$$

so that the diagonal identity (6.18) matches the first-kind Volterra equation (6.4), which proves the claim. Both integrals are singular at the endpoint $t=z$ —their integrands behave as $(z-t)^{-E_z}$ and $(z-t)^{-1-E_z}$— and must be read in a regularised, finite-part sense; carrying out that regularisation rigorously is precisely the analytical obstruction that leaves Model-B open. □

Model-B is beyond the scope of this thesis and, therefore, left open. Its role is to illustrate the mathematical obstacle —the Volterra-type integral equation. Model-B_2 and Model-C_2 avoid such a problem by shutting down all derivative terms in y. This allows us to take our boundary function $P_y(y)$ outside the integral, as we would only integrate with respect to x, as opposed to $y(x)$.

Remark 3 (The full Model-X is *a fortiori* intractable). Model-B already shows the central problem of taking this route. As one can imagine, reinstating the abandonment mechanism —$\theta_1,\theta_2>0$— does not help simplify the problem; quite the opposite. It deepens the difficulty in two ways: first, the characteristics cease to be straight lines; second, we lose the diagonal anchor —Corollary 2 no longer holds—, so the diagonal $P(z,z)$ is no longer known in closed form.

7 Analysis of Model-B_2 ($\gamma_1>0$, $\gamma_2=0$, $\theta_1=\theta_2=0$)

Model-B_2 keeps one-way jockeying from class-1 to class-2 ($\gamma_1>0$, $\gamma_2=0$) and no abandonment ($\theta_1=\theta_2=0$). This model is more tractable than Model-B, as all the terms with derivatives in y disappear, leaving us a first-order linear ODE in x, for each fixed $y\in(0,1]$. We will solve this using the integrating factor (§3.6.1).

Stability. As in Models A and B, $\rho=(\lambda_1+\lambda_2)/\mu<1$. Jockeying conserves the total customer count and does not affect the stability condition. Figure 6 shows the queue layout.

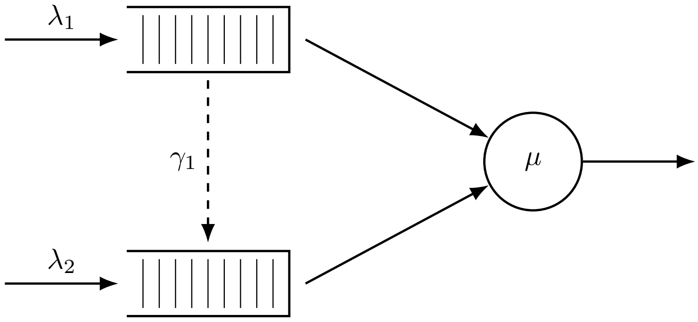

Figure 6: Queue layout for Model-B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): class-1 customers may defect to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

7.1 Reduced fundamental equation

Setting $\gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ in the general fundamental equation (4.14) (Lemma 1) annihilates the derivative terms in y, leaving:

$$\begin{aligned}\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big] P + \gamma_1 xy(x-y)\,\frac{\partial P}{\partial x} \\ = \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0).\end{aligned} \tag{7.1}$$

For each fixed $y \in (0,1]$ this is a first-order linear ODE in x; the entire analysis of this section is built on it.

Theorem 2. *Consider the two-class non-preemptive priority queueing system of Model-B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the boundary generating function is*

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{7.2}$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{7.3}$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1-y) + \gamma_1}{\gamma_1}, \tag{7.4}$$

the auxiliary integrals are

$$\mathcal{H}_1(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha-1}(y-t)^{\beta}\, dt, \tag{7.5}$$

$$\mathcal{H}_2(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha}(y-t)^{\beta-1}\, dt, \tag{7.6}$$

and the empty-state probability $\pi(0,0) = \rho(1-\rho)$ is given by Corollary 3 below. On $y < x \le 1$ the same expression (7.3) holds with $\mathcal{H}_1, \mathcal{H}_2$ replaced by their finite-part continuations; that the right-hand side is genuinely analytic across the interior point $x = y$ is the content of the smoothness condition established in Step (ii) of the proof.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models.

Corollary 3. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model-B_2 are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho),$$

identical to the Model-A baseline.

As in Model-B, the proof follows trivially by setting $x = y = z$ in our fundamental equation (Equation (7.1)). The surviving equation is identical to the homologous in Model-A, and the result is proven in Corollary 1.

The proof of Theorem 2 consists of three steps: (i) obtaining the standard form and the integration factor, (ii) determining $P_y(y)$ by analyticity in $x = y$, and (iii) recovering $P(x, y)$.

Proof of Theorem 2. **Step (i): Standard form and the integrating factor.**

Fix $y \in (0, 1]$, cancel a factor y on both sides and divide the reduced fundamental equation (7.1) by the derivative term in x. The equation takes standard form $\frac{dP}{dx} + p(x; y)\, P = q(x; y)$ of §3.6.1 (Equation (3.12)), with

$$p(x; y) = \frac{(\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x - y)}, \tag{7.7}$$

$$q(x; y) = \frac{\mu\, P_y(y)}{\gamma_1 xy} - \frac{\mu(1 - y)\, \pi(0, 0)}{\gamma_1 y(x - y)}. \tag{7.8}$$

Write the numerator of p as a polynomial in x,

$$N(x) = -\lambda_1 x^2 + \big[(\lambda_1 + \lambda_2 + \mu) - \lambda_2 y\big]x - \mu = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big),$$

where the roots $x^*(y), x^+(y)$ are those found for the kernel in Model-A (Equation (5.8)), with $\lambda_1 x^* x^+ = \mu$ following from Vieta's formula. The Partial Fraction Decomposition of $p = N(x)/\big[\gamma_1 x(x-y)\big]$ reads as follows:

$$\begin{aligned}
A &= -\lambda_1,\\
B &= \frac{N(0)}{0 - y} = \frac{-\mu}{-y} = \frac{\mu}{y},\\
C &= \frac{N(y)}{y} = \frac{-(\lambda y - \mu)(y - 1)}{y} = \frac{(1 - y)(\lambda y - \mu)}{y},
\end{aligned}$$

with $N(y) = -\lambda y^2 + (\lambda + \mu)y - \mu = -(\lambda y - \mu)(y - 1)$. Thus

$$p(x; y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1 - y)(\lambda y - \mu)/y}{x - y}\right]. \tag{7.9}$$

Since we will have a factor $(x - y)$ due to jockeying, we work on the region $0 \le x \le y$, where $x - y \le 0$; the complementary region $y < x \le 1$ is recovered later. On this region, a term $\beta \ln|x - y|$ becomes simply $\beta \ln(y - x)$, so integrating the partial fraction (7.9) yields, up to a constant,

$$\begin{aligned}
\int^x p(t; y)\, dt &= \frac{1}{\gamma_1}\Big[-\lambda_1 x + \frac{\mu}{y}\ln x + \frac{(1 - y)(\lambda y - \mu)}{y}\ \ln(y - x)\Big]\\
&= -\frac{\lambda_1}{\gamma_1}\, x + \alpha(y) \ln x + \beta(y) \ln(y - x),
\end{aligned}$$

with the exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1 - y)(\lambda y - \mu)/(\gamma_1 y)$of (7.4). Exponentiating, and dropping the multiplicative constant to which an integrating factor is indifferent, gives our integrating factor

$$\mu_I(x; y) = e^{-\lambda_1 x/\gamma_1}\, x^{\alpha(y)}\, (y - x)^{\beta(y)},$$

real-valued in our region because we take the base $(y - x) > 0$ rather than $(x - y)$; its reciprocal $1/\mu_I$ is the homogeneous solution, playing here the role that P_h played for Model-B (§3.6.3). It indeed solves $\mu_I'/\mu_I = p$:

$$\begin{aligned}
\frac{\mu_I'(x; y)}{\mu_I(x; y)} &= -\frac{\lambda_1}{\gamma_1} + \frac{\alpha}{x} + \beta\, \frac{d}{dx} \ln(y - x)\\
&= -\frac{\lambda_1}{\gamma_1} + \frac{\alpha}{x} + \frac{\beta}{x - y} = p(x; y).
\end{aligned}$$

The general solution of §3.6.1 (Equation (3.13)) now applies, and its arbitrary function is fixed immediately: since $\alpha > 0$ we have $\mu_I \to 0$ as $x \to 0^+$, so the boundary contribution $P(0, y)\, \mu_I(0; y)$ vanishes and integrating $\frac{d}{dx}[P\, \mu_I] = q\, \mu_I$ over $(0, x]$ leaves

$$P(x, y)\, \mu_I(x; y) = \int_0^x q(t; y)\, \mu_I(t; y)\, dt.$$

Multiplying q from (7.8) by μ_I and applying the identity $(y-t)^\beta/(t-y) = -(y-t)^{\beta-1}$ to the second term, the integrand separates into two pieces,

$$q(t;y)\,\mu_I(t;y) = \frac{\mu}{\gamma_1 y}\Big[P_y(y)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha-1}(y-t)^\beta + (1-y)\pi(0,0)\,e^{-\lambda_1 t/\gamma_1}\,t^\alpha(y-t)^{\beta-1}\Big],$$

With the auxiliary integrals $\mathcal{H}_1, \mathcal{H}_2$ of (7.5)–(7.6) and dividing through by $\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}x^\alpha(y-x)^\beta$, the solution on the principal region $0 \le x \le y$ is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{7.10}$$

which still contains the unknown boundary function $P_y(y)$; Step (ii) determines it.

Step (ii): Determine $P_y(y)$ by analyticity at $x = y$.

Under stability (i.e. $\rho < 1$) $\mu - \lambda y > 0$ for $y \in (0,1)$, so the second exponent is negative:

$$\beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{7.11}$$

So the prefactor $(y-x)^{-\beta} = (y-x)^{|\beta|}$ in (7.10) *vanishes* as $x \to y^-$, while $\mathcal{H}_1, \mathcal{H}_2$ may *diverge*: the product is an indeterminate $0\cdot\infty$ form, whose resolution fixes $P_y(y)$.

On the diagonal the value is known outright. Setting $x = y = z$ in the fundamental equation (7.1) drops the jockeying and boundary terms (each containing a factor $x - y$) and leaves Model A's balance, so

$$P(z,z) = \frac{\rho(1-\rho)}{1-\rho z}, \tag{7.12}$$

the diagonal value of Corollary 1.

Smoothness condition and $P_y(y)$. Pull the analytic, non-vanishing prefactor of (7.10) into $\Psi(x) = \mu\, e^{\lambda_1 x/\gamma_1}x^{-\alpha}/(\gamma_1 y)$, so that

$$P(x,y) = \Psi(x)\,(y-x)^{|\beta|}\big[\,P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\,\big], \qquad |\beta| = -\beta > 0.$$

We need only the leading behaviour of the two integrals as $x \to y^-$, not a full expansion. Each splits into its endpoint value $\mathcal{H}_i(y,y) = \int_0^y(\cdots)\,dt$ and a remainder set by the integrand near $t = y$ — $\mathcal{H}_1$ carries $(y-t)^\beta$, $\mathcal{H}_2$ carries $(y-t)^{\beta-1}$ — so that, with constants c_1, c_2 whose values we never need,

$$\mathcal{H}_1(x,y) = \mathcal{H}_1(y,y) + c_1\,(y-x)^{\beta+1} + \cdots, \qquad \mathcal{H}_2(x,y) = \mathcal{H}_2(y,y) + c_2\,(y-x)^\beta + \cdots.$$

Multiplied by $(y-x)^{-\beta}$, the c_1, c_2 remainders turn into a smooth $O(y-x)$ term and the constant regular value (7.12); only the endpoint values retain the bare, non-integer power $(y-x)^{|\beta|}$ ($|\beta| \notin \mathbb{Z}$ generically). Since $P(\cdot, y)$ is a power series in x — hence smooth at the interior point $x = y < 1$, whereas that power is not — its coefficient must vanish:

$$P_y(y)\,\mathcal{H}_1(y,y) + (1-y)\,\pi(0,0)\,\mathcal{H}_2(y,y) = 0, \tag{7.13}$$

the endpoint integrals being read, since $\beta < 0$, as the Euler–Beta continuations found next. To evaluate these endpoint integrals, substitute $t = ys$ ($dt = y\,ds$) in (7.5)–(7.6), which rescales the two integrands and pulls out a common $y^{\alpha+\beta}$,

$$\begin{aligned} t^{\alpha-1}(y-t)^\beta\,dt &= y^{\alpha+\beta}\,s^{\alpha-1}(1-s)^\beta\,ds, \\ t^\alpha(y-t)^{\beta-1}\,dt &= y^{\alpha+\beta}\,s^\alpha(1-s)^{\beta-1}\,ds, \end{aligned}$$

so that, with $c := \lambda_1 y/\gamma_1$,

$$\mathcal{H}_1(y,y) = y^{\alpha+\beta}\int_0^1 s^{\alpha-1}(1-s)^\beta e^{-cs}\,ds, \qquad \mathcal{H}_2(y,y) = y^{\alpha+\beta}\int_0^1 s^\alpha(1-s)^{\beta-1}e^{-cs}\,ds.$$

These are Euler integrals (3.10), $\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt = B(a,b)\,{}_1F_1(a;a+b;z)$, with $(a,b,z) = (\alpha, \beta+1, -c)$ for $\mathcal{H}_1$ and $(\alpha+1, \beta, -c)$ for $\mathcal{H}_2$, so $a + b = b^\star(y)$ in both. As $\beta < 0$ by (7.11), the right-hand side is read as its analytic continuation in b beyond the convergence range $b > 0$, giving

$$\mathcal{H}_1(y,y) = B(\alpha,\,\beta+1)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big), \tag{7.14}$$

$$\mathcal{H}_2(y,y) = B(\alpha+1,\,\beta)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big). \tag{7.15}$$

Solving the smoothness condition (7.13) for $P_y(y)$ and inserting the endpoint values (7.14)–(7.15), the common factor $y^{\alpha+\beta}$ cancels and

$$P_y(y) = -(1-y)\,\pi(0,0)\,\frac{\mathcal{H}_2(y,y)}{\mathcal{H}_1(y,y)} = -(1-y)\,\pi(0,0)\,\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)}\,\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}. \tag{7.16}$$

The Beta ratio reduces, via $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ and $\Gamma(\beta+1) = \beta\Gamma(\beta)$, to

$$\begin{aligned}\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)} &= \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}\cdot\frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)}\\ &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\cdot\frac{\Gamma(\beta)}{\Gamma(\beta+1)} = \frac{\alpha}{\beta},\end{aligned} \tag{7.17}$$

and the surviving prefactor simplifies, using $\alpha/\beta = \mu/\big[(1-y)(\lambda y-\mu)\big]$, as

$$\begin{aligned}-(1-y)\,\frac{\alpha}{\beta} &= -(1-y)\cdot\frac{\mu}{(1-y)(\lambda y-\mu)}\\ &= -\frac{\mu}{\lambda y-\mu} = \frac{\mu}{\mu-\lambda y}.\end{aligned} \tag{7.18}$$

Substituting (7.17)–(7.18) into (7.16) gives the boundary generating function

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha(y)+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}, \tag{7.19}$$

confirming (7.2). The prefactor equals the regular value $P(y,y)$ of (7.12), and the Kummer ratio is the boundary correction. As $y \to 0^+$ the argument $-\lambda_1 y/\gamma_1 \to 0$, so the ratio tends to 1 and $P_y(0) = \pi(0,0)$, consistent with $P_y(0) = \sum_{n_2}\pi(0,n_2)\cdot 0^{n_2} = \pi(0,0)$.

Step (iii): Recovery of $P(x,y)$.

Substituting (7.19) into the real-form expression (7.10) gives, for $0 \le x \le y$,

$$\begin{aligned}P(x,y) = \frac{\mu\,e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[&\frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}\cdot\mathcal{H}_1(x,y)\\ &+(1-y)\pi(0,0)\cdot\mathcal{H}_2(x,y)\Bigg],\end{aligned}$$

which is the statement of the theorem (Equation (7.3)). The smoothness condition (7.13) guarantees that this extends analytically across $x = y$, with diagonal value $P(y,y) = \rho(1-\rho)/(1-\rho y)$ as computed in (7.12). On $y < x \le 1$ the same formula holds with $\mathcal{H}_1, \mathcal{H}_2$ read as their finite-part continuations; $\pi(0,0) = \rho(1-\rho)$ from Corollary 3 makes the solution fully explicit. The recovery of Model-A in the limit $\gamma_1 \to 0^+$ is deferred to Remark 4 below, as the limit is singular and merits separate discussion. □

Remark 4 (The singular limit $\gamma_1 \to 0^+$). The no-jockeying limit $\gamma_1 \to 0^+$ is a *singular* perturbation: the coefficient $\gamma_1 xy(x-y)$ of the highest derivative $\partial_x P$ vanishes, so one must check that the convection term itself, not merely its coefficient, disappears. It does. Each P is a probability generating function, so $|P| \le 1$ on the closed unit polydisc uniformly in γ_1, and Cauchy's estimate then bounds $\partial_x P$ uniformly on every compact subset of the open polydisc; hence $\gamma_1 xy(x-y)\,\partial_x P \to 0$ there. The fundamental equation (7.1) therefore degenerates to the algebraic Model-A equation (5.2), and the uniformly bounded family $\{P\}_{\gamma_1}$, being normal, converges (in full, by uniqueness of the analytic solution) to the Model-A PGF (5.3). In particular the boundary function reduces to the Model-A value (5.9),

$$P_y(y) \xrightarrow[\gamma_1\to 0^+]{} \pi(0,0)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}, \tag{7.20}$$

with $x^*(y)$ the Model-A kernel root (5.8); this needs no special-function asymptotics.

The same limit can be read off the closed form (7.19). Its prefactor $\mu\,\pi(0,0)/(\mu-\lambda y) = \pi(0,0)/(1-\rho y)$ is γ_1-independent, so (7.20) is equivalent to the Kummer ratio tending to

$$\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\lambda_1 y/\gamma_1\big)} \xrightarrow[\gamma_1\to 0^+]{} (1-\rho y)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}. \tag{7.21}$$

The factor $(1-\rho y)$ shows the ratio does *not* tend to the bare $x^*(y)(1-y)/(x^*(y)-y)$; the discrepancy is exactly the prefactor. This matches the first-parameter three-term recurrence for ${}_1F_1$ [DLM, §13.3]: as $\gamma_1 \to 0^+$ the parameters α, $b^\star$ and $c = \lambda_1 y/\gamma_1$ all grow like $1/\gamma_1$, and the recurrence reduces at leading order to a quadratic whose admissible root is (7.21).

8 Analysis of Model-C_2 ($\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

In this section we derive $P(x, y)$ for the model in which only class-1 customers abandon, with no class-2 abandonment and no jockeying. Figure 7 shows the queue layout.

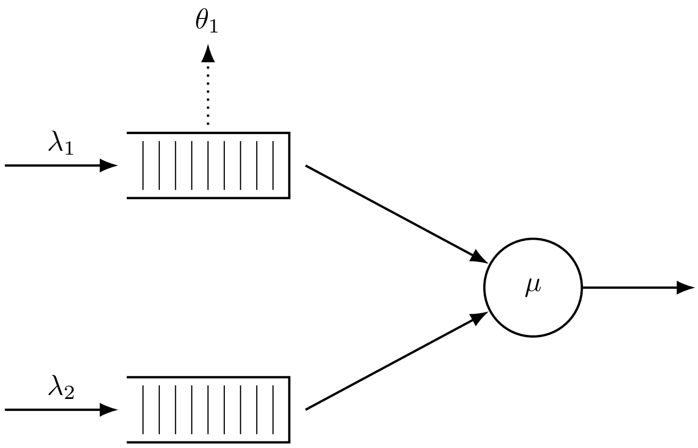

Figure 7: Queue layout for Model-C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$). There is no jockeying.

This choice of parameters is motivated by two key aspects of the resulting model: first, class-2 still sees Poisson arrivals, so the probabilistic argument of Model-A carries over; second, substituting these values into the general fundamental equation (4.14) leaves a first-order linear ODE in x with no y-derivatives, so the boundary function $P_y(y)$ comes out of the integral, sparing us the Volterra equation that left Model-B open (§6).

Stability. Unlike the models without abandonment, stability here is non-trivial and rests on two key observations that also drive the analysis below. First, the class queue 1 alone is an $M/M/1{+}M$ (Erlang-A) subsystem with rates λ_1, μ, θ_1, which is stable for *every* load (§3.3); its busy period B_C is always well defined. Second, class-2 sees an $M/G/1$ queue whose service time is one such class-1 busy period B_C (made precise in §8.3). The only constraint on stability comes from this second queue, which is stable exactly when its utilisation stays below one:

$$\lambda_2\,\mathbb{E}[B_C] \;<\; 1 \qquad \Longleftrightarrow \qquad \rho_2 \cdot {}_1F_1\Big(1;\, \tfrac{\mu}{\theta_1}+1;\, \tfrac{\lambda_1}{\theta_1}\Big) \;<\; 1. \tag{8.1}$$

8.1 Reduced fundamental equation

Setting $\theta_1 > 0$ and $\gamma_1 = \gamma_2 = \theta_2 = 0$ in the general fundamental equation (4.14) clears most of the derivative terms, with the class-1 abandonment term as the only survivor. What remains is, for each fixed y, a first-order linear ODE in x:

$$\begin{aligned}
&\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]\, P(x,y) + \theta_1\, xy(x-1)\,\frac{\partial}{\partial x} P(x,y) \\
&= \quad \mu(x-y)\, P_y(y) - \mu x(1-y)\, \pi(0,0).
\end{aligned} \tag{8.2}$$

Equation (8.2) has two unknowns: $P(x, y)$ and its boundary trace $P_y(y)$.

The following theorem states the closed form. Its proof, given immediately afterwards in §8.2, is purely analytical; §8.3 then re-derives the boundary function $P_y(y)$ probabilistically as an independent method to determine the boundary function.

Theorem 3. *Consider the two-class non-preemptive priority system of Model-C_2 ($\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$): Poisson arrivals at rates λ_1, λ_2, common exponential service at rate μ, and per-customer abandonment of* waiting *class-1 customers at rate θ_1. Under the stability condition (8.1),* *the probability generating function is*

$$P(x,y) = \frac{e^{\lambda_1 x/\theta_1}\, \mu\, \pi(0,0)\,(1-y)}{\theta_1\, x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y)) - y\big)} \times \Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y) - \big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big],$$

where

$$\mathcal{H}_1(x,y) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1 - 1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\, dt,$$
$$\mathcal{H}_2(x,y) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1 - 1}\, dt,$$

$\widetilde{B}_C(s)$ is the LST of the busy period of the class-1 $M/M/1{+}M$ subsystem given by (3.3), and $\pi(0,0)$ is given by Corollary 4 below.

The following corollary is invoked:

Corollary 4. *The limiting probability $\pi(0,0)$ of the busy state with both queues empty, and the empty-system probability π_0, are*

$$\pi(0,0) = \frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)} = (\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{8.3}$$

$$\pi_0 = \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{8.4}$$

where $\mathbb{E}[B_C]$ is the mean class-1 busy period (3.4). As $\theta_1 \to 0^+$, $\mathbb{E}[B_C] \to (\mu-\lambda_1)^{-1}$ and these recover the Model-A baseline $\pi(0,0) = \rho(1-\rho)$, $\pi_0 = 1-\rho$.

8.2 Analytical derivation of $P(x,y)$

Proof of Theorem 3. **Step (i): standard form and integrating factor.**

The surviving abandonment factor $(x-1)$ places the operative singularity on the boundary $x=1$. Dividing (8.2) by the coefficient $\theta_1 xy(x-1)$ of the derivative puts it in the standard form (Equation (3.12)):

$$\frac{\partial P}{\partial x} + \underbrace{\frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\theta_1\, x(x-1)}}_{p(x;y)}\, P = \underbrace{\frac{\mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0)}{\theta_1\, xy(x-1)}}_{q(x;y)}.$$

The numerator of $p(x;y)$ is the quadratic $A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy$, the kernel polynomial common to all seen models. Applying Partial Fraction Decomposition to $A(x;y)/(x(x-1))$ yields

$$\begin{aligned} &A = \lambda_1,\\ &B = A(0,y)/(0-1) = (-\mu)/(-1) = \mu,\\ &C = A(1,y) = \lambda_2(1-y); \end{aligned}$$

so,

$$p(x;y) = \frac{1}{\theta_1}\cdot\frac{A(x;y)}{x(x-1)} = \frac{1}{\theta_1}\left[-\lambda_1 + \frac{\mu}{x} + \frac{\lambda_2(1-y)}{x-1}\right]. \tag{8.5}$$

Integrating term by term, $\int p\,dx = \theta_1^{-1}\big[-\lambda_1 x + \mu\ln x + \lambda_2(1-y)\ln(1-x)\big]$, which exponentiates to the integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\theta_1}\, x^{\alpha}\,(1-x)^{\beta(y)}, \tag{8.6}$$

with

$$\alpha = \frac{\mu}{\theta_1}, \qquad \beta(y) = \frac{\lambda_2(1-y)}{\theta_1}. \tag{8.7}$$

Both exponents are non-negative ($\alpha > 0$ always; $\beta(y) \geq 0$ for $y \in [0,1]$), so μ_I vanishes at $x = 0$ and at $x = 1$. By the integrating-factor solution (3.13) —with the boundary term $P(0,y)\,\mu_I(0;y)$ killed by $\mu_I(0;y) = 0$— we obtain

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)} \int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{8.8}$$

Step (ii): determination of $P_y(y)$ by boundedness at $x = 1$.

We determine $P_y(y)$ by imposing boundedness of $P(x,y)$ as $x \to 1$, for fixed $y \in [0,1)$: we have $\beta(y) > 0$, so $\mu_I(x;y) \to 0$. Since the left-hand side $P(x,y)$ of (8.8) stays bounded while $1/\mu_I(x;y) \to \infty$, the integral must vanish in the limit:

$$\int_0^1 q(t;y)\,\mu_I(t;y)\,dt \;=\; 0. \tag{8.9}$$

Substituting the explicit q and μ_I, the integrand is

$$q(t;y)\,\mu_I(t;y) \;=\; \frac{\mu\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]}{\theta_1\, y} \cdot \frac{e^{-\lambda_1 t/\theta_1}\, t^{\alpha}(1-t)^{\beta(y)}}{t\,(t-1)},$$

where $t^\alpha/t = t^{\alpha-1}$ and $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$ flips the sign and lowers the exponent. The constant prefactor $-\mu/(\theta_1 y)$, nonzero for $y \in (0,1]$, divides out of (8.9), leaving

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]\,dt \;=\; 0.$$

Introduce the auxiliary integrals

$$\begin{aligned}
\mathcal{J}_0(y) \;&=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\,dt,\\
\mathcal{J}_1(y) \;&=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt.
\end{aligned}$$

Distributing the bracket over $t^{\alpha-1}$, we get the term $t \cdot t^{\alpha-1} = t^\alpha$ in $\mathcal{J}_1$ and $t^{\alpha-1}$ in $\mathcal{J}_0$, becoming

$$\begin{aligned}
0 \;&=\; P_y(y)\int_0^1 e^{-\lambda_1 t/\theta_1}\big(t^\alpha - y\,t^{\alpha-1}\big)(1-t)^{\beta(y)-1}\,dt \;-\; (1-y)\,\pi(0,0)\int_0^1 e^{-\lambda_1 t/\theta_1}\, t^\alpha(1-t)^{\beta(y)-1}\,dt\\
&=\; P_y(y)\big[\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)\big] \;-\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y),
\end{aligned}$$

so

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)}. \tag{8.10}$$

It remains to evaluate the ratio $\mathcal{J}_1/\mathcal{J}_0$. Setting $a = \alpha = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$, the Euler representation (3.10) writes each $\mathcal{J}_i$ as a Beta multiple of a ${}_1F_1$ value, and the closed-form summation (3.7) of §3.3 identifies their ratio with the busy-period LST:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^a(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{8.11}$$

Taking $s = \lambda_2(1-y)$ —so that $b = \beta(y)$ matches the exponent in (8.7)— gives $\widetilde{B}_C(\lambda_2(1-y)) = \mathcal{J}_1(y)/\mathcal{J}_0(y)$. Dividing the numerator and denominator of (8.10) by $\mathcal{J}_0(y)$ yields the boundary generating function

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{8.12}$$

This reaches $P_y(y)$ analytically through the Beta-integral (Equation (8.11)) alone, without requiring the probabilistic argument in §8.3.

Step (iii): recovery of $P(x,y)$.

We return to the ODE (Equation (8.2)) and substitute $P_y(y)$ (Equation (8.12)) into its rhs. Writing $\zeta(y) \equiv \widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} &= \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0) \\
&= \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y} - x\right] \\
&= \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y) - x\big[\zeta(y)-y\big]}{\zeta(y)-y} \\
&= \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which has the same form as the RHS of Model-A, now with the more involved service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in (8.5) and (8.6), the standard-form source becomes

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]},$$

and (8.8) reads $P(x,y) = \mu_I(x;y)^{-1}\int_0^x q(t;y)\,\mu_I(t;y)\,dt$. To carry out the integral, use the partial fraction

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1)+t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\begin{aligned}
\mathcal{H}_1(x,y) &= \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt, \\
\mathcal{H}_2(x,y) &= \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt,
\end{aligned}$$

with $\alpha, \beta(y)$ as in (8.7). They are related to the full integrals of Step-(ii) in the following way: $\mathcal{H}_2(1,y) = \mathcal{J}_1(y)$ exactly, and $\mathcal{H}_1(1,y) \neq \mathcal{J}_0(y)$ (its $(1-t)$ exponent is $\beta(y)$, not $\beta(y)-1$).

Multiplying $q(t;y)$ by $\mu_I(t;y) = e^{-\lambda_1 t/\theta_1}t^{\alpha}(1-t)^{\beta(y)}$ and inserting the partial fraction, the integrand splits into the two pieces defining $\mathcal{H}_1, \mathcal{H}_2$:

$$\begin{aligned}
q(t;y)\,\mu_I(t;y) &= \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\left[\frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1}\right] e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)} \\
&= \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha-1}(1-t)^{\beta(y)} - \big(1-\zeta(y)\big)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\Big],
\end{aligned}$$

the second term again using $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$, which produces both the minus sign and the exponent shift on $\mathcal{H}_2$. Since $\pi(0,0)$ and $\zeta(y)$ leave the integral,

$$\int_0^x q(t;y)\,\mu_I(t;y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,\mathcal{H}_1(x,y) \;-\; \big(1-\zeta(y)\big)\,\mathcal{H}_2(x,y)\Big].$$

Dividing by $\mu_I(x;y)$ and restoring $\zeta(y) = \widetilde{B}_C(\lambda_2(1-y))$, we arrive at the closed form

$$\begin{aligned}
P(x,y) \;=\; &\frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \\
&\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y) - \big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big],
\end{aligned} \tag{8.13}$$

which is the expression in the statement of the theorem. □

The non-trivial proof for the corollary invoked in Theorem 3 comes next

Proof of Corollary 4. The idle system state, with limiting probability π_0, only communicates to the busy-server-empty-queues state with probability $\pi(0,0)$. No abandonments can take place in any of the two states, as the queues are empty in both cases. We have, as Equation (4.7),

$$(\lambda_1+\lambda_2)\,\pi_0 \;=\; \mu\,\pi(0,0), \tag{8.14}$$

so $\pi(0,0) = (\rho_1 + \rho_2)\,\pi_0$. We again use the $M/G/1$ structure identified in §8.3. The long run fraction this queue is busy is $\rho_B = \lambda_2\,\mathbb{E}[B_C]$, and "idle" refers exclusively to no class-2 customers being present. So

$$\mathbb{P}(\text{no class-2 present}) \;=\; 1 - \lambda_2\,\mathbb{E}[B_C].$$

Conditional on not having class-2 customers, the priority subsystem evolves as the $M/M/1+M$ queue $(\lambda_1, \mu, \theta_1)$ defining B_C alternates idle and busy periods in a renewal structure, whose regenerations are the moments it empties. An idle period lasts until the next class-1 arrival takes place, i.e. at an exponential rate λ_1. We have

$$\mathbb{P}(\text{class-1 absent} \mid \text{no class-2 present}) \;=\; \frac{1/\lambda_1}{1/\lambda_1 + \mathbb{E}[B_C]} \;=\; \frac{1}{1 + \lambda_1\,\mathbb{E}[B_C]}.$$

The fully idle event is the intersection of $A = \{\text{class-1 absent}\}$ and $B = \{\text{no class-2 present}\}$, so $\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$ (not independence),

$$\pi_0 \;=\; \big(1 - \lambda_2\,\mathbb{E}[B_C]\big) \cdot \frac{1}{1 + \lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1 - \lambda_2\,\mathbb{E}[B_C]}{1 + \lambda_1\,\mathbb{E}[B_C]},$$

which is (8.4); substituting into (8.14) gives (8.3). □

8.3 Alternative derivation of $P_y(y)$: the $M/G/1$ / Pollaczek–Khinchine route

The boundary function $P_y(y)$ admits a second derivation, using the same probabilistic argument seen in §5.1.2 that does not require imposing boundedness. By the *law of total expectation*, splitting $P_y(y) = \mathbb{E}[\mathbf{1}\{\text{busy}, N_1 = 0\}\, y^{N_2}]$ over the event $\{\text{busy}, N_1 = 0\}$ factors it into the mass of that event and the conditional PGF of N_2 given it:

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0) \,\cdot\, \mathbb{E}\big[y^{N_2} \,\big|\, \text{busy},\, N_1 = 0\big]\,. \tag{8.15}$$

We adopt the point of view of a class-2 customer while the server is busy. Unlike jockeying, class-1 customers abandon the system, so the class-2 Poisson arrival stream still holds. The class-2 *effective service time* is therefore one full busy period B_C of the class-1 subsystem — an $M/M/1+M$ queue with rates λ_1, μ, θ_1, rather than the pure $M/M/1$ of Model-A. So class-2 sees an $M/G/1$ queue with Poisson input λ_2 and service distributed as B_C, whose LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ come from the first-step analysis of §3.3 (Equations (3.3) and (3.4)). This $M/G/1$ is stable if and only if its utilisation $\lambda_2\,\mathbb{E}[B_C] < 1$ (Equation (8.1)), a condition strictly weaker than the no-abandonment requirement $\rho = \rho_1 + \rho_2 < 1$.

Applying the Pollaczek–Khinchine formula (3.8) with the identifications $\lambda \mapsto \lambda_2$, $\widetilde{B} \mapsto \widetilde{B}_C$, $\rho_B = \lambda_2\,\mathbb{E}[B_C]$ gives the PGF of the stationary number of class-2 customers in that $M/G/1$:

$$L_2(y) \;=\; \frac{(1 - \lambda_2 \mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}.$$

Since class-2 arrivals are Poisson, by *PASTA* (§3.2) the number of class-2 customers an arrival finds equals the time-average; conditioning on $\{\text{busy},\, N_1 = 0\}$ identifies that count with N_2:

$$\mathbb{E}\big[y^{N_2} \,\big|\, \text{busy},\, N_1 = 0\big] \;=\; L_2(y). \tag{8.16}$$

Substituting (8.16) into the decomposition (8.15) gives

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0)\, \frac{(1 - \lambda_2 \mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{8.17}$$

The unknown $\mathbb{P}(\text{busy}, N_1 = 0)$ is determined by $P_y(0) = \pi(0,0)$: evaluating (8.17) at $y = 0$, where $\widetilde{B}_C(\lambda_2)$ cancels,

$$\begin{aligned}
\pi(0,0) = P_y(0) &= \mathbb{P}(\text{busy},\, N_1 = 0)\, \frac{(1 - \lambda_2 \mathbb{E}[B_C]) \cdot 1 \cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)} \\
&= \mathbb{P}(\text{busy},\, N_1 = 0)\,(1 - \lambda_2 \mathbb{E}[B_C]),
\end{aligned}$$

so $\mathbb{P}(\text{busy}, N_1 = 0) = \pi(0,0)/(1 - \lambda_2 \mathbb{E}[B_C])$. Substituting back, the factor $(1 - \lambda_2 \mathbb{E}[B_C])$ cancels and we recover exactly (8.12), in agreement with the analytical route. The result also matches the probabilistic argument for Model-A (Equation (5.9)); the only difference is the effective service time.

Finally, this route shows the Kummer fraction on the probabilistic side. The busy-period LST that the Pollaczek–Khinchine formula feeds on is, from the Erlang-A analysis (§3.3, Equation (3.7)), the ratio of confluent hypergeometric functions

$$\widetilde{B}_C(s) \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}, \qquad a=\frac{\mu}{\theta_1}, \quad b=\frac{s}{\theta_1}, \quad c=\frac{\lambda_1}{\theta_1}.$$

This is the very expression (8.11) that the analytical route produced as the Beta-integral ratio $\mathcal{J}_1(y)/\mathcal{J}_0(y)$ at $s=\lambda_2(1-y)$. The two derivations therefore meet at this Kummer fraction: what boundedness delivers as a ratio of Euler integrals is precisely the LST of the class-1 busy period.

9 Analysis of Model-B_2^{H} ($\gamma_1>0$ head-of-line, $\gamma_2=\theta_1=\theta_2=0$)

Inspired by the work of Devos et al. [DWFB20], this model —together with Model-C_2^{H}— eliminates the state-dependent jockeying rate for the class-1, only allowing the head-of-line customer from class-1 to transition at a fixed rate γ_1 whenever $n_1\ge 1$, giving an aggregate rate $\gamma_1\,\mathbf{1}_{\{n_1\ge 1\}}$.

This significantly simplifies the model. The flat rate replaces the derivative term in x found in §7 by an algebraic factor $\gamma_1 y(x-y)$ so the fundamental equation remains algebraic and can be solved by the Kernel Method exactly as in Model-A (§5).

Stability. Jockeying conserves the total number of customers; since there is no abandonment, the chain is positive recurrent if and only if the total traffic intensity satisfies

$$\rho=\frac{\lambda_1+\lambda_2}{\mu}<1. \tag{9.1}$$

9.1 Balance equations

The jockeying transition $(n_1,n_2)\to(n_1-1,n_2+1)$ happens at the fixed rate γ_1 for $n_1\ge 1$. On state space S the global balance equations read

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu+\gamma_1]\,\pi(n_1,n_2)\\ &\quad=\quad \mu\,\pi(n_1+1,n_2)+\lambda_1\pi(n_1-1,n_2)+\lambda_2\pi(n_1,n_2-1)+\gamma_1\pi(n_1+1,n_2-1)\end{aligned}\qquad n_1\ge 1,\ n_2\ge 1, \tag{9.2}$$

$$[\lambda_1+\lambda_2+\mu+\gamma_1]\,\pi(n_1,0) \;=\; \mu\,\pi(n_1+1,0)+\lambda_1\pi(n_1-1,0) \qquad n_1\ge 1,\ n_2=0, \tag{9.3}$$

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu]\,\pi(0,n_2)\\ &\quad=\quad \mu\,\pi(1,n_2)+\gamma_1\pi(1,n_2-1)+\lambda_2\pi(0,n_2-1)+\mu\,\pi(0,n_2+1)\end{aligned}\qquad n_1=0,\ n_2\ge 1, \tag{9.4}$$

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu]\,\pi(0,0)=(\lambda_1+\lambda_2)\pi_0+\mu\,\pi(1,0)+\mu\,\pi(0,1),\\ &(\lambda_1+\lambda_2)\pi_0=\mu\,\pi(0,0).\end{aligned} \tag{9.5}$$

9.2 Fundamental PGF equation

Lemma 4 (Fundamental equation for Model-B_2^{H}). *Assume positive recurrence, so that the stationary distribution exists and $P(x,y)=\sum_{n_1,n_2\ge 0}\pi(n_1,n_2)x^{n_1}y^{n_2}$ is analytic on the closed unit polydisc. Then $P(x,y)$ and the boundary function $P_y(y)=P(0,y)$ satisfy*

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\gamma_1y(x-y)\big]\,P(x,y)\\ &\quad=\quad [\mu(x-y)+\gamma_1y(x-y)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned} \tag{9.6}$$

Proof. Besides the jockeying terms, equations (9.2)–(9.5) coincide with those of Model-A. Multiplying each balance equation by $x^{n_1}y^{n_2}$, summing over its domain and multiplying through by xy, we obtain the fundamental equation for Model-A (Equation (5.2)):

$$\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\big]\,P(x,y)=\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0),$$

so it remains only to collect the jockeying contribution. Its *out*-flow appears in every state with $n_1\ge 1$:

$$\gamma_1\sum_{n_1\ge 1}\sum_{n_2\ge 0}\pi(n_1,n_2)\,x^{n_1}y^{n_2}=\gamma_1\big[P(x,y)-P_y(y)\big],$$

while its *in*-flow enters every state with $n_2 \ge 1$ from (n_1+1, n_2-1):

$$\gamma_1 \sum_{n_1\ge 0}\sum_{n_2\ge 1} \pi(n_1+1,n_2-1)\, x^{n_1} y^{n_2} = \gamma_1\, \frac{y}{x} \sum_{m_1\ge 1}\sum_{m_2\ge 0} \pi(m_1,m_2)\, x^{m_1} y^{m_2} = \gamma_1\, \frac{y}{x}\big[P(x,y) - P_y(y)\big],$$

where $m_1 = n_1+1,\ m_2 = n_2-1$. The jockeying contribution (out minus in) is hence $\gamma_1\big(1-\frac{y}{x}\big)\big[P(x,y) - P_y(y)\big]$; multiplying by xy gives $\gamma_1 y(x-y)\big[P(x,y)-P_y(y)\big]$. Adding this to the Model-A equation and moving the P_y part to the right-hand side yields (9.6). □

9.3 Closed-form solution

The following theorem is the main result of this section; its proof is based on the analytical (Kernel Method) treatment of Model-A in §5.1.1.

Theorem 4 (Closed-form PGF for Model-B_2^{H})**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 jockeying at rate $\gamma_1 > 0$ (with $\gamma_2 = \theta_1 = \theta_2 = 0$). Under the stability condition (9.1), the joint PGF $P(x,y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is the rational function*

$$P(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y)-x\big)}, \tag{9.7}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the kernel quadratic

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + (\mu + \gamma_1 y) = 0, \tag{9.8}$$

with $x^*(y)$ the root inside the unit disc.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models.

Corollary 5. *Under (9.1), the empty-state probabilities of Model-B_2^{H} coincide with those of Model-A,*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho).$$

In particular $P(1,1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

The proof is the same as for Corollary 1.

9.3.1 Analytical approach for Model-B_2^{H}

Proof of Theorem 4. We apply the *Kernel Method* to the fundamental equation (9.6). Fix $y \in (0,1]$ and seek the values of x at which the coefficient of $P(x,y)$ vanishes. Since $P(x,y)$ is a PGF it is bounded on the closed unit disc, so at any such root the right-hand side of (9.6) must also vanish. Dividing the kernel by $y \neq 0$,

$$0 = (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy + \gamma_1(x-y),$$

and rearranging into a quadratic in x gives exactly (9.8),

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + (\mu + \gamma_1 y) = 0,$$

whose two roots are

$$x^\pm(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu+\gamma_1 y)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1.$$

We label the root carrying the negative sign $x^*(y)$ and the other $x^+(y)$. By Vieta's formulas the product of the roots is $x^*(y)\,x^+(y) = (\mu+\gamma_1 y)/\lambda_1$.

To locate the roots relative to the unit disc we evaluate the kernel quadratic, written as $f(x) := \lambda_1 x^2 - A(y)x + (\mu+\gamma_1 y)$, at $x = 1$:

$$f(1) = \lambda_1 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big] + (\mu+\gamma_1 y) = -(1-y)\,(\lambda_2+\gamma_1) < 0 \qquad \text{for } y \in [0,1).$$

We have that$x^*(y)$ is the unique root inside the unit disc, and at $y = 1$ one checks $f(1) = 0$ with $x^*(1) = 1$ and $x^+(1) = (\mu+\gamma_1)/\lambda_1 > 1$. For later use, we record $\frac{d}{dy}x^*(1)$. The cleanest route is implicit differentiation of $f(x,y) := \lambda_1 x^2 - A(y)x + (\mu + \gamma_1 y) = 0$ at $(x,y) = (1,1)$, where $f_y = \lambda_2 x + \gamma_1$ and $f_x = 2\lambda_1 x - A(y)$ evaluate to $f_y(1,1) = \lambda_2 + \gamma_1$ and $f_x(1,1) = \lambda_1 - \mu - \gamma_1$, so $\frac{d}{dy}x^*(1) = -f_y/f_x$:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2 + \gamma_1}{\mu + \gamma_1 - \lambda_1}. \tag{9.9}$$

Unlike Model-A's (5.7), the numerator carries an extra γ_1: the constant term $\mu + \gamma_1 y$ of the kernel (9.8) is itself y-dependent here.

Setting $x = x^*(y)$ in (9.6) forces the right-hand side to vanish:$(x^*(y) - y)(\mu + \gamma_1 y)\, P_y(y) = \mu x^*(y)(1-y)\,\pi(0,0)$, whence the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1-y)\,\pi(0,0)}{(\mu + \gamma_1 y)\,\big(x^*(y) - y\big)}. \tag{9.10}$$

Finally, substitute (9.10) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 5) back into (9.6). Writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$ and simplifying the right-hand side,

$$\begin{aligned}\big[\mu(x-y) + \gamma_1 y(x-y)\big] P_y(y) - \mu x(1-y)\,\pi(0,0) &= \mu(1-y)\,\pi(0,0)\left[\frac{(x-y)\,x^*(y)}{x^*(y)-y} - x\right]\\ &= \mu(1-y)\,\pi(0,0)\,\frac{y\,\big(x - x^*(y)\big)}{x^*(y)-y},\end{aligned}$$

where the bracket collapses because $(x-y)x^* - x(x^* - y) = y(x - x^*)$. Dividing by the factored kernel, the common factor $\big(x - x^*(y)\big)$ cancels out and we have

$$P(x,y) = \frac{\mu(1-y)\,\pi(0,0)\,y\,(x - x^*(y))/(x^*(y)-y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y) - x\big)},$$

which matches Equation (9.7) and concludes the proof. □

10 Analysis of Model-C_2^{H} ($\theta_1 > 0$ head-of-line, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

Model-C_2^{H} is the abandonment counterpart of Model-B_2^{H}, and the head-of-line simplification of the class-1 abandonment model Model-C_2. Only the *customer at the head of Queue 1* abandons, at the fixed rate θ_1 whenever $n_1 \geq 1$. As in Model-B_2^{H}, the flat rate $\theta_1\,\mathbf{1}_{\{n_1 \geq 1\}}$ contributes an *algebraic* term rather than a derivative, and the Kernel Method applies directly. True departure breaks the conservation of the total number of customers that we had in models without abandonment.

Stability. Because only the head-of-line from class-1 abandons, the aggregate abandonment rate is bounded by θ_1 regardless of queue length as long as we have a queue, so a genuine stability condition is required. As shown in Corollary 6 below, the chain is positive recurrent if and only if $\pi_0 > 0$, that is

$$(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1 \;>\; 0, \qquad \lambda = \lambda_1 + \lambda_2. \tag{10.1}$$

This is *weaker* than $\rho < 1$: if $\rho < 1$ then $\mu - \lambda > 0$ and the condition holds trivially, but it can also hold with $\rho \geq 1$ provided θ_1 is large enough. Moreover (10.1) implies $\rho_C := \lambda_1/(\mu + \theta_1) < 1$. Hence $\mu + \theta_1 > \lambda_1$, i.e. $\rho_C < 1$.

10.1 Balance equations

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. Since both a service completion and a head abandonment from $(n_1 + 1, n_2)$ land in (n_1, n_2), the two mechanisms combine into the coefficient $\mu + \theta_1$ on the in-flows. On S,

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1,n_2)\\ &\quad = (\mu+\theta_1)\pi(n_1+1,n_2) + \lambda_1\pi(n_1-1,n_2) + \lambda_2\pi(n_1,n_2-1) && n_1 \geq 1,\ n_2 \geq 1, \qquad (10.2)\\ &[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1,0) = (\mu+\theta_1)\pi(n_1+1,0) + \lambda_1\pi(n_1-1,0) && n_1 \geq 1,\ n_2 = 0,\\ &[\lambda_1 + \lambda_2 + \mu]\,\pi(0,n_2)\\ &\quad = (\mu+\theta_1)\pi(1,n_2) + \lambda_2\pi(0,n_2-1) + \mu\,\pi(0,n_2+1) && n_1 = 0,\ n_2 \geq 1,\\ &[\lambda_1 + \lambda_2 + \mu]\,\pi(0,0) = (\lambda_1+\lambda_2)\pi_0 + (\mu+\theta_1)\pi(1,0) + \mu\,\pi(0,1),\\ &(\lambda_1+\lambda_2)\pi_0 = \mu\,\pi(0,0). && \qquad (10.3)\end{aligned}$$

10.2 Fundamental PGF equation

We introduce the following result:

Lemma 5 (Fundamental equation for Model-C_2^{H})**.** *Assume positive recurrence. Then the joint PGF $P(x,y)$ and the boundary function $P_y(y)=P(0,y)$ satisfy*

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\theta_1y(x-1)\big]\,P(x,y)\\ =\quad&\big[\mu(x-y)+\theta_1y(x-1)\big]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{10.4}$$

Proof. As in Lemma 4, the non-abandonment terms of (10.2)–(10.3) reproduce the Model-A fundamental equation (5.2) after multiplication by $x^{n_1}y^{n_2}$, summation, and multiplication by xy. The abandonment *out*-flow appears in every state with $n_1\ge1$,

$$\theta_1\sum_{n_1\ge1}\sum_{n_2\ge0}\pi(n_1,n_2)\,x^{n_1}y^{n_2}=\theta_1\big[P(x,y)-P_y(y)\big],$$

and its *in*-flow enters every state (n_1,n_2) from (n_1+1,n_2),

$$\theta_1\sum_{n_1\ge0}\sum_{n_2\ge0}\pi(n_1+1,n_2)\,x^{n_1}y^{n_2}=\theta_1\,x^{-1}\sum_{m_1\ge1}\sum_{n_2\ge0}\pi(m_1,n_2)\,x^{m_1}y^{n_2}=\theta_1\,x^{-1}\big[P(x,y)-P_y(y)\big],$$

with $m_1=n_1+1$. The net contribution is $\theta_1\big(1-x^{-1}\big)\big[P(x,y)-P_y(y)\big]$; multiplying by xy gives $\theta_1y(x-1)\big[P(x,y)-P_y(y)\big]$. Adding to the Model-$A$ equation and rearranging yields (10.4). $\square$

10.3 Closed-form solution

Theorem 5 (Closed-form PGF for Model-C_2^{H})**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1,λ_2, common exponential service rate μ, and head-of-line class-1 abandonment at rate $\theta_1>0$ (with $\gamma_1=\gamma_2=\theta_2=0$). Under the stability condition (10.1), the joint PGF $P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]$ is the rational function*

$$P(x,y)\;=\;\frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\big(x^+(y)-x\big)\,\big[(\mu+\theta_1y)\,x^*(y)-y(\mu+\theta_1)\big]},\tag{10.5}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the kernel quadratic

$$\lambda_1x^2-\big[\lambda_1+\lambda_2(1-y)+\mu+\theta_1\big]x+(\mu+\theta_1)=0,\tag{10.6}$$

with $x^*(y)$ the root inside the unit disc, and $\pi(0,0)$ is given by Corollary 6. As $\theta_1\to0^+$ the formula recovers Model-A (Theorem 1) exactly.

The theorem invokes the following Corollary:

Corollary 6. *Under (10.1), the empty-state probabilities of Model-C_2^{H} are*

$$\pi_0=\frac{(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1}{\mu(\mu+\theta_1)+\lambda_1\theta_1},\qquad\pi(0,0)=\frac{\lambda\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]}{\mu\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]},\tag{10.7}$$

with $\lambda=\lambda_1+\lambda_2$. At $\theta_1\to0^+$ these reduce to $\pi_0\to1-\rho$ and $\pi(0,0)\to\rho(1-\rho)$, recovering Model-A.

Proof of Corollary 6. Two evaluations of the fundamental equation (10.4) suffice. First set $y=1$: the term $\mu x(1-y)\pi(0,0)$ vanishes, the coefficient of $P(x,1)$ factors as $-(x-1)\big(\lambda_1x-(\mu+\theta_1)\big)$, and the RHS becomes $(x-1)(\mu+\theta_1)P_y(1)$. Cancelling $(x-1)$ we obtain

$$P(x,1)\;=\;\frac{(\mu+\theta_1)\,P_y(1)}{\mu+\theta_1-\lambda_1x}\;=\;\frac{P_y(1)}{1-\rho_C\,x},\qquad\rho_C:=\frac{\lambda_1}{\mu+\theta_1}.\tag{10.8}$$

This geometric marginal shows that the priority queue clears at the effective rate $\mu+\theta_1$. Evaluating at $x=1$ and using $P(1,1)=1-\pi_0$ gives

$$1-\pi_0\;=\;\frac{P_y(1)}{1-\rho_C}\quad\Longrightarrow\quad P_y(1)\;=\;(1-\pi_0)(1-\rho_C).\tag{10.9}$$

Second, set $x = 1$ in (10.4). The coefficient of $P(1, y)$ collapses to $\lambda_2 y(1-y)$ and the right-hand side to $\mu(1-y)\big[P_y(y) - \pi(0,0)\big]$, yielding the following relation

$$\lambda_2\, y\, P(1,y) \;=\; \mu\big[P_y(y) - \pi(0,0)\big]. \tag{10.10}$$

Letting $y \to 1$ and using (10.9) and the idle balance (10.3), $\pi(0,0) = \lambda\pi_0/\mu$, we have

$$\lambda_2(1-\pi_0) \;=\; \mu\big[P_y(1) - \pi(0,0)\big] \;=\; \mu(1-\pi_0)(1-\rho_C) - \lambda\pi_0.$$

Expanding and collecting the π_0 terms,

$$\big[\mu(1-\rho_C) + \lambda - \lambda_2\big]\pi_0 = \mu(1-\rho_C) - \lambda_2, \qquad \text{i.e.} \qquad \pi_0 = \frac{\mu(1-\rho_C) - \lambda_2}{\mu(1-\rho_C) + \lambda_1},$$

where $\lambda - \lambda_2 = \lambda_1$. Substituting $\mu(1-\rho_C) = \mu(\mu+\theta_1-\lambda_1)/(\mu+\theta_1)$ and multiplying numerator and denominator by $\mu+\theta_1$, the numerator becomes $\mu(\mu+\theta_1-\lambda_1) - \lambda_2(\mu+\theta_1) = (\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1$ and the denominator $\mu(\mu+\theta_1-\lambda_1) + \lambda_1(\mu+\theta_1) = \mu(\mu+\theta_1) + \lambda_1\theta_1$, yielding the first identity in (10.7); the second follows directly from $\pi(0,0) = \lambda\pi_0/\mu$. The denominator $\mu(\mu+\theta_1) + \lambda_1\theta_1$ is strictly positive, so $\pi_0 > 0$ is equivalent to the stability condition (10.1). $\square$

10.3.1 Analytical approach for Model-C_2^{H}

Proof of Theorem 5. We again apply the Kernel Method to (10.4). Dividing the coefficient of $P(x,y)$ by $y \neq 0$ and rearranging gives the kernel quadratic (10.6),

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \theta_1\big]x + (\mu+\theta_1) = 0,$$

Its roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu+\theta_1)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \theta_1, \tag{10.11}$$

with product $x^*(y)\, x^+(y) = (\mu+\theta_1)/\lambda_1$ by Vieta's formulas. Evaluating the quadratic $f(x) := \lambda_1 x^2 - A(y)x + (\mu+\theta_1)$ at $x = 1$,

$$f(1) = \lambda_1 - A(y) + (\mu+\theta_1) = -\lambda_2(1-y) < 0 \qquad \text{for } y \in [0,1),$$

So $x = 1$ lies between the two roots and $x^*(y) < 1 < x^+(y)$ for $y \in [0,1)$; at $y = 1$, $x^*(1) = 1$ and $x^+(1) = (\mu+\theta_1)/\lambda_1 > 1$ (the latter by $\rho_C < 1$). Hence $x^*(y)$ is the unique root inside the unit disc. Differentiating (10.11) at $y = 1$,

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2}{\mu+\theta_1-\lambda_1}. \tag{10.12}$$

Setting $x = x^*(y)$ in (10.4) forces the RHS to vanish,

$$\big[\mu(x^*(y) - y) + \theta_1 y(x^*(y) - 1)\big]P_y(y) = \mu x^*(y)(1-y)\,\pi(0,0),$$

and, regrouping the bracket as $\mu(x^* - y) + \theta_1 y(x^* - 1) = (\mu + \theta_1 y)x^* - y(\mu+\theta_1)$, the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1-y)\,\pi(0,0)}{(\mu+\theta_1 y)\,x^*(y) - y(\mu+\theta_1)}. \tag{10.13}$$

Write $\Delta(y) := (\mu+\theta_1 y)x^*(y) - y(\mu+\theta_1)$ for the denominator of (10.13); at $y = 0$, $\Delta(0) = \mu\, x^*(0)$ (with $x^*(0) > 0$), so $P_y(0) = \pi(0,0)$, exactly as for Model-B_2^{H} at (9.10).

Substituting (10.13) into Equation (10.4) and writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$, the right-hand side becomes

$$\big[\mu(x-y)+\theta_1 y(x-1)\big]P_y(y) - \mu x(1-y)\,\pi(0,0) = \frac{\mu(1-y)\,\pi(0,0)}{\Delta(y)}\Big\{\big[\mu(x-y)+\theta_1 y(x-1)\big]x^*(y) - x\,\Delta(y)\Big\}.$$

Substituting $\Delta(y)$, it simplifies to $(\mu+\theta_1)\,y\,(x - x^*(y))$. Dividing by the factored kernel cancels $(x - x^*(y))$, yielding

$$P(x,y) = \frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)\,y\,(x - x^*(y))/\Delta(y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)}{\lambda_1\,(x^+(y) - x)\,\Delta(y)},$$

which is Equation (10.5) and concludes the proof. $\square$

11 Results

Every closed-form expression derived above is first validated against an exact continuous-time Markov chain (CTMC), then used to compare the solved models and quantify the two mechanisms.

11.1 Validation against the CTMC

The ground truth is a truncated CTMC solver: it solves the linear system $\pi Q = 0$ directly, with no approximation beyond the finite truncation at $N_{\max} = 40$. Each theoretical result — two lemmas, three theorems, two corollaries, one approximation theorem, and two experiment-model theorems — is verified against the exact stationary distribution at base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$, with the mechanism parameter (γ_1 or θ_1) set to 0.5 where applicable.

Table 4 collects the verdict for every result. All mathematical results are accepted to within CTMC truncation noise: the lemma residuals and closed-form PGF errors fall between 10^{-7} and 10^{-15}, and the corollary scatter lies on the diagonal to 10^{-4} across 60 random configurations each. A discrete-event simulation of the master chain cross-validates the CTMC itself to the $O(1/\sqrt{n})$ sampling floor.

Table 4: Validation summary. All mathematical results are accepted; the PPGF approximation is conditionally accepted (accurate only for $\rho_1/\rho \geq 0.6$ and $\rho \leq 0.6$). Max errors reflect CTMC truncation, not formula error.

ID	**Result**	**Model(s)**	**Check**	**Max error**	**Verdict**
L1	Lemma 1	A, B, B_2, C_2	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B_2, C_2	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem 1	A	$P(x, y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 1	A, B, B_2	π_0, $\pi(0, 0)$, 60 configurations	$< 10^{-4}$	Accept
T-B2	Theorem 2	B_2	$P(x, y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem 3	C_2	$P(x, y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 4	C_2	π_0, $\pi(0, 0)$, 60 configurations	$< 10^{-4}$	Accept
T-S	Approximation 1	A	ε_∞ over (ρ_1, ρ_2)	0–100%	Conditional
E-B	Theorem 4 — Model-B_2^{H}	B_2^{H}	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 5 — Model-C_2^{H}	C_2^{H}	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

A single entry is only conditionally accepted. The partial-PGF recursion on the alternative state space $\widetilde{S}$ (Approximation 1) is approximate by design: it drops the boundary-diagonal forcing term, so it is exact at $y = 1$ but degrades elsewhere. It is reliable only in the priority-dominated, moderately loaded regime ($\rho_1/\rho \gtrsim 0.6$, $\rho \lesssim 0.6$).

11.2 Structural comparison of the solved models

With every closed form validated, we compare the five solved models: the baseline (A), the jockeying extension (B_2), the abandonment extension (C_2), and their head-of-line variants (B_2^{H}, C_2^{H}). The first three are the tractable specialisations of Model-X; the head-of-line variants are the standalone simplifications of §§9–10. Each produces a closed-form or integral-form expression for $P(x, y)$. Table 5 collects the key structural facts across all five models.

Table 5: Structural comparison of the five solved models. Replacing a length-proportional mechanism rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a head-of-line rate ($\gamma_1 \mathbf{1}_{\{n_1\geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1\geq 1\}}$) collapses the governing ODE back to the algebraic Model-A form, so $P(x,y)$ becomes rational again.

	Model-A	**Model-B_2**	**Model-B_2^{H}**	**Model-C_2**	**Model-C_2^{H}**
Extra mechanism (rate)	none	jockey $1 \to 2$, $\gamma_1 n_1$	jockey $1 \to 2$, $\gamma_1 \mathbf{1}_{\{n_1\geq 1\}}$	abandon, $\theta_1 n_1$	abandon, $\theta_1 \mathbf{1}_{\{n_1\geq 1\}}$
Fundamental equation	algebraic	1st-order ODE	algebraic	1st-order ODE	algebraic
Pinning of $P_y(y)$	kernel root $x^*(y)$	analyticity at $x = y$	kernel root $x^*(y)$	boundedness at $x = 1$	kernel root $x^*(y)$
$P(x,y)$ form	rational	integral $+\,{}_1F_1$	rational	integral $+\,{}_1F_1$	rational
Conserves total N?	yes	yes	yes	no	no
$\pi(0,0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	(8.3)	(10.7)
π_0	$1-\rho$	$1-\rho$	$1-\rho$	$> 1-\rho$, (8.4)	(10.7)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\lambda_2 \mathbb{E}[B_C] < 1$	(10.1)

Model-A: the rational baseline

Model-A imposes no extra departure mechanism. Class-1 operates as an independent $M/M/1(\lambda_1,\mu)$ queue; class-2 sees an $M/G/1$ with service time equal to a class-1 busy period (by PASTA and the priority decomposition of §3.2). The Pollaczek–Khinchine formula (3.8) delivers $P_y(y)$ as a rational function of $x^*(y)$, and $P(x,y)$ is a ratio of polynomials in x, y, and $x^*(y)$. All performance measures — π_0, $\pi(0,0)$, $\mathbb{E}[N_i]$, $\mathbb{E}[W_i]$ — are in closed form. Model-A also furnishes the structural identities that persist across the hierarchy: $P(z,z) = \rho(1-\rho)/(1-\rho z)$ (total queue is $M/M/1$), $\pi_0 = 1-\rho$, and $\pi(0,0) = \rho(1-\rho)$.

Effect of jockeying: Model-A $\to$ Model-B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in Queue 1 may defect to Queue 2 at rate γ_1. The consequences are asymmetric.

What is preserved. Jockeying transfers customers but does not remove them, so the total-customer process remains a standard $M/M/1(\lambda,\mu)$ birth–death chain. Consequently $P(z,z)$, π_0, $\pi(0,0)$, and $\mathbb{E}[N]$ are identical to Model-A for all $\gamma_1 > 0$.

What is destroyed. Jockeying breaks the per-class Poisson/renewal structure: a class-2 customer can no longer identify its effective service time with a class-1 busy period. The P-K route is therefore unavailable.

What is new. The fundamental PGF equation gains the convection term $\gamma_1 xy(x-y)\partial_x P$, which reduces it to a first-order ODE in x for each fixed y (since $\gamma_2 = 0$ kills the ∂_y term). The solution closes via an integrating factor with a branch singularity at the *interior* point $x = y$; the branch coefficient must vanish for P to be holomorphic, and this analyticity condition pins $P_y(y)$ as a Kummer ratio. The individual means $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ change with γ_1 (class-1 becomes shorter, class-2 longer), with no closed form for the split.

Effect of abandonment: Model-A $\to$ Model-C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at rate θ_1.

What is preserved. Class-2 customers still see Poisson arrivals (abandonment only affects class-1; the class-2 arrival process is undisturbed). PASTA therefore still applies, and the $M/G/1$ subsystem for class-2 is intact — now with effective service time B_C (the busy period of the $M/M/1+M$ queue of §3.3) replacing B. The P-K formula (3.8) applies with B_C and delivers $P_y(y)$.

What is destroyed. Abandonment is a true departure: it reduces the total number of customers and breaks the conservation property. Consequently $P(z,z)$ no longer has the simple $M/M/1$ form of Corollary 1, $\pi_0 > 1-\rho$, $\pi(0,0)$ requires $\mathbb{E}[B_C]$, and $\mathbb{E}[N] < \rho^2/(1-\rho)$. The stability condition is also generalised: $\rho < 1$ is no longer necessary; the relevant condition is $\lambda_2\, \mathbb{E}[B_C] < 1$, which is strictly weaker for all $\theta_1 > 0$.

What is new. The fundamental equation gains the convection term $\theta_1 xy(x-1)\partial_x P$, which already reduces it to a first-order ODE in x for each fixed y (with $\theta_2 = 0$ killing the ∂_y term). The integrating factor now has its singularity at the *boundary* $x = 1$ with a positive exponent, so μ_I

vanishes there and mere boundedness of P forces $\int_0^1 q\,\mu_I\,dt = 0$. This determines $P_y(y)$ without any reference to holomorphy. Both $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ decrease relative to Model-A as θ_1 increases: class-2 benefits indirectly because class-1 abandonments free the server sooner.

Mechanism contrast: Models B_2 and C_2

Both models reduce the fundamental equation to a first-order linear ODE in x and both close in confluent hypergeometric functions; yet the mechanism that pins $P_y(y)$ is structurally different.

	Model-C_2 ($\theta_1 > 0$)	**Model-B_2** ($\gamma_1 > 0$)
Convection coefficient	$\theta_1\, x(x-1)$	$\gamma_1\, x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ ($\mu_I \to 0$)	$\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y) < 0$ ($\mu_I \to \infty$)
Pinning principle	*boundedness*: $\int_0^1 q\,\mu_I\,dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ exactly (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1{+}M$ (reneging $= \gamma_1$); no P-K

In words: in Model-C_2 the singularity sits on the boundary $x = 1$ with a positive exponent, so μ_I vanishes and mere boundedness pins $P_y(y)$; the class-2 stream stays Poisson, so P-K and an $M/G/1$ renewal argument deliver $P_y(y)$ and $\pi(0,0)$. In Model-B_2 the singularity sits at the interior point $x = y$ with a negative exponent, so μ_I blows up and only holomorphy — the absence of the branch $(x-y)^{|\beta|}$ — pins $P_y(y)$. Jockeying leaves no P-K identity; the $M/M/1{+}M$ reading of the class-1 marginal is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in both B_2 and C_2:

$$ {}_1F_1\left(1;\ \frac{\mu}{r}+1;\ \frac{\lambda_1}{r}\right), \tag{11.1}$$

with $r = \gamma_1$ (Model-B_2) or $r = \theta_1$ (Model-C_2). This is the Erlang-A utilisation factor: in C_2 it equals $\mu\,\mathbb{E}[B_C]$, while in B_2 it controls the Kummer ratio in $P_y(y)$ through the exponent $\alpha(y) = \mu/(\gamma_1 y)$. In both cases the function evaluates at a negative argument $(-\lambda_1/r)$ in C_2 and $(-\lambda_1 y/\gamma_1)$ in B_2, and the ratio ${}_1F_1(\alpha{+}1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the Model-A rational formula.

Model-B_2 degenerates to Model-A only in the singular limit $\gamma_1 \to 0^+$, in which $\alpha, \beta \to \infty$ and the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model-A. This limit is not uniform, consistent with γ_1 multiplying the highest-order term of the ODE. Model-C_2 degenerates to Model-A uniformly as $\theta_1 \to 0^+$: $\mathbb{E}[B_C] \to (\mu-\lambda_1)^{-1}$, and all C_2 formulas recover their Model-A counterparts.

Head-of-line simplifications: Models B_2^{H} and C_2^{H}

The head-of-line variants of §§9–10 restrict each mechanism to the single customer at the head of Queue 1, replacing the length-proportional rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a constant one ($\gamma_1\,\mathbf{1}_{\{n_1\ge 1\}}$ or $\theta_1\,\mathbf{1}_{\{n_1\ge 1\}}$). This annihilates the derivative term and converts the governing ODE into an algebraic equation, so $P(x,y)$ becomes rational — exactly as in Model-A (last two columns of Table 5). For Model-B_2^{H} the result is structurally identical to Model-A, the sole change being $\mu \to \mu + \gamma_1 y$ in the kernel, and conservation is retained ($\pi_0 = 1-\rho$, $\pi(0,0) = \rho(1-\rho)$). For Model-C_2^{H} the broken conservation introduces the extra factor $(\mu+\theta_1)$ and the modified boundary denominator $\Delta(y)$, and shifts the stability boundary to (10.1). Each head-of-line model is thus the $k = 1$ rung of a ladder that interpolates between Model-A and its full-rate parent (B_2 or C_2) as the head-of-line restriction is relaxed.

11.3 Quantitative comparison across load

Holding $\mu = 1$ and the arrival split fixed at $\lambda_1{:}\lambda_2 = 3{:}4$, we sweep the offered load ρ across $[0.10, 0.95]$ and compute the exact stationary performance of the five solved models on the truncated CTMC. Figure 8 plots $\mathbb{E}[N]$, $\mathbb{E}[N_1]$, $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ (Little's Law), and π_0; Table 6 reads off the full descriptor set at three reference loads.

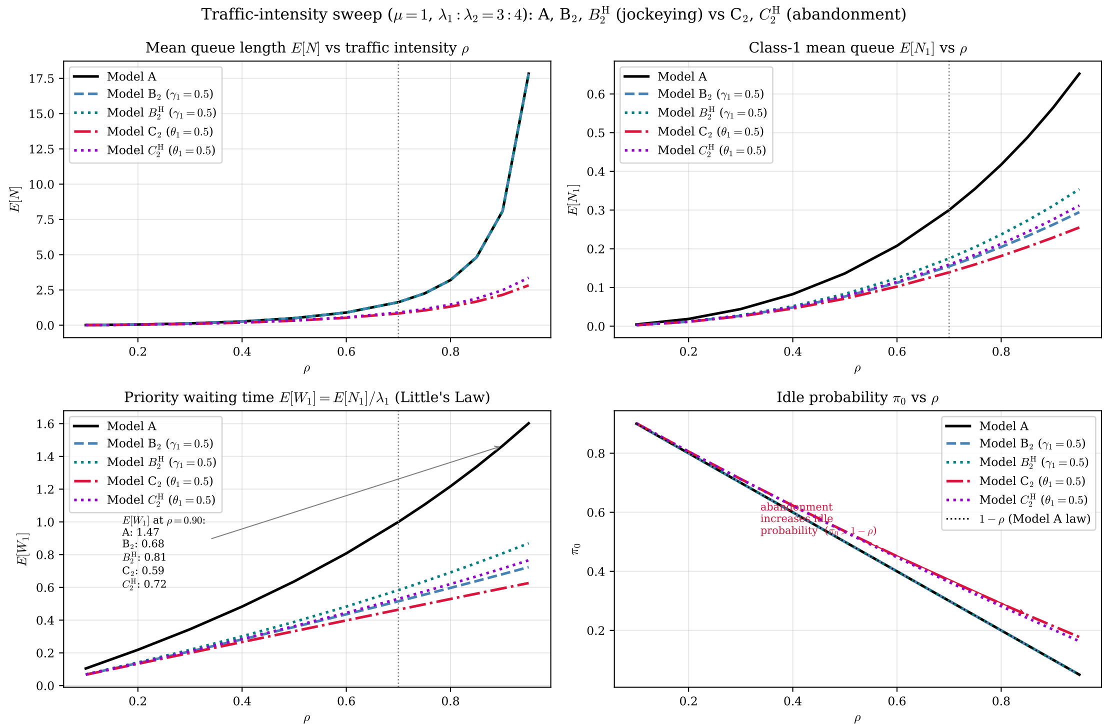

Figure 8: Performance versus traffic intensity ρ for the five solved models — the baseline A, the jockeying pair B_2, B_2^{H} ($\gamma_1 = 0.5$), and the abandonment pair C_2, C_2^{H} ($\theta_1 = 0.5$) — at $\mu = 1$ and $\lambda_1{:}\lambda_2 = 3{:}4$; the vertical line marks the canonical $\rho = 0.70$. *Top-left:* total queue $\mathbb{E}[N]$ — A, B_2 and B_2^{H} coincide because jockeying conserves N (Corollaries 1, 5), while C_2 and C_2^{H} lie strictly below because abandonment removes customers (C_2^{H} by less). *Top-right:* priority queue $\mathbb{E}[N_1]$ — the head-of-line variants drain it less than their full-rate cousins. *Bottom-left:* priority waiting time $\mathbb{E}[W_1]$. *Bottom-right:* idle probability π_0, with the $1-\rho$ law (Models A, B_2, B_2^{H}) shown dotted; C_2 and C_2^{H} sit above it for every ρ, as predicted by Corollaries 4, 6.

The two mechanisms produce two qualitatively different responses. Jockeying is internal: the $1 \to 2$ transition relabels a waiting customer without changing the total count. The B_2 and A curves for $\mathbb{E}[N]$ and π_0 are therefore indistinguishable across the entire sweep — the server sees the same workload, and the busy-but-empty probability stays on the $\rho(1-\rho)$ parabola of Corollary 3. Abandonment is a true departure at rate $\theta_1 n_1$, so C_2 carries a strictly lighter queue and a strictly larger idle fraction: the priority backlog is bled off before it can be served. At $\rho = 0.70$ (Table 6) this gives $\pi_0 = 0.3696$ for C_2 against 0.3000 for A and B_2, and $\mathbb{E}[N] = 0.8358$ against 1.6333. The divergence widens with load: at $\rho = 0.90$ the baseline class-2 queue reaches $\mathbb{E}[N_2] = 7.5349$, whereas C_2 holds it to 1.9245 — a factor of nearly four. The self-regulating valve $\theta_1 n_1$ acts most strongly precisely when the queue is long. The two head-of-line variants interpolate monotonically between Model-A and their full-rate parents in every descriptor: B_2^{H} conserves N exactly like B_2 and departs only in the $\mathbb{E}[N_1]$ split, while C_2^{H} sheds less load than C_2 ($\pi_0 = 0.3636$, $\mathbb{E}[N] = 0.9015$ at $\rho = 0.70$).

Two columns warrant a reviewer's caution. First, $\mathbb{E}[W_1]$ is smallest for C_2 at every load, but $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ counts residence in queue i, which every departure mode shortens — service *and* attrition. A smaller $\mathbb{E}[W_1]$ thus does not certify faster service. Second, jockeying actively *harms* class 2: at $\rho = 0.70$, B_2 raises $\mathbb{E}[W_2]$ to 3.6970 from the baseline 3.3333 (+10.9%). Finally, near criticality the dense CTMC truncation leaves a boundary tail mass of order 6×10^{-4} for A and B_2 at $\rho = 0.95$ (against 1.5×10^{-7} for C_2), so the $\rho \geq 0.92$ points for the non-abandoning models are lower bounds.

Table 6: Comparison of Models A, B_2 and B_2^{H} ($\gamma_1 = 0.5$), and C_2 and C_2^{H} ($\theta_1 = 0.5$) at three traffic intensities with $\lambda_1{:}\lambda_2 = 3{:}4$, $\mu = 1$. The primed models use the head-of-line rate $\mathbf{1}_{\{n_1 \geq 1\}}$; the unprimed use the full rate n_1.

ρ	Model	$E[N_1]$	$E[N_2]$	$E[N]$	$E[W_1]$	$E[W_2]$	π_0	$\pi(0,0)$
0.50	A	0.1364	0.3636	0.5000	0.6364	1.2727	0.5000	0.2500
	B_2	0.0767	0.4233	0.5000	0.3578	1.4817	0.5000	0.2500
	B_2^{H}	0.0833	0.4167	0.5000	0.3889	1.4583	0.5000	0.2500
	C_2	0.0712	0.2521	0.3233	0.3323	0.8823	0.5356	0.2678
	C_2^{H}	0.0778	0.2593	0.3370	0.3630	0.9074	0.5333	0.2667
0.70	A	0.3000	1.3333	1.6333	1.0000	3.3333	0.3000	0.2100
	B_2	0.1545	1.4788	1.6333	0.5151	3.6970	0.3000	0.2100
	B_2^{H}	0.1750	1.4583	1.6333	0.5833	3.6458	0.3000	0.2100
	C_2	0.1392	0.6967	0.8358	0.4639	1.7417	0.3696	0.2587
	C_2^{H}	0.1591	0.7424	0.9015	0.5303	1.8561	0.3636	0.2545
0.90	A	0.5651	7.5346	8.0997	1.4651	14.6506	0.1000	0.0900
	B_2	0.2626	7.8371	8.0997	0.6808	15.2388	0.1000	0.0900
	B_2^{H}	0.3115	7.7882	8.0997	0.8077	15.1437	0.1000	0.0900
	C_2	0.2292	1.9245	2.1537	0.5941	3.7421	0.2146	0.1931
	C_2^{H}	0.2760	2.2084	2.4844	0.7157	4.2941	0.2025	0.1823

11.4 Convergence to the baseline

Each specialised model must recover Model-A as its distinguishing parameter vanishes; the comparison above establishes this algebraically, and the CTMC pipeline confirms it numerically. Measured in the L_1 distance $\|\pi - \pi_A\|_1$, all four specialised models — B_2, C_2, B_2^{H}, C_2^{H} — approach the baseline at first order in their distinguishing parameter, with fitted log–log slopes of 0.85–0.93. For the conserving models B_2 and B_2^{H} the empty-state descriptors π_0 and $\pi(0,0)$ equal their Model-A values *identically* for every $\gamma_1 > 0$ (Corollaries 3, 5), so the approach is carried entirely by interior redistribution of mass.

12 Conclusion and Future Work

This thesis analysed a two-class non-preemptive priority $M/M/1$ queue under two waiting-room mechanisms — jockeying and abandonment. It obtained the exact stationary bivariate PGF $P(x,y)$ in closed or integral form for five models: the baseline A, one-way jockeying B_2, class-1 abandonment C_2, and the head-of-line variants B_2^{H} and C_2^{H}. Underpinning these is a single structural finding: one object governs tractability throughout — the algebraic form of the fundamental PGF equation of Lemma 1. The hierarchy delivers less a catalogue of closed forms than a map of *why* each model is solvable or not, and that map is the central contribution.

12.1 The obstruction taxonomy

Remark 1 reads the fundamental equation as a competition between an algebraic coefficient and two convection terms, and the analysis bears out a clean dichotomy, summarised model by model in the comparison of §11.2 (Table 5). Where the equation is solvable, the unknown boundary function is determined by exactly one of four *pinning mechanisms*. The first is evaluation at a kernel root lying strictly inside the unit disc, the purely algebraic route of the baseline Model-A (Section 5). The second is interior analyticity: when one-way jockeying survives, the factor $(x - y)$ places the operative singularity at the diagonal point $x = y$ inside the disc, and the boundary function is fixed by demanding holomorphy there (Model-B_2, Section 7). The third is boundary boundedness: when class-1 abandonment survives, the factor $(x - 1)$ moves the singularity onto the unit-circle boundary $x = 1$, where mere boundedness of the PGF suffices to pin the unknown (Model-C_2, Section 8). The fourth is head-of-line algebraic collapse: restricting a mechanism to the customer at the head of the queue replaces a length-proportional rate by a constant one, annihilates the derivative term, and returns the equation to the algebraic Model-A form, solvable by the same kernel argument (Models B_2^{H} and C_2^{H}, Sections 9–10).

Where the equation is not solvable, exactly two irreducible obstructions appear, and they are what Models X and B exist to expose. The first is *transcendental characteristics*: in the full model the method of characteristics no longer follows straight lines but integral curves of a nonlinear ODE, so the kernel root is unavailable in closed form. The second is *diagonal gradient coupling*: closing the reduced equation requires a Volterra integral equation whose forcing depends on a boundary or diagonal trace of the very solution being sought. In Model-B the straight-line characteristics survive but the boundary function still satisfies a first-kind Volterra equation, while in Model-X both obstructions compound, the anchoring diagonal value $P(z,z)$ itself becoming unknown. The hierarchy is therefore not a loose collection of cases but a complete enumeration: four ways the functional equation can close, and two reasons it cannot.

12.2 The mechanism dichotomy

The two mechanisms act in opposite registers. Jockeying *redistributes* delay at fixed total work: relocating a waiting customer conserves the total count, so $\mathbb{E}[N]$, π_0 and $\pi(0,0)$ are identical to the baseline for every γ_1. The mechanism's entire effect is to move delay between classes — the priority premium $\mathbb{E}[W_2]/\mathbb{E}[W_1]$ inflating to 22.38 at $\rho = 0.90$ against the Model-A value 10.00.

Abandonment instead *sheds* load: the idle probability rises, every queue shortens, and total work falls.

12.3 Operational reading and the deterioration-direction caveat

These results close the healthcare loop opened in the Introduction. The rates γ_1 and θ_1 are the model's two clinical dials: θ_1 is the rate at which a waiting high-priority patient leaves the list by death, transfer, or deterioration past the point of benefit, and γ_1 is the rate at which a waiting class-1 patient is moved into the class-2 queue. Here an honest caveat is owed. The Introduction motivates jockeying as *deterioration* — a waiting patient becoming more urgent, that is an *upgrade* from the lower to the higher class. The solved model, Model-B_2, instead implements one-way jockeying in the *opposite* direction, $1 \to 2$, a downgrade out of the priority queue. The readings partly reconcile if the $1 \to 2$ transition is read as the events that genuinely move demand downward: administrative reclassification, clinical recovery, or a triage demotion. But the deterioration direction proper — an upgrade $2 \to 1$, that is $\gamma_2 > 0$ — is exactly the bidirectional case that Model-B leaves open as a Volterra equation (Section 6). The mechanism the clinical motivation most wants is thus the one the analysis identifies as irreducibly hard, which sharpens rather than weakens the case for the future work below.

For a waiting-list manager the takeaway is concrete: monitor the second-tail mass $\mathbb{P}(N_1 \geq 2)$, the proxy for the head-of-line gap of Section 9.

12.4 Limitations

Several assumptions bound the contribution. Every result rests on exponential interarrival, service, jockeying, and abandonment times. Relaxing any of these — general service times in particular — breaks the Markov structure on which the PGF method depends, as the $M/G/1$ intractability behind de Waal's approximate analysis already illustrates. The treatment is single-server throughout, and it reports means rather than full waiting-time distributions. The partial-PGF result on the alternative state space is an *approximation*, not a theorem (Approximation 1): it is reliable only in the priority-dominated, moderate-load region, and unreliable elsewhere. The truncated-CTMC validation carries a small downward bias for the conserving (non-abandoning) models at $\rho \geq 0.92$, already flagged in §11.2; those near-critical points should be read as lower bounds. Finally, the class-2 waiting time reported for Model-B_2 is a per-entrant queue residence recovered through Little's law, not the end-to-end delay of a tagged customer followed across a jockeying transition — a gap the tagged-customer item below is meant to resolve.

12.5 Future work

Several extensions follow directly from the machinery already built, ranked here from the most immediate.

1. *Head-of-line versions of the remaining models.* The head-of-line collapse that made Models B_2^{H} and C_2^{H} algebraic applies verbatim to the still-unsolved members of the hierarchy — a head-of-line B_1 ($2 \to 1$ jockeying), C and C_1 (class-2 and two-class abandonment), and even a head-of-line X carrying *both* mechanisms at the queue head. Each replaces a length-proportional rate by a constant $\mathbf{1}_{\{n_1 \geq 1\}}$ rate, removes the derivative term, and is solvable

by the Kernel Method exactly as in Sections 9–10. This is near-term low-hanging fruit that needs no new tools.

2. *The k-customer head-of-line ladder.* The head-of-line variants let only the single customer at the head of Queue 1 jockey or abandon; the natural next rung lets the first *two* waiting class-1 customers do so, then the first three, and so on — replacing the rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ by $\gamma_1 \min(n_1, k)$ for $k = 2, 3, \ldots$ (and likewise $\theta_1 \min(n_1, k)$). Writing $\min(n_1, k) = \sum_{j=1}^{k} \mathbf{1}_{\{n_1 \geq j\}}$, the bounded rate keeps the fundamental equation *algebraic* — no derivative term appears — at the price of k unknown boundary functions $\widetilde{P}(n_1, y)$, $n_1 = 0, \ldots, k-1$, in place of the single $P_y(y)$; whether the Kernel Method supplies enough pinning conditions for $k \geq 2$ is precisely the open question. Since $\min(n_1, k) \uparrow n_1$, the ladder interpolates monotonically between B_2^{H} (or C_2^{H}) at $k = 1$ and the length-proportional Models B_2 and C_2 in the limit $k \to \infty$, so each solvable rung would yield a constructive approximation to the full-rate models — a discrete counterpart to the parameter limits studied in Section 11, whose convergence in k the CTMC pipeline of that section could quantify directly.

3. *Numerical inversion of the Model-B Volterra equation.* Unlike Model-X, Model-B yields a first-kind Volterra equation with an *explicit* kernel and a *known* forcing (Section 6); numerical quadrature of the boundary function is therefore immediately feasible and would deliver the class-2 distribution under bidirectional jockeying without a closed form.

4. *Boundary-value methods for Model-B.* A closed-form attack on the same equation belongs to the Riemann–Hilbert and boundary-value tradition of [FIM99], a programme that continues to yield explicit two-class results — see the analytic-continuation analysis of Devos, Walraevens, Fiems & Bruneel [DWFB20]; the straight-line characteristics of Model-B make it the natural first test case for importing that machinery into the present framework.

5. *Tagged-customer sojourns.* The end-to-end delay of a class-2 customer in Model-B_2, followed across a possible jockeying transition, and the conditional time-to-service in Model-C_2 — the latter already estimated by simulation as 0.3863 against the count-based 0.4639 — both call for an analytic tagged-customer treatment that the mean-based analysis here does not provide.

6. *Waiting-time distributions.* The present means could be lifted to full sojourn-time distributions through a distributional Little's law, supplying the tail and quantile information that a mean cannot.

7. *Multi-server extension.* Carrying the mechanisms to $M/M/c$ would connect this exact single-server line to the approximate multi-server scheduling results of [HCD22], trading the exactness the single server buys for operational realism.

8. *Synchronised abandonment.* The independent patience clocks of Model-C_2 admit an orthogonal generalisation in which waiting customers abandon *simultaneously*, each with probability p, at Poisson opportunity epochs. Kapodistria [Kap11] solves the single-class case exactly through basic (q-)hypergeometric series with deformation parameter $q = 1 - p$, interpolating between the $M/M/1{+}M$ ($p \to 0^+$, the Kummer regime of the class-1 subsystem here) and an $M/M/1$ with total catastrophes ($p \to 1^-$). Whether the binomial bulk-departure structure survives the two-class non-preemptive setting — in effect a q-deformation of the Model-C_2 analysis — is open, and would add a synchronisation axis to the mechanism hierarchy of this thesis.

9. *Calibration.* Fitting γ_1, θ_1 and the loads to real elderly-care waiting-list data [Zen99] would test whether the regimes identified here describe operational reality.

A final, self-contained item is the closure of the partial-PGF recursion on the alternative state space $\widetilde{S}$, deferred from the analysis of Model-A (Section 5.2). A rigorous treatment would retain the boundary-diagonal forcing that Approximation 1 neglects and solve the resulting non-homogeneous recurrence by variation of constants in n. As anticipated there, this reintroduces the unknown diagonal sequence $\{\pi(0, n)\}$ as the quantity to be determined self-consistently. Settling whether that recursion closes, and proving the conjectured monotonicity of $\mathbb{E}[N_2]$ in θ_1, would tie off the loose ends this thesis has deliberately left explicit.

References

[AR02] Ivo Adan and Jacques Resing. Queueing theory. *Eindhoven University of Technology*, 180:10, 2002.

[Coh56] Jacob Willem Cohen. Certain delay problems for a full availability trunk group loaded by two traffic sources. *Communication News*, 16:105–113, 1956.

[DLM] NIST digital library of mathematical functions. https://dlmf.nist.gov/. Release 1.1.12, 2023. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

[dW92] Peter de Waal. *Approximate analysis of an M/G/1 priority queue with priority changes due to impatience*. CWI Report BS-R9202. CWI, Department of Operations Research, Statistics, and System Theory, Amsterdam, 1992.

[DWFB20] Arnaud Devos, Joris Walraevens, Dieter Fiems, and Herwig Bruneel. Analysis of a discrete-time two-class randomly alternating service model with Bernoulli arrivals. *Queueing Systems*, 96:133–152, 2020.

[FIM99] Guy Fayolle, Roudolf Iasnogorodski, and Vadim Malyshev. *Random Walks in the Quarter-Plane: Algebraic Methods, Boundary Value Problems and Applications*. Springer, 1999.

[Gau67] Walter Gautschi. Computational aspects of three-term recurrence relations. *SIAM Review*, 9(1), 1967.

[GMR02] Ofer Garnett, Avishai Mandelbaum, and Martin I. Reiman. Designing a call center with impatient customers. *Manufacturing & Service Operations Management*, 4(3), 2002.

[HCD22] Yue Hu, Carri W. Chan, and Jing Dong. Optimal scheduling of proactive service with customer deterioration and improvement. *Management Science*, 2022.

[JR14] Oualid Jouini and Alex Roubos. On multiple priority multi-server queues with impatience. *Journal of the Operational Research Society*, 65(5):616–632, 2014.

[Kap11] Stella Kapodistria. The M/M/1 queue with synchronized abandonments. *Queueing Systems*, 68(1):79–109, 2011.

[STZ14] David A. Stanford, Peter Taylor, and Ilze Ziedins. Waiting time distributions in the accumulating priority queue. *Queueing Systems*, 77:297–330, 2014.

[WBSW15] Jianfu Wang, Opher Baron, and Alan Scheller-Wolf. M/M/c queue with two priority classes. *Operations Research*, 63(3):733–749, 2015.

[Zen99] Stefanos A. Zenios. Modeling the transplant waiting list: A queueing model with reneging. *Operations Research*, 47(3), 1999.

[ZK23] Josef Zuk and David Kirszenblat. Explicit results for the distributions of queue lengths for a non-preemptive two-level priority queue. *arXiv preprint arXiv:2309.09428*, 2023.

[ZM05] Sergey Zeltyn and Avishai Mandelbaum. Call centers with impatient customers: Many-server asymptotics of the M/M/n+G queue. *Queueing Systems*, 51, 2005.